TOPIC NAME :

DAY :

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#Introduction To Electrical Circuits

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Voltage, Current, Resistance

Ch:2, and Ch:3

OHM's LAW, power, Energy, Series Resistors

Ch:4, and Ch:5

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Circuit:—

⇒ Circuit means a path way of electrical current due to applying a potential difference (or voltage) consisting of conductors or semi-conductors with suitable other components.

Current:—

⇒ Electrical current is defined as rate of charge or amount of charge per unit time.

$$\therefore \text{Current } \boxed{I = \frac{\text{charge}}{\text{time}} = \frac{Q}{t}}$$

Unit of current is Ampere(A)

Q:- 1A, what does it means?

⇒ 1A means the capacity or rate of 1 Coulomb (1C) charge per unit second(s)

$$\therefore 1A = \frac{1C}{1sec} \text{ A or c/sec}$$

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Note:-

⇒ Here to carry the current means flow the charge or free electrons.

Order of Electron flow:-

1. Conductor → Cu → High tensile strength
2. Semi-Conductor → Si → Medium tensile strength
3. Insulator → Rubber → Low tensile strength

Voltage or potential Difference:-

⇒ It is defined as the work done to move 1 Coulomb (1c) charge from one point to another point with in the electric field.

1. Voltage or potential difference is always defined for one point with respect to another (or a reference point)

Reference point:-

⇒ As sum of all charges are zero at ground hence surface is always considered 0V and used as reference point.

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② For electrical charges:-

(i) Like charges will repel each other.

(ii) Opposite charges will attract each other.

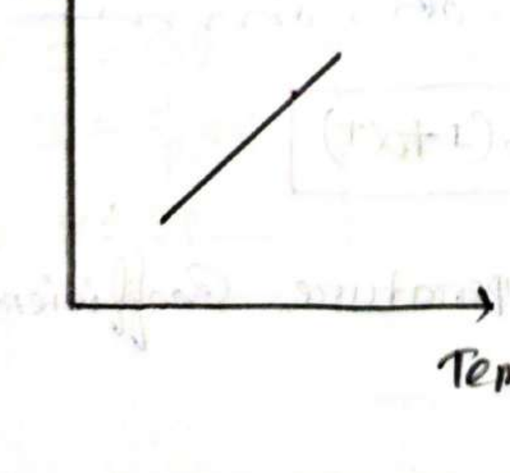
Resistance:-

⇒ It is the ability to restrict the flow of current.

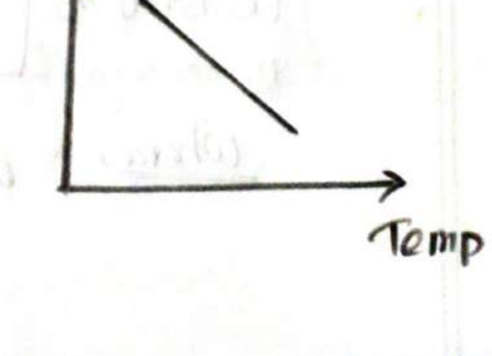
1. Due to free electrons, the electrical current is Established in a circuit. These charges face restriction due to presence of other charges in the material.

Graph of Resistance Vs Temperature:-

(i) For Conductors:-



(ii) For Semiconductors:-



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Order of Resistance:-

① For Conductors - Materials:-

⇒ Resistance is the lowest, (ρ is low), (G is high)

② For Insulators:-

⇒ Resistance is the Highest (ρ is high), (G is Low)

Relationship between Resistance, Resistivity, Length & Area

∴ Resistance $\boxed{R = \frac{\rho l}{A} \ \Omega \text{(ohm)}}$

Where, ρ = resistivity in Ω(m)

l = length of material (m)

A = Cross Section Area of material (m^2)

In terms of Temperature:-

Resistance $\boxed{R = R_0(1+\alpha T)}$

Where, α = temperature Coefficient

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Conductivity:—

⇒ It in the exact opposite of resistivity and Conductance is the inverse of resistance. denote by G

$$\boxed{G = \frac{1}{R} \text{ mho}(\mho)/\text{Siemens}(S)}$$

(1) If there in more resistance:—

⇒ The Conductance will be low and current will be low. or Vice Versa.

Ohm's Law:—

⇒ At a Constant temperature current passing through a Conductor in directly proportional to it potential difference between two point on it.

$\therefore I \propto V$

$$\boxed{\begin{aligned} &\Rightarrow I = GV \\ &\Rightarrow I = \frac{1}{R} \times V \\ &\Rightarrow V = IR \end{aligned}}$$

where,
G = Conductivity
R = Resistance

Note:-

1. Voltage and Current are active parameters.
2. Resistance or Conductance is passive parameters.

Combination of Resistance:-

⇒ Resistors can be combined as following

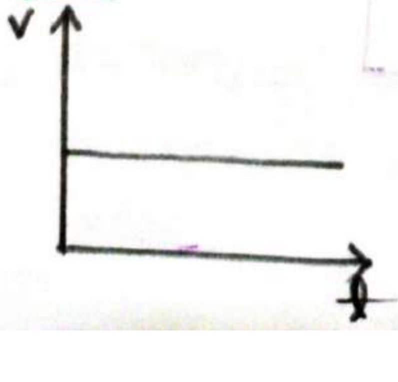
1. Series Combination.
2. Parallel Combination.

Source of Voltage, Current & Resistor

1. Voltage Source (unit → Volt)

or

Graph:-



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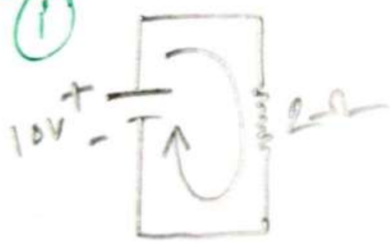
(2) Current Sources (Unit → A)

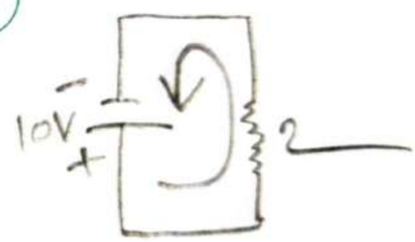
(3) Resistor Sources (Unit → Ω)

Ex:- (i)



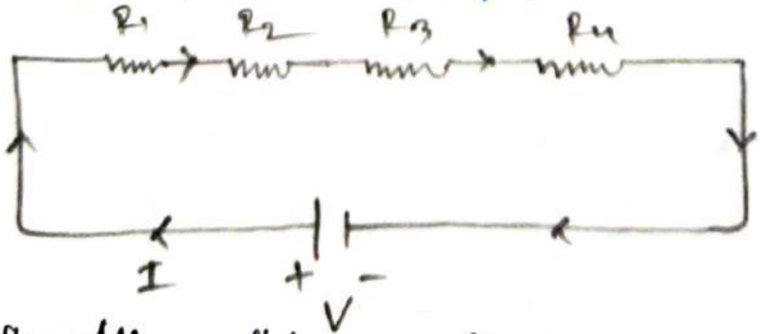
$I = \frac{V}{R} = \frac{10}{2} = 5A$

(ii)



$I = \frac{V}{R} = \frac{10}{2} = 5A$

Series Combination:-



⇒ For Series:- All resistor will carry the same current.

∴ Total equivalent resistance

$$\boxed{R_T = R_1 + R_2 + R_3 + R_4 + \cdots\cdots}$$

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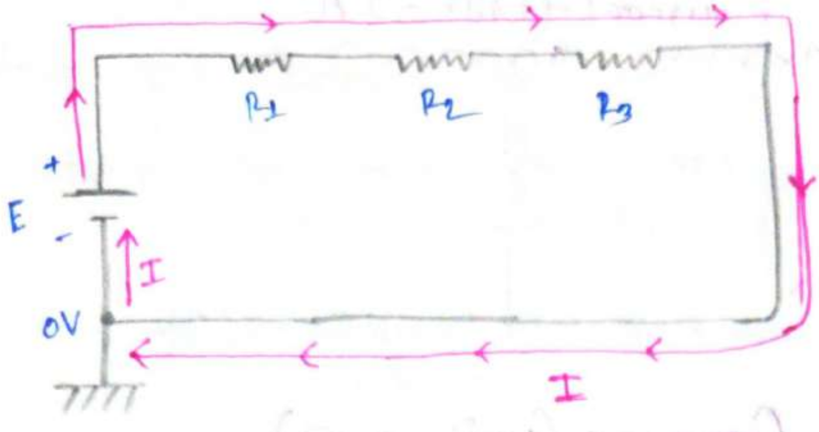


Here,
E = Battery or DC source Voltage.

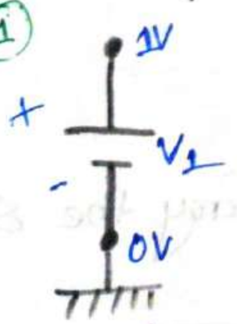
I = Source current in Amp

$\overline{////}$ or $\underline{\perp}$ = Symbol for ground or Earthing represents '0'V.

How to Define Voltage:-

1. V_1 or V_a = Voltage with respect to ground (0V)
2. V_{ab} = Voltage with respect to point 'b' of point 'a'
3. V_{12} = Voltage of point '1' with respect to point '2'

Ex:- ①



$+V_1$ Volt

②

$-V_1$ Volt

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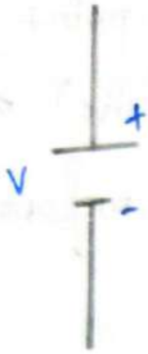
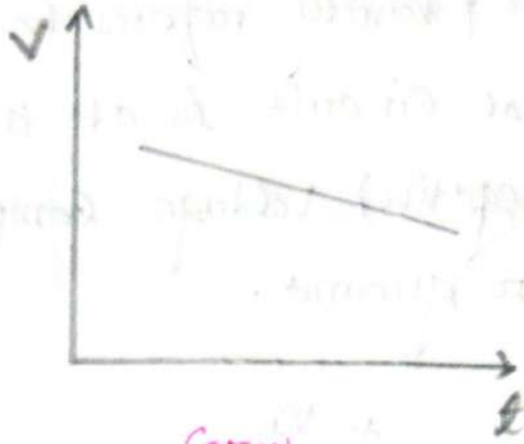
Source of DC-Current:-



Symbol.



Graph.

1. One-Time battery:- Ex: (AA, AAA, C type, D type, etc)
2. Rechargable batteries:- Ex: (Pb-acid, NiMH, Li-ion, etc)
3. DC generator (Rotary machine):- Ex: (PV cells, fuel cell, etc)

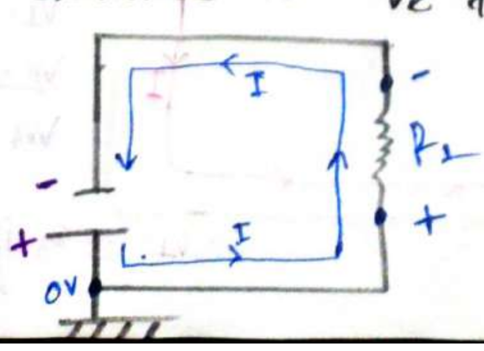
Direction of electrical Current:-

⇒ Electrical Current Direction is always opposite to e^- flow direction

For Source/Battery:-

⇒ +Ve terminal to -Ve terminal.

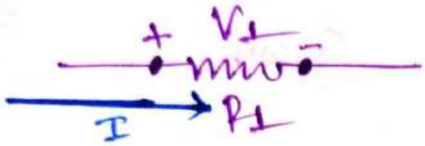
Ex:-



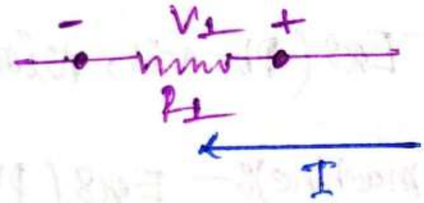
Voltage polarity:—

⇒ Voltage polarity refers to whether a point in an electrical circuit is at a higher (positive) or lower (negative) voltage compared to a reference point or ground.

(1)



(2)



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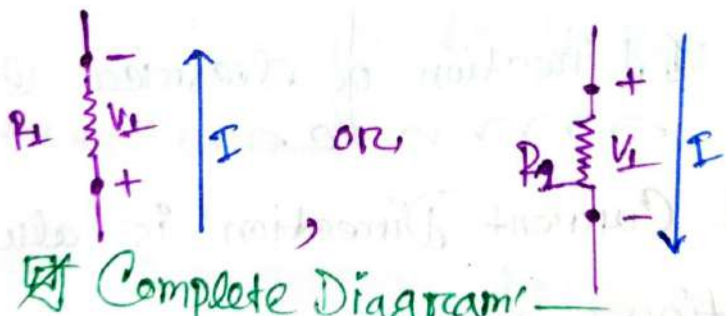
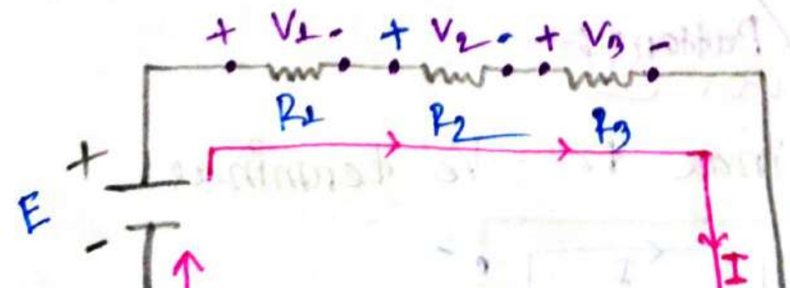


Complete Diagram:—



<u>Where</u>

$V_1 = IR_1$

$V_2 = IR_2$

$V_3 = IR_3$

$V_1 = V_2 = V_3$ if $R_1 = R_2 = R_3$

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power:—

⇒ Rate of work done is defined as power. denote by P. and unit W (watts).

$$\therefore \boxed{P = \frac{W}{t} \; Js^{-1}}$$

unit = Watt = Js^{-1}

power with respect to electrical circuit:—

we know,

$$V = \frac{W}{Q}$$

$$I = \frac{Q}{t}$$

$$I = \frac{V}{R} \text{ or } V = IR$$

$$\therefore \boxed{P = \frac{W}{t} = \frac{QV}{t} = VI = I^2R = \frac{V^2}{R}}$$

where,

Q = Charge.
V = Voltage.
I = Current
R = Resistor

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Power delivered by the Source:-

$$\boxed{P = VI}$$

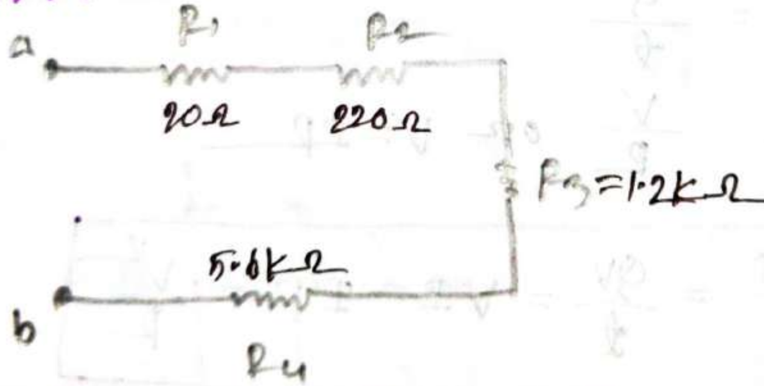
Power consumed by a resistor/load:-

$$\boxed{P = I^2R = V^2/R}$$

Note:-

⇒ Resistor convert consumed electrical energy to heat.

Problem: 5.1:-



⇒ Determine the total resistance of the series connection in the Fig. Note that all the resistors appearing in this network are standard values.

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In given:—

$R_1 = 20\,\Omega$

$R_2 = 220\,\Omega$

$R_3 = 1200\,\Omega$

$R_4 = 5600\,\Omega$

We know,

For Series Connections:—

$R_T = R_1 + R_2 + R_3 + R_4$

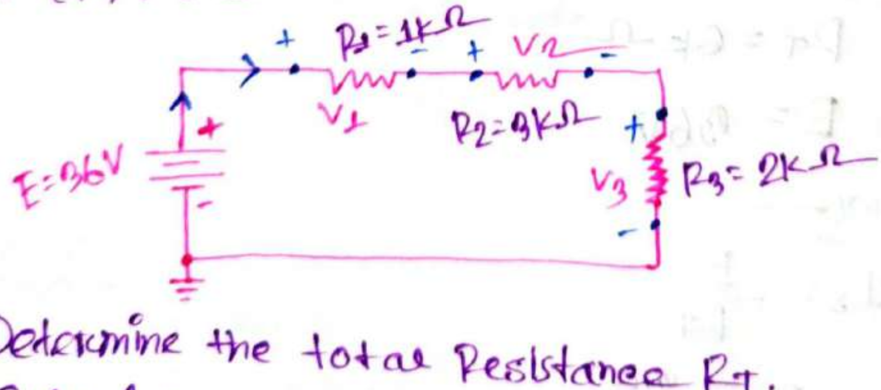
$= (20 + 220 + 1200 + 5600)\,\Omega$

$= 7040\,\Omega$

$= 7.04\,k\Omega$

Ans:—

Problem No. 5.7



1. Determine the total Resistance R_T.
2. Calculate the Current I_S.
3. Determine the Voltage across each resistor.
4. Find the power Supplied by the battery.
5. Determine the power dissipated by each resistor.
6. Prove that the total power supplied equals the total power dissipated.

(1)

In given:—

$R_1 = 1000\,\Omega = 1k\Omega$

$R_2 = 3000\,\Omega = 3k\Omega$

$R_3 = 2000\,\Omega = 2k\Omega$

We know:—

For Series Connection:—

$R_T = R_1 + R_2 + R_3$

$= (1000 + 3000 + 2000)\,\Omega$

$= 6000\,\Omega$

$= 6k\Omega$

(2)

We have:—

$R_T = 6k\Omega$

$E = 36V$

We know:—

$I_S = \frac{E}{R_T}$

$= \frac{36}{6 \times 10^3}\,A$

$= 6 \times 10^{-3}\,A$

$= 6\,mA$

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(3)

We have:—

$I_s = 6 \times 10^{-3} A$

$R_1 = 1 k\Omega$

$R_2 = 3 k\Omega$

$R_3 = 2 k\Omega$

We know:—

$V = IR$

$\therefore V_1 = I_s R_1 = 6 \times 10^{-3} \times 1 \times 10^{3}$

$= 6$ Volt

$V_2 = I_s R_2 = 6 \times 10^{-3} \times 3 \times 10^{3}$

$= 18$ Volt

$V_3 = I_s R_3 = 6 \times 10^{-3} \times 2 \times 10^{3}$

$= 12$ Volt.

Ans:—

(4)

We have:—

$E = 36 V$

$I_s = 6 \times 10^{-3} A$

We know:—

power supplied by the battary

$P = EI$

$= 36 \times 6 \times 10^{-3}$

$= 0.216 W$

$= 216 mW$

Ans:—

5

We have:—

$E = 36V$

$R_1 = 1k\Omega$

$R_2 = 3k\Omega$

$R_3 = 2k\Omega$

$I_S = 6\times10^{-3}A$

We know:—

power dissipated by resistor:—

$P = V_x^2/R_x = I_S^2 R_x = V_x I_S$

$\therefore P_1 = I_S^2 R_1 = (6.0\times10^{-3})^2 \times 1000 = 0.036\,W = 36\,mW$

Now $V_2 = I_S R_2$

$= 3000 \times 6\times10^{-3}$

$= 18$ Volt

$\therefore P_2 = \frac{V_2^2}{R_2} = \frac{(18)^2}{3000} = 0.108W = 108\,mW$

Again:— $V_3 = I_S R_3$

$= 6\times10^{-3} \times 2000$

$= 12$ Volt

$\therefore P_3 = V_3 I_S = 12 \times 6.\times10^{-3} = 0.072\,W = 72\,mW$

Ans:—

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(6)

we found:—

power supplied by the battary = 216 mw

power dissipated by the resistors :—

$P_1 = 36\ mw$

$P_2 = 105\ mw$

$P_3 = 72\ mw$

$\therefore\ P_E = P_1 + P_2 + P_3$

$216 = 36 + 108 + 72$

$216 = 216$

L.H.S = R.H.S

so (proved)

Ans:—

problem 4.1:—

⇒ Determine the current resulting from the application of a 9V battary with a resistance of 2.2 Ω

⇒ Is given:—

$V = 9V$

$R = 2.2\ \Omega$

we know:—

$I = \frac{V}{R} = \frac{9}{2.2} = 4.09A.$

Ans:—

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problem: 4.2:—

⇒ Calculate the resistance of a 60w bulb if a current of 500mA result from an applied voltage of 120V

⇒ Is given:—

$P = 60w$

$I = 500mA = 500 \times 10^{-3} A$

$V = 120v$

we know:—

$P = I^2 R$

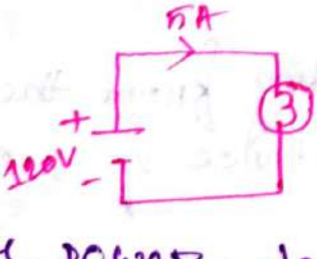
$\therefore R = \frac{P}{I^2}$

$= \frac{60}{(500 \times 10^{-3})^2}$

$= \underline{240 \, \Omega}$

Ans:—

Problem: 4.6:—



(i)

⇒ Find the power delivered by the Source.

(ii)

⇒ What is the power dissipated by a 5Ω resistor if the current is 4A ?

(i)

Is given:—

$V = 120V$

$I = 5A$

We know:—

power delivered by the Source:—

$P = VI$

$= 120 \times 5$

$= 600W$

Ans:—

(ii)

In given:—

$R = 5\Omega$

$I_s = 4A$

We know:—

power dissipated by the resistor:—

$P = I^2R$

$= (4)^2 \times 5$

$= 80W$

Ans:—

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problem: 20:

⇒ If 420J of energy are absorbed by a resistor In 4 min. What is the power of resistor?

⇒ Is given:

$W = 420J$

$t = 4$ min

$= 4 \times 60$ sec

$= 200 + 40$ sec

$= 240$ sec

We know:

$$P = \frac{W}{t}$$

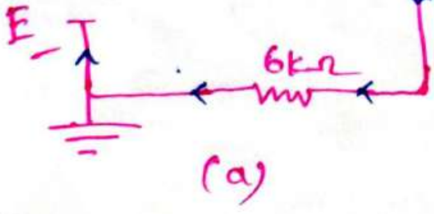
$$= \frac{420}{240}$$

$$= 1.75\,W$$

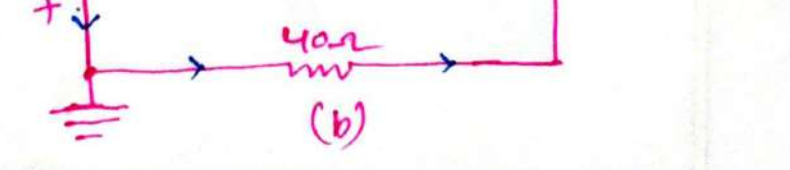
Ans:

problem 89:

⇒ Find the applied Voltage necessary to develop the Current specified in each circuit



(a)



(b)

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(a)

In given:—

Let:—

$R_1 = 12 k\Omega$

$R_2 = 4 k\Omega$

$R_3 = 6 k\Omega$

$I_s = 4 mA = 4.0 \times 10^{-3} A$

We know:—

$R_s = R_1 + R_2 + R_3$

$= (12 + 4 + 6) k\Omega$

$= 22 k\Omega$

$E = V = I_s R_s$

$= (4.0 \times 10^{-3} \times 22 k\Omega)$

$= 4 \times 10^{-3} \times 22 \times 10^{3}$

$= 88$ Volt

Ans:—

(b)

In given:—

Let:—

$R_1 = 18 \Omega$

$R_2 = 14 \Omega$

$R_3 = 8 \Omega$

$R_4 = 40 \Omega$

$I_s = 250 mA = 250 \times 10^{-3} A$

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We know:—

$$R_S = R_1 + R_2 + R_3 + R_4$$
$$= 18 + 14 + 8 + 40$$
$$= 80\ \Omega$$

$$E = V = I_S R_S$$
$$= 80 \times 250 \times 10^{-3}$$
$$= 20\ \text{Volt}.$$

Ans:—

problem:

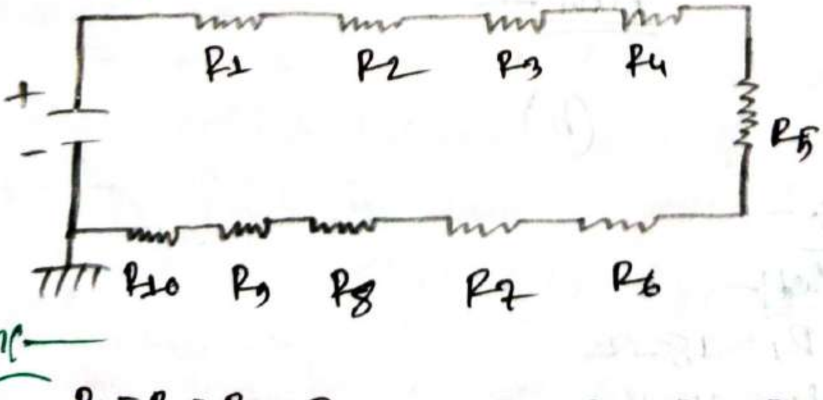
⇒ A 9V DC source is connected in series with ten $2M\Omega$ resistor, Find

(i) Total resistance

(ii) Current

(iii) Voltage for each resistance.

⇒



In given:—

$$R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = R_7 = R_8 = R_9 = R_{10} = 2M\Omega$$

(i)

We know:—

$$R_T = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7 + R_8 + R_9 + R_{10}$$
$$= 2 \times 10 = 20 M\Omega$$

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(ii)

We have:—

$R_T = 20 M\Omega$

$E = 9V$

We know:—

$E = IR_T$

$I = \frac{E}{R_T}$

$= \frac{9}{20M}$

$= \frac{9}{20 \times 10^6}$

$= 4.5 \times 10^{-7} A$

(iii)

We have:—

$R_T = 20 M\Omega$

$E = 9V$

$I = 4.5 \times 10^{-7} A$

Since, All resistors are same value.
So, Voltage for each resistor are same.

∴ We know:—

$V_1 = IR_1$

$= 4.5 \times 10^{-7} \times 2.0 \times 10^6$

$= 0.9$ Volt

∴ Voltage for each resistor $= 0.9$ Volt.

Ans:—

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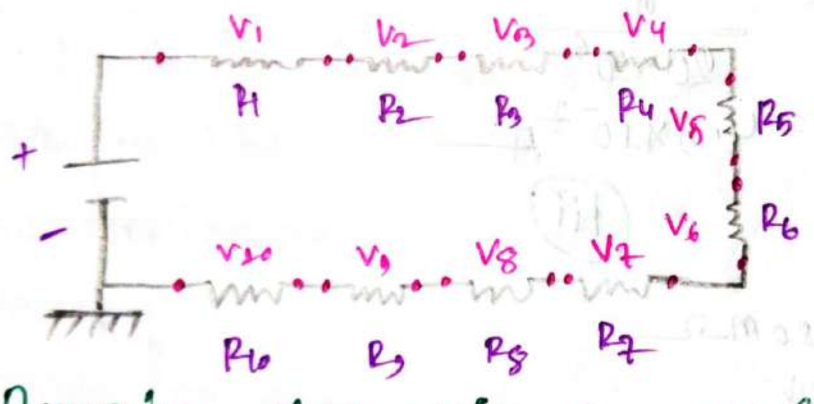
Kirchoff's Voltage law (KVL)

⇒ In an enclosed circuit, sum of all Voltages are zero. or Sum of Voltage rise equals sum of Voltage drop.

(1) For KVL, calculation should start from ground to Source, to each resistors and other Components and back to ground.

Ex:-



☒ Assuming clock-wise Apporoch:-(ACWA):-

(1) '–' to '+' → Indicates Voltage rise

(2) '+' to '–' → Indicates Voltage Drop

☒ Assuming Anti-Clock-wise Apporoch:-(AACWA):-

(1) '–' to '+' → Indicates Voltage Drop

(2) '+' to '–' → Indicates Voltage rise.

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Note!—

⇒ The Voltage of the battary/Source will always rise.

From the KVL Law!—

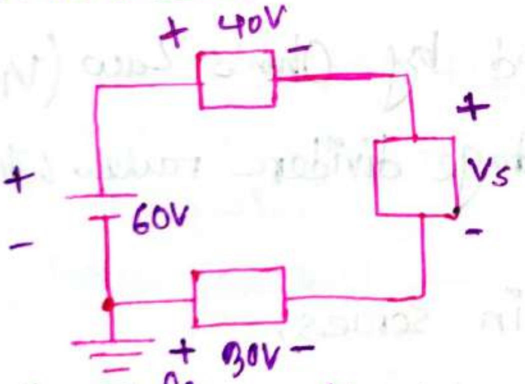
$\therefore E + V_1 + V_2 + V_3 + V_4 + V_5 + V_6 + V_7 + V_8 + V_9 + V_{10} = 0$

$$\boxed{E = -(V_1 + V_2 + V_3 + V_4 + V_5 + V_6 + V_7 + V_8 + V_9 + V_{10})}$$ ['−' sign indicate Voltage drop]

From the Last Law of KVL!—

$$\boxed{E = V_1 + V_2 + V_3 + V_4 + V_5 + V_6 + V_7 + V_8 + V_9 + V_{10}}$$

problem: 5.11—



⇒ Using kirchhoff's Voltage law, determine the unknown Voltage for the circuit.

⇒ is given!—

$E = 60V$
$V_1 = 40V$
$V_s = ?$
$V_2 = 30V$

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we know:—

$\Rightarrow E - V_1 - V_s + V_2 = 0$ [from KVL Law]

$\Rightarrow V_s = E - V_1 + V_2$

$= 60 - 40 + 30$

$= 50V$

$\therefore V_s = 50V$.

Ans:—

Voltage Divider rule (VDR)

⇒ For a series circuit, the current remains same through all components but the voltage of each component can be found by Ohm's Law ($V_1 = IR_1$, $V_2 = IR_2$ etc)

⇒ Another option is Voltage divider rule, which is the follows:—

- For 'n'th resistor in series,

Voltage for 'n'th resistor

$$\boxed{V_n = \frac{R_n}{R_T} \times E}$$

R_T = Equivalent total resistance for series circuit.

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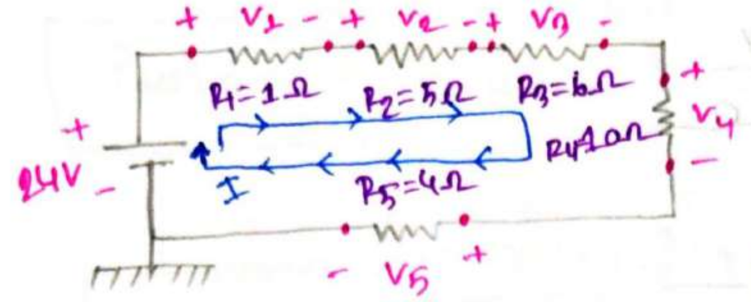
problem:—



① Find the voltage Drops for 6Ω and 4Ω Resistor

② Find the total Current

③ Find the rest of the Voltage drop and verify KVL

①

⇒ Is given:—

$R_1 = 1\Omega,\ R_2 = 5\Omega,\ R_3 = 6\Omega$

$R_4 = 10\Omega\ \ R_5 = 4\Omega,\ E = 24V$

We know:—

$R_T = R_1 + R_2 + R_3 + R_4 + R_5$

$= 1+5+6+10+4$

$= 26\Omega$

Voltage Drops for 6Ω:—

$$V_3 = \frac{R_3}{R_T} \times E$$

$$V_3 = \frac{6}{26} \times 24$$

$$= 5.538 \text{ Volt}$$

Voltage Drops for 4Ω:—

$$V_5 = \frac{R_5}{R_T} \times E$$

$$= \frac{4}{26} \times 24 = 3.692 \text{ Volt}$$

(Ans:)

(2)

we have:—

$E = 24\,V$

$R_T = 26\,\Omega$

we know:—

$E = V = IR$

$\therefore I_s = \frac{E}{R_T}$

$= \frac{24}{26}$

$= 0.923\,A$

Ans:—

(3)

we have:—

$V_3 = 5.538\,Volt$

$V_5 = 3.692\,Volt$

we know.—

$V_n = I^2 R_n$

$\therefore V_1 = I_s^2 R_1 = (0.923)^2 \times 1 = 0.923\,Volt$

$\therefore V_2 = I_s R_2 = (0.923) \times 5 = 4.615\,Volt$

$\therefore V_4 = I_s R_4 = (0.923) \times 10 = 9.23\,Volt$

From KVL—

$E = V_1 + V_2 + V_3 + V_4 + V_5$

$24V = (0.923 + 4.615 + 5.538 + 9.23 + 3.692)\,V$

$24V = 24V$ (verified)

Ans:—

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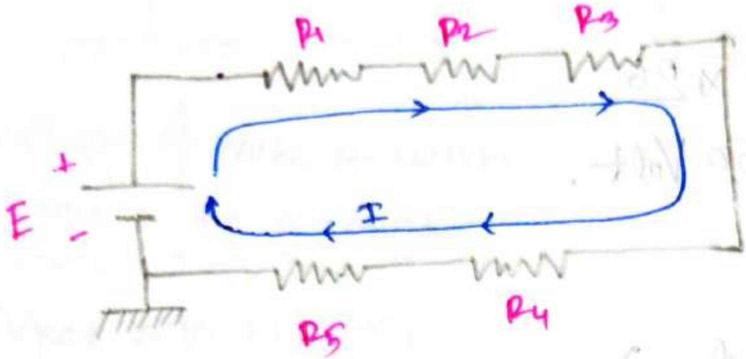
Tellegen's Law (for power):—

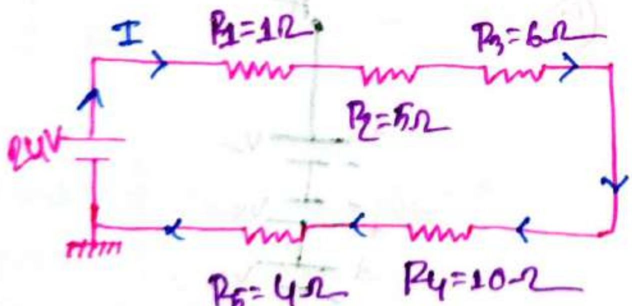
$$\boxed{P_{delivered} = P_{consumed}}$$



$\therefore$ $\boxed{P = EI}$ = power delivered by the source

$\boxed{P = I^2R_1 + I^2R_2 + I^2R_3 + I^2R_4 + I^2R_5}$ → power consumed

problem:—



(1) Verify the Tellegen's Law?

⇒ We know,—

$R_T = R_1 + R_2 + R_3 + R_4 + R_5$

$= 1 + 5 + 6 + 10 + 4$

$= 26\Omega$

$I_s = \frac{E}{R_T} = \frac{24}{26} = 0.923A$

$\therefore$ Power delivered—

$P_d = EI = 24 \times 0.923 = 22.152W$

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Power Consumed:—

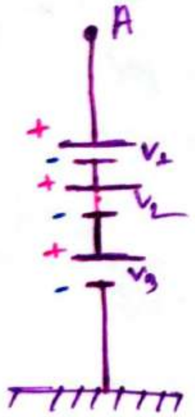
$$P_c = I^2R_1 + I^2R_2 + I^2R_3 + I^2R_4 + I^2R_5$$
$$= I^2(R_1 + R_2 + R_3 + R_4 + R_5)$$
$$= I^2R_T$$
$$= (0.923)^2 \times 26$$
$$= 22.150 \text{ Volt}.$$

$$\therefore \boxed{P_d = P_c}$$

(verified)

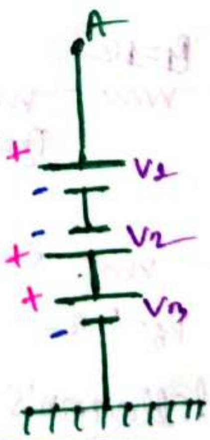
Voltage Source in Series:—

(1)



$\therefore \boxed{V_{net} = V_3 + V_2 + V_1}$ → Voltage of node A with respect to ground

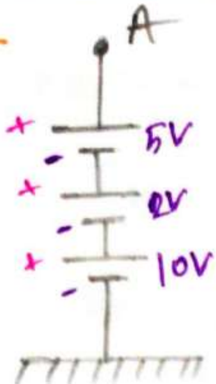
(2)



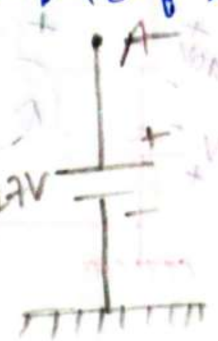
$$\boxed{V_{net} = V_3 - V_2 + V_1}$$

Ex:-

(i)



Dominating battery/Source



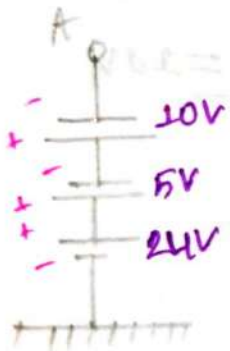
∴ Voltage of Node A with respect to ground:—

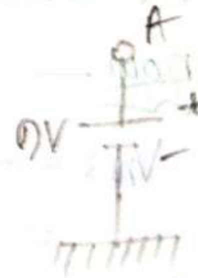
$V_{net} = V_1 + V_2 + V_3$

$= 10 + 5 + 2$

$= 17V$

(ii)



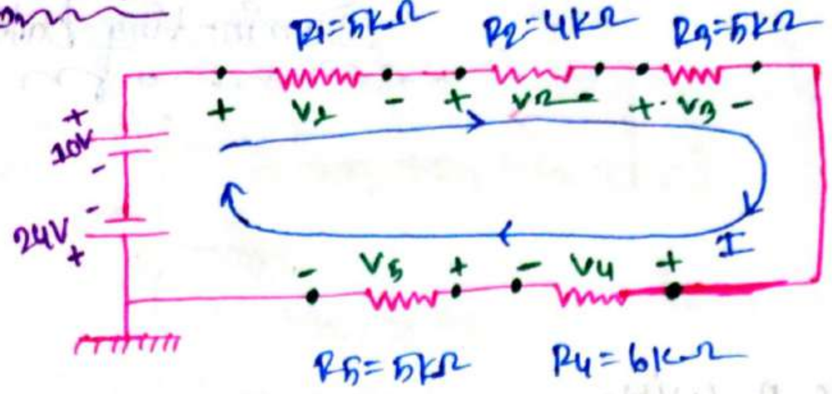


∴ Voltage of Node A with respect to ground:—

$V_{net} = (24 - 5 - 10)V$

$= 9V$

problems–



① Find the total current

② Voltage drops for R_3 & R_4 using VDR

①

⇒ In given:—

$R_1 = 5k\Omega,\ R_2 = 4k\Omega,\ R_3 = 5k\Omega$

$R_4 = 5k\Omega,\ R_5 = 6k\Omega,\ E_1 = 24V,\ E_2 = 10V$

We know—

$V_{net} = -24 + 10$

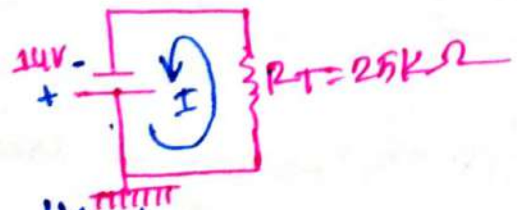
$= -14V$

$R_T = R_1 + R_2 + R_3 + R_4 + R_5$

$= 5 + 4 + 5 + 5 + 6$

$= 25k\Omega$

Re Draw the Circuit:—



$\therefore\ I = \dfrac{|V_{net}|}{R_T} = \dfrac{14}{25\times10^3} = 5.6\times10^{-4}A$

Ans:—

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(ii)

We know—

$$I_n = \frac{|V_{net}|}{R_T} = \frac{14}{25\times10^3} = 5.6\times10^{-4} A$$

$$V_n = \frac{R_n}{R_T}\times E$$

$$\therefore V_3 = \frac{R_3}{R_T}\times|E|$$

$$= \frac{5k\Omega}{25k\Omega}\times 14V$$

$$= 2.8V$$

$$V_4 = \frac{R_4}{R_T}\times|E|$$

$$= \frac{6k}{25k}\times 14$$

$$= 3.36V$$

Note:

⇒ If Direction of current is opposite to the battery/Source Voltage polarity it will be −Ve otherwise +Ve.

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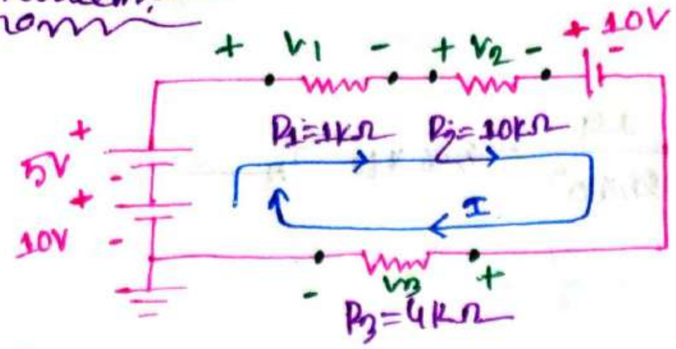
problem:-



(i) Find the total current

(ii) Find the each of voltage drops and verify the KVL

⇒ Is given:-

$V_1 = 10V$ | $R_1 = 1k\Omega$
$V_2 = 5V$ | $R_2 = 10k\Omega$
$V_3 = 10V$ | $R_3 = 4k\Omega$

(i)

We know:-

$$E_{net} = V_1 + V_2 - V_3$$
$$= 10 + 5 - 10$$
$$= 5V$$

$$R_T = R_1 + R_2 + R_3$$
$$= (1 + 10 + 4) k\Omega$$
$$= 15 k\Omega$$

$$I_s = \frac{E_{net}}{R_T} = \frac{5V}{15 \times 10^3 \Omega}$$
$$= 3.33 \times 10^{-4} A$$

Ans:-

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we know,

$V_n = IR_n$

$\therefore V_1 = IR_1$

$= 1\times10^3 \times 3.33\times10^{-4}$

$= 0.333\ Volt$

$V_2 = IR_2$

$= 3.33\times10^{-4}\times10\times10^3$

$= 3.33\ Volt$

$V_3 = IR_3$

$= 3.33\times10^{-4}\times4\times10^3$

$= 1.332$

from KVL:

$E = V_1 + V_2 + V_3$

$5 = 1.332 + 3.33 + 0.333$

$5V = 5V$ (verified)

Ans:

TOPIC NAME : DAY : TIME : DATE : / /

problem 2.1:—

⇒ Find the voltage between 2 points if 60 J of energy are required to move a charge of 20 C between two points.

⇒ Is given:—

$W = 60J$

$Q = 20C$

we know,—

$$V = \frac{W}{Q}$$

$$= \frac{60}{20}$$

$= 3$ Volt.

Ans:—

problem 2.2:—

⇒ Determine the Energy expended moving a charge of $20\mu C$ between 2 points if the Voltage is 6V.

⇒ Is given:—

$Q = 20 \times 10^{-6} C$

$V = 6V$

we know,—

$$V = \frac{W}{Q}$$

$$\therefore W = VQ = 6 \times 20 \times 10^{-6} = 120 \times 10^{-6} J$$

$$= 120 \mu J$$

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problem : 2.3.1 :-

=> Find the charge that requires 120 J of energy to be moved through a potential difference of 20V.

=> Is given:-

$$W = 120\,J$$
$$V = 20\,V$$

We know -

$$V = \frac{W}{Q}$$

$$\therefore Q = \frac{W}{V} = \frac{120}{20} = 6C$$

Ans:-

problem : 2.3 :-

=> The charge flowing through the imaginary surface is 0.16C every 64ms. Determine current in Amperes.

=> Is given:-

$$Q = 0.16C$$
$$t = 64 \times 10^{-3} s$$

We know -

$$I = \frac{Q}{t}$$

$$= \frac{0.16}{64 \times 10^{-3}}$$

$$= 2.5A.$$

Ans:-

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Example 8 2.4:—

⇒ Determine how long it will be take $4\times10^{16}\,e^-$ to pass through the imaginary surface if the current is 5mA.

⇒ Is given:—

$$e^- = 4\times10^{16}$$

$$I = 5mA = 5\times10^{-3}A$$

we know:—

$$1C = \frac{1}{1.602\times10^{-3}} = 6.24\times10^{18}\,e^-$$

$$\therefore 6.24\times10^{18}\,e^- = 1C$$

$$\text{So, } 4\times10^{16}\,e^- = \frac{1\times4\times10^{16}}{6.24\times10^{18}}\,C$$

$$= 6.41\times10^{-3}\,C$$

Again:—

$$I = \frac{Q}{t}$$

$$\therefore t = \frac{Q}{I} = \frac{6.41\times10^{-3}}{5\times10^{-3}}$$

$$= 1.282\ \text{Sec}$$

Ans:—

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Problem 2.4.1 :-

⇒ If a current 100 mA exists for 1.5 min. How many

(i) Coulomb of charge, and

(ii) Electrons have passed through the wire?

⇒ In given :-

$$I = 100\,mA = 100 \times 10^{-3} A$$

$$t = 1.5 sec = 90 sec$$

We know :-

$$I = \frac{Q}{t}$$

$$Q = It = 100 \times 10^{-3} \times 90$$
$$= 9C$$

Again :-

$$1C = 6.24 \times 10^{18} e^-$$

$$\therefore 9C = 9 \times 6.24 \times 10^{18}$$
$$= 56.16 \times 10^{18} e^-$$

Ans :-

Problem 2.4.2 :-

⇒ If a conductor with a current of 120 mA passing through it converts 20 J of electrical energy into heat in 20 s. What is the drop of potential across the conductor?

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⇒ Is given:-

$I' = 120\ mA = 120 \times 10^{-3} A$

$W = 20 J$

$t = 20 Sec$

We know:-

$I = \frac{Q}{t}$

$\therefore Q = It$

$= 120 \times 10^{-3} \times 20$

$= 2.4 C$

Again:-

$V = \frac{W}{Q}$

$= \frac{20}{2.4}$

$= 8.33 Volt.$

Ans:-

problem: 2.4.36

⇒ charge is flowing through a Conductor at the rate of 200 c/min. If 750 J of electrical Energy are Converted to heat in 45s, what is the potential diffrence across the Conductor.?

⇒ Is given:-

$I = 200\ C/min = (200/60)\ C/s$ or A

$= 3.333\ C/s$ or A

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$W = 750J$

$t = 45 Sec$

we know:-

$$I = \frac{Q}{t}$$

$$\therefore Q = It = 3.33 \times 45 = 149.85 C$$

Again:-

$$V = \frac{W}{Q}$$

$$V = \frac{750}{149.85} = 5.005 V$$

Ans:-

problem: 2.4.4:-

⇒ The potential difference between two points in an electric circuit is 48V. if 1.2J of energy were dissipated in a period of 15ms. what would be the current?

⇒ Is given:-

$V = 48 Volt$

$W = 1.2 J$

$t = 15ms = 15 \times 10^{-3} Sec$

we know:-

$$V = \frac{W}{Q}$$

$$\therefore Q = \frac{W}{V} = \frac{1.2}{48} = 0.025 C$$

Again:- $I = \frac{Q}{t} = \frac{0.025}{15 \times 10^{-3}} = 1.667 A$

Ans:-

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problem: 3.7:-

⇒ Determine the resistance of 100 ft Copper telephone wire if the diameter is 0.0126 in. where $\rho = 1.723 \times 10^{-6} \Omega$-cm

⇒ Is given:-

$$l = 100 ft = 100 \times 0.3048 = 30.48 m$$

$$d = 0.0126 \text{ in} = 0.0126 \times 0.0254 = 3.20040 \times 10^{-4} m$$

$$\rho = 1.723 \times 10^{-6} \Omega\text{-cm} = 1.723 \times 10^{-6} \times 10^{-2} = 1.723 \times 10^{-8} \Omega m$$

We know:-

Area of circle $A = \pi r^2$ | $r = \frac{d}{2}$

$$= \pi \left(\frac{d}{2}\right)^2$$

$$= \pi \times \left(\frac{3.2 \times 10^{-4}}{2}\right)^2$$

$$= 8.04 \times 10^{-8} m^2$$

Again:-

$$R = \rho \times \frac{l}{A}$$

$$= 1.723 \times 10^{-8} \times \frac{30.48}{8.04 \times 10^{-8}}$$

$$= 6.532 \Omega$$

Ans:-

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problem: 3·2·1

⇒ Determine the value of resistance of an aluminum conductor if the area is increased by a factor of 3. The original resistance was 12 Ω.

⇒ In given:—

$R_i = 12\,\Omega$

Let:—

$\rho_i = \rho_f = \rho$

Area $A_i = A$

$A_f = 3A$

Length $l_i = l_f = l$

We know:—

$R = \rho \frac{l}{A}$

$\therefore \frac{R_1}{R_2} = \frac{l_i \rho_i / A_i}{l_f \rho_f / A_f}$

$= \frac{l\rho / A}{l\rho / 3A}$

$= \frac{l\rho}{A} \times \frac{3A}{l\rho}$

$= 3$

$\therefore R_f = \frac{1}{3} \times R_i$

$= \frac{1}{3} \times 12$

$= 4\,\Omega$

Ans:—

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problem 83.2.21

⇒ Determine the value of resistance of a silver Conductor if the length is doubled. the Orginal resistance was 5Ω.

⇒ In given:

$R_i = 5\Omega$

Let:

$\rho_i = \rho_f = \rho$

$A_i = A_f = A$

$l_i = l$

$l_f = 2l$

we know

$$R = \rho \frac{l}{A}$$

$$\therefore \frac{R_f}{R_i} = \frac{\rho l_f / A_f}{\rho_i l_i / A_i}$$

$$= \frac{\rho l_f}{A_f} \times \frac{A_i}{\rho_i l_i}$$

$$= \frac{\rho 2l}{A} \times \frac{A}{\rho \times l}$$

$$R_f = 2 \times R_i$$

$$= 2 \times 5$$

$$= 10\Omega$$

Ans:

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problem: 3.9:—

⇒ If the resistance of a copper wire is 50Ω at 50°C, What is the resistance at 100°C where $T_c = 234.5°C$

⇒ In given:—

$R_1 = 50\Omega$, $R_2 = ?$

$T_1 = 50°C$
$T_2 = 100°C$
$T_c = 234.5°C$

We know—

$$\frac{T_c + T_1}{R_1} = \frac{T_c + T_2}{R_2}$$

$$\therefore R_2 = \frac{R_1(T_c + T_2)}{T_c + T_1}$$

$$= \frac{50(234.5 + 100)}{(234.5 + 50)}$$

$$= 58.78\Omega$$

problem: 3.10:—

⇒ If the resistance of a copper wire at freezing (0°C) is 30Ω what is the resistance at −40°C? $T_c = 234.5°C$

⇒ In given:—

$R_1 = 30\Omega$, $R_2 = ?$

$T_1 = 0°C$
$T_2 = -40°C$
$T_c = 234.5°C$

We know:—

$$\frac{T_c + T_1}{R_1} = \frac{T_c + T_2}{R_2}$$

$$\therefore R_2 = \frac{R_1(T_c + T_2)}{T_c + T_1} = \frac{30(234.5 + (-40))}{234.5 + 0} = 24.88\Omega$$

Ans:—

problem 9.9.1:—

⇒ Determine the Conductance of 3048 cm length of copper telephone wire which resistivity is $\rho = 1.723\times10^{-6}\,\Omega$-cm and there Area in $8.04\times10^{-4}cm^2$.

⇒ Is given:—

$l = 3048\times$ cm

$\rho = 1.723\times10^{-6}\,\Omega$-cm

$A = 8.04\times10^{-4}cm^2$

We know:—

$R = \rho\times\frac{l}{A}$

$= 1.723\times10^{-6}\times\frac{3048}{8.04\times10^{-4}}$

$= 6.532\,\Omega$

Again:—

$G = \frac{1}{R}$

$= \frac{1}{6.532}$

$= 0.153$ mho $(\mho)$/S

problem 9.15:—

⇒ Determine the Conductance of 1Ω, $50k\Omega$, $100M\Omega$ resistor.

⇒ We know:—

$G = \frac{1}{R}$

$\therefore G_1 = \frac{1}{R_1} = \frac{1}{1} = 1S$

$G_2 = \frac{1}{R_2} = \frac{1}{50k} = 0.02mS$

$G_3 = \frac{1}{R_3} = \frac{1}{100M} = 0.01\mu S$ Ans:—

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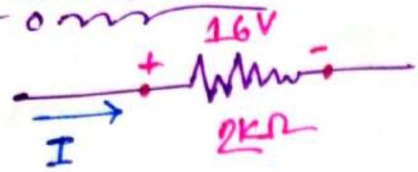
problems 4.3:-



⇒ Calculate the current?

⇒ In given:-

$E = 16V$

$R = 2k\Omega = 2\times10^{3}\Omega$

we know:-

$E = V = IR$

$\therefore I = \frac{V \text{ or } E}{R}$

$= \frac{16}{2\times10^{3}}$

$= 0.008A$

$= 8mA$

problem: 4.9:-

⇒ Determine the current through a 5kΩ resistor when the power dissipated by the element in 20 mW.

⇒ In given:-

$R = 5k\Omega = 5\times10^{3}\Omega$

$P = 20mW = 20\times10^{-3}W$

we know:-

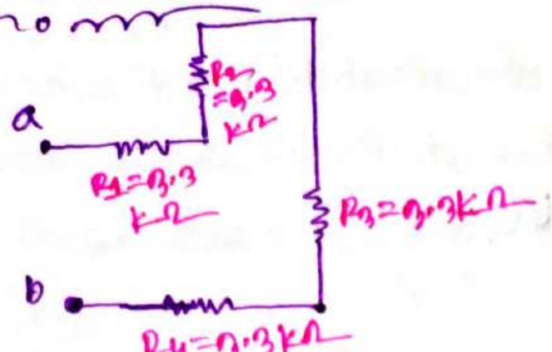
$P = I^{2}R$

$\therefore I = \sqrt{\frac{P}{R}} = \sqrt{\frac{20\times10^{-3}}{5\times10^{3}}} = 0.022A$

$= 2mA$

Ans:-

problem: 5.2



⇒ Determine the total resistance.

⇒ Is given:—

$R_1 = R_2 = R_3 = R_4 = 3.3\,k\Omega$

We know:—

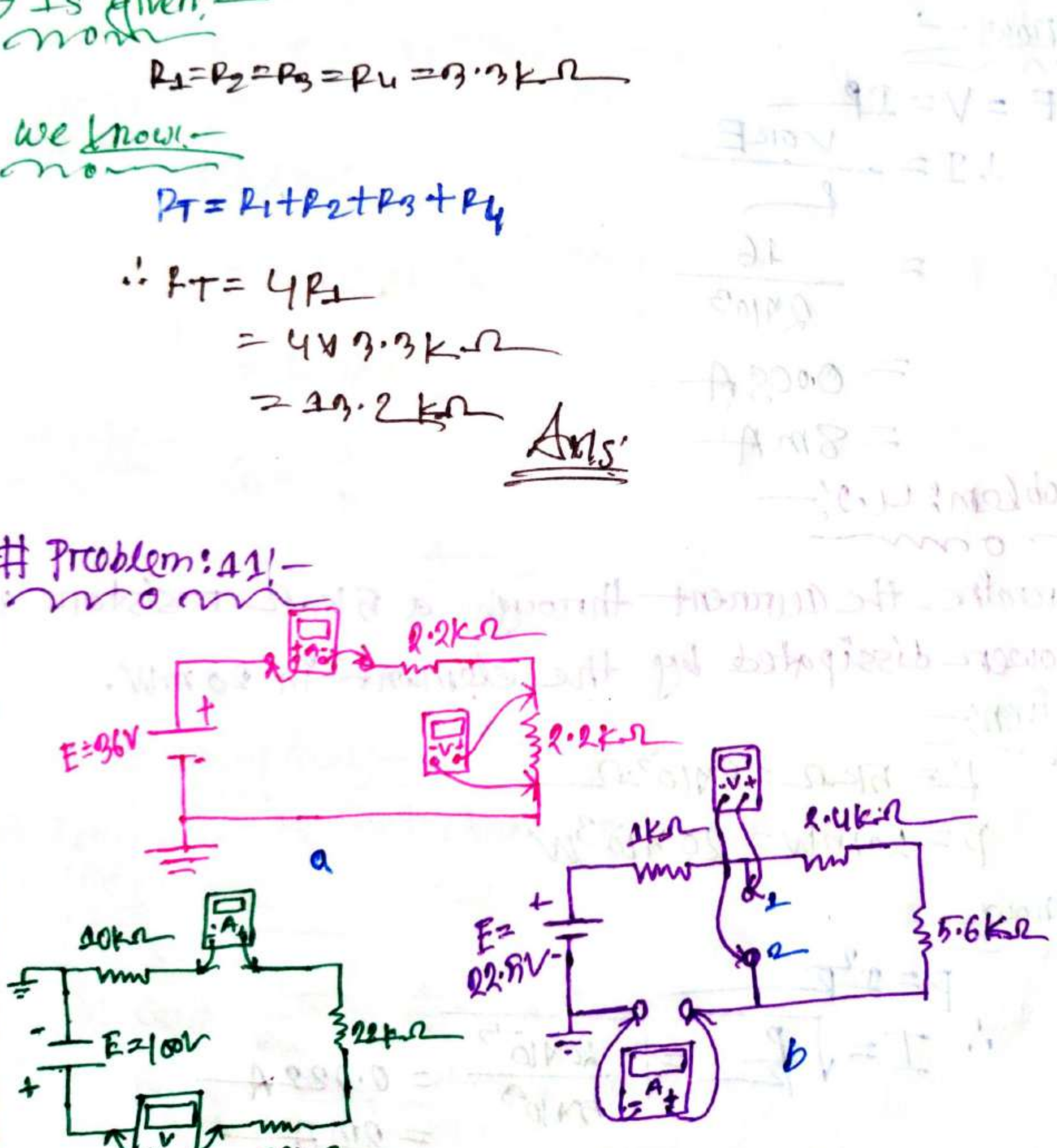
$R_T = R_1 + R_2 + R_3 + R_4$

$\therefore R_T = 4R_1$

$= 4 \times 3.3\,k\Omega$

$= 13.2\,k\Omega$ **Ans**

Problem: 41:—



$R_{net} = 10\,M\Omega$

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⇒ Find the Current And Voltage ?.

(a)

Is given:—

$E = 36V$
$R_1 = 2.2k\Omega$
$R_2 = 2.2k\Omega$

We know:—

$R_T = R_1 + R_2$
$= 2.2 + 2.2 = 4.4k\Omega$

$I_s = \frac{E}{R} = \frac{36}{4.4k} = 8.18 \times 10^{-3} A$
$= 8.18 mA$

$V_2 = I_s R_2 = (8.18 \times 10^{-3} \times 2.2 \times 10^{3})$
$= 18V$

Ans:—

(b)

Is given:—

$E = 22.5V$
$R_1 = 1k\Omega$
$R_2 = 2.4k\Omega$
$R_3 = 5.6k\Omega$

We know:—

$R_T = R_1 + R_2 + R_3$
$= 1 + 2.4 + 5.6 = 9k\Omega$

$I_s = \frac{E}{R_T} = \frac{22.5}{9k} = 2.5mA$

$V_{12} = I_s R' = (2.5m \times 8k) = 20V$

Ans:—

(c)

Is given:-

$E = 100V$ $\quad$ $R_1 = 10k\Omega$

$R_{net} = 10M\Omega$ $\quad$ $R_2 = 22k\Omega$

$R_3 = 33k\Omega$

we know:-

$$R'_T = R_1 + R_2 + R_3$$
$$= (10 + 22 + 33)k\Omega$$
$$= 65k\Omega$$
$$= 65 \times 10^3 k\Omega$$

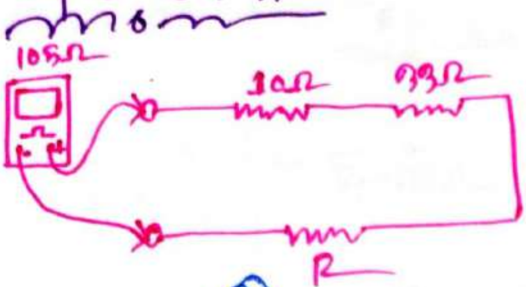
$$\therefore R = R'_T + R_{net}$$
$$= 10 \times 10^6 + 65 \times 10^3$$
$$= 10.065M\Omega$$

$$I = \frac{E}{R}$$
$$= \frac{100}{10.065M}$$
$$= 9.935\mu A$$

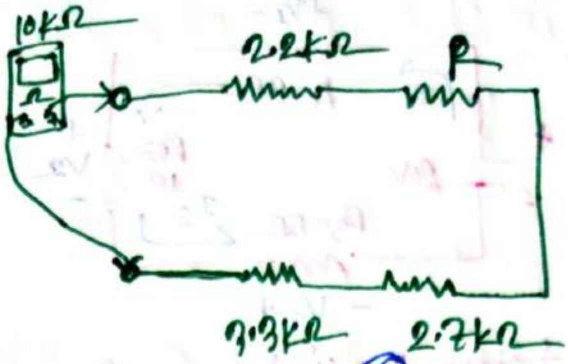
$$V = IR_{net}$$
$$= (9.935\mu) \times (10M)$$
$$= 99.354 \text{ Volt}$$

Ans:-

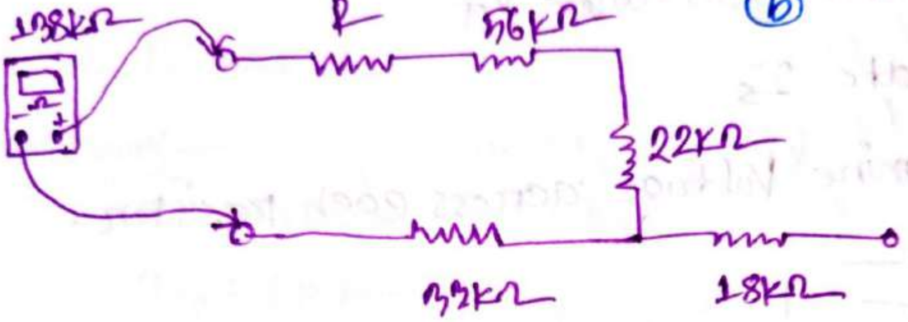
problem 5:



(a)



(b)



(c)

5) Find the value of R

(a)

$R_T = R_1 + R + R_2$

$R = R_T - R_1 - R_2$

$= 105 - 10 - 33 = 62\Omega$

(b)

$R_T = R_1 + R_2 + R_3 + R$

$\therefore R = 10k - (2.2k + 3.3k + 2.7k)$

$= 10k - 8.2k$

$= 1.8k$

(c)

Is given:

$R_1 = 56k\Omega$

$R_2 = 22k\Omega$

$R_3 = 18k\Omega$

$R_4 = 33k\Omega$

we know

$R_T = R + R_1 + R_2 + R_4$

$\therefore R = 138k - (56k + 22k + 33k)$

$= 27k$

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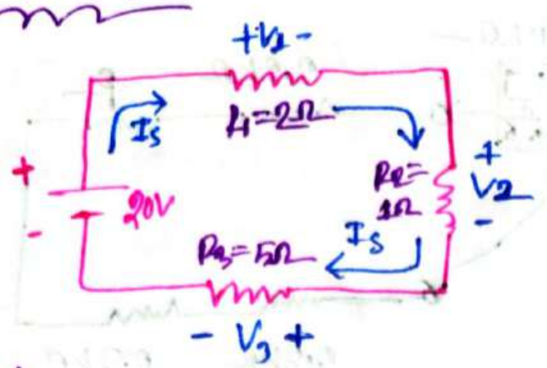
problem: 5.4:—



(i) Find total resistance R_T

(ii) Calculate I_s

(iii) Determine Voltage across each resistor.

⇒ Is given—

$R_1 = 2\Omega$

$R_2 = 1\Omega$

$R_3 = 5\Omega$

$E = 20V$

(i)

We know—

$R_T = R_1 + R_2 + R_3$

$= 1 + 2 + 5 = 8\Omega$

(ii)

We know—

$I_s = \frac{E}{R_T} = \frac{20}{8} = 2.5A$

(iii)

We know—

$V_1 = I_s R_1 = 2.5 \times 2 = 5V$

$V_2 = I_s R_2 = 2.5 \times 1 = 2.5V$

$V_3 = I_s R_3 = 2.5 \times 5 = 12.5V$

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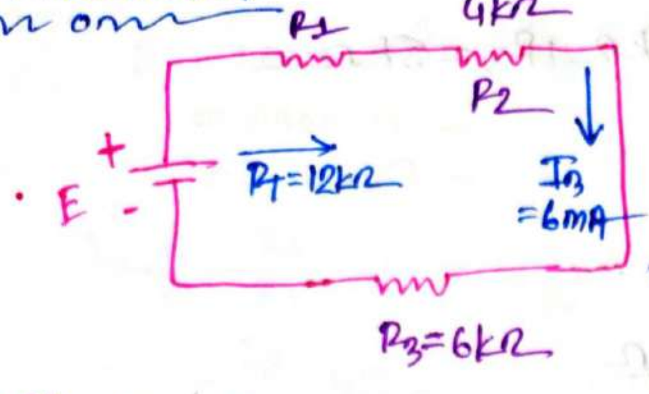
problem: 5.6!—



(1) Find the E and R_1

⇒ we have!—

$R_T=12k\Omega$

$I_3=6mA=I_S$

$R_2=4k\Omega$

$R_3=6k\Omega$

we know,—

$R_T=R_1+R_2+R_3$

$\therefore R_1=(12-4-6)k=2k\Omega$

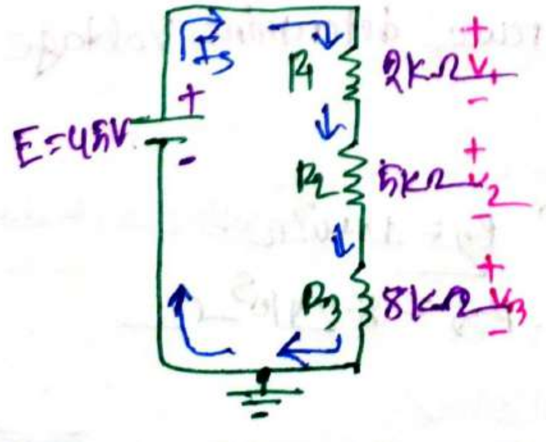
$E=I_S R_T$

$=6mA \times 12 \times k\Omega$

$=72V$

Ans!—

problem: 5.16!—



⇒ using the Voltage divider rule, determine voltage V_1, V_2, V_3 for the circuit.

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we have:—

$R_1 = 2k\Omega, R_2 = 5k\Omega, R_3 = 8k\Omega$

$E = 45V$

we know —

$R_T = R_1 + R_2 + R_3$

$= (2+5+8)\,k\Omega$

$= 15k\Omega$

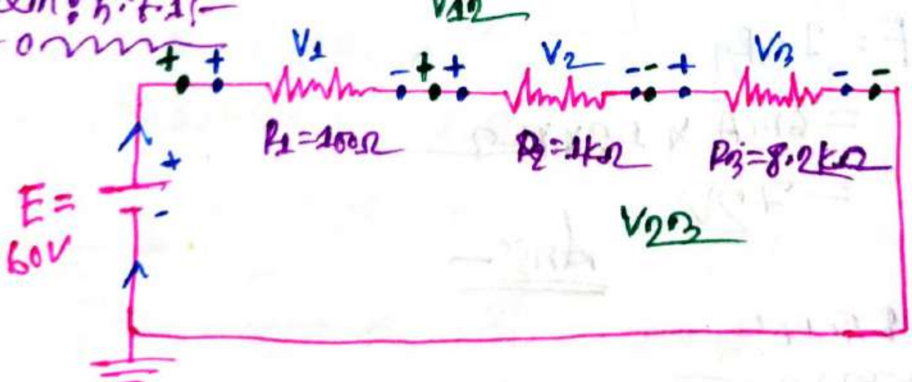
$V_1 = \frac{R_1}{R_T} \times E = \frac{2k \times 45}{15k} = 6V$

$V_2 = \frac{R_2}{R_T} \times E = \frac{5k \times 45}{15k} = 15V$

$V_3 = \frac{R_3}{R_T} \times E = \frac{8k \times 45}{15k} = 24V$

Ans:

problem: 5.7.1:—



⇒ Using voltage divider rule determine voltage V_1, V_2, V_3, V_{12} and V_{23}.

⇒ Is given:—

$E = 60V$, $R_1 = 100\Omega$ | $R_2 = 1\times10^3\Omega$, $R_3 = 8.2\times10^3\Omega$

we know,-

$$R_T = 100 + 1\times10^3 + 8.2\times10^3$$
$$= 9300\ \Omega$$
$$= 9.3k\Omega$$

$$V_1 = \frac{R_1}{R_T}\times E = \frac{100\times60}{9300} = 0.6451\ \text{Volt}$$

$$V_2 = \frac{R_2}{R_T}\times E = \frac{1K\times60}{9.3K} = 6.451V$$

$$V_3 = \frac{R_3}{R_T}\times E = \frac{8.2K\times60}{9.3K} = 52.903V$$

Now,

$$R_{12} = R_1 + R_2 = 100 + 1000 = 1100\ \Omega$$

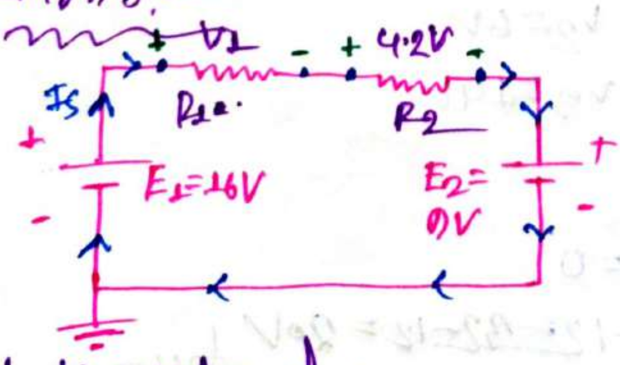
$$R_{23} = R_2 + R_3 = 1000 + 8200 = 9200\ \Omega$$

$$V_{12} = \frac{R_{12}}{R_T}\times E = \frac{1100\times60}{9300} = 7.096V$$

$$V_{23} = \frac{R_{23}}{R_T}\times E = \frac{9200\times60}{9300} = 59.354V$$

Ans:-

#Problem: 5.8:-



(1) Find the value of V_1

(1)

we have:-

$E_1 = 16V$

$E_2 = 0V$

$V_2 = 4.2V$

We know:-

$E_1 - V_1 - V_2 - E_2 = 0$

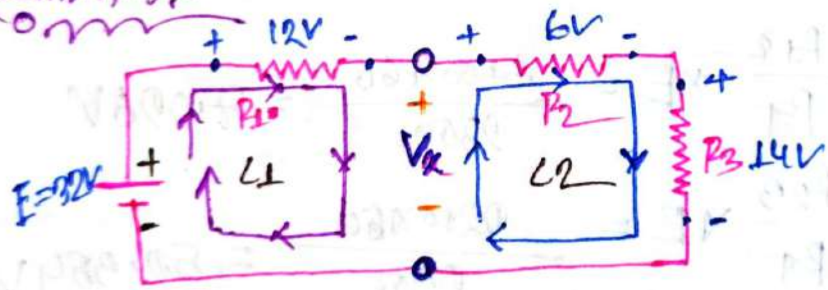
$\therefore V_1 = E_1 - V_2 - E_2$

$= 16 - 4.2 - 0$

$= 2.8V$

Ans:-

problem 5.9:-



⇒ Determine the unknown Voltage V_x.

⇒ Is given:-

$E = 32V$, $V_1 = 12V$ | $V_2 = 6V$, $V_3 = 14V$

From 1st loop:-

$E - 12 - V_x = 0$

$\therefore V_x = E - 12 = 32 - 12 = 20V$ Ans:-

From 2nd loop:-

$V_x - 6 - 14 = 0$

$\therefore V_x = 14 + 6 = 20V$

Ans:-

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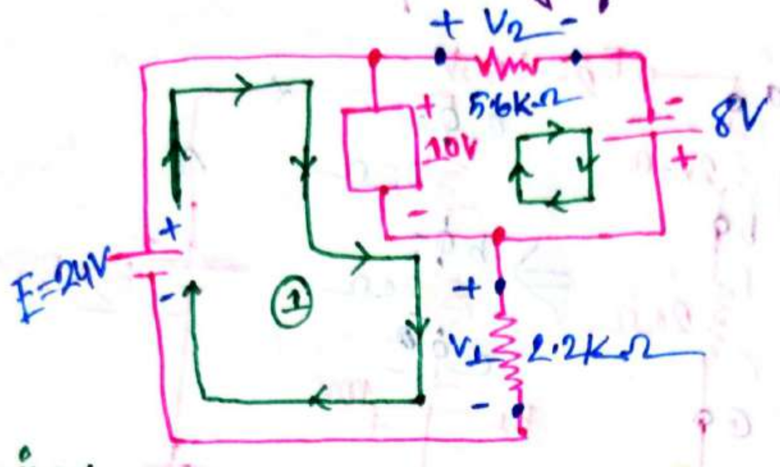
problem:- 5.6.1:-

=> Determine the unknown Voltage for the following circuit



=> In given:-

$E = 24V$

$R_1 = 2.2 k\Omega$

$R_2 = 5.6 k\Omega$

We know:-

from 1st loop:-

$24 - 10 - V_1 = 0$

$V_1 = 14V$

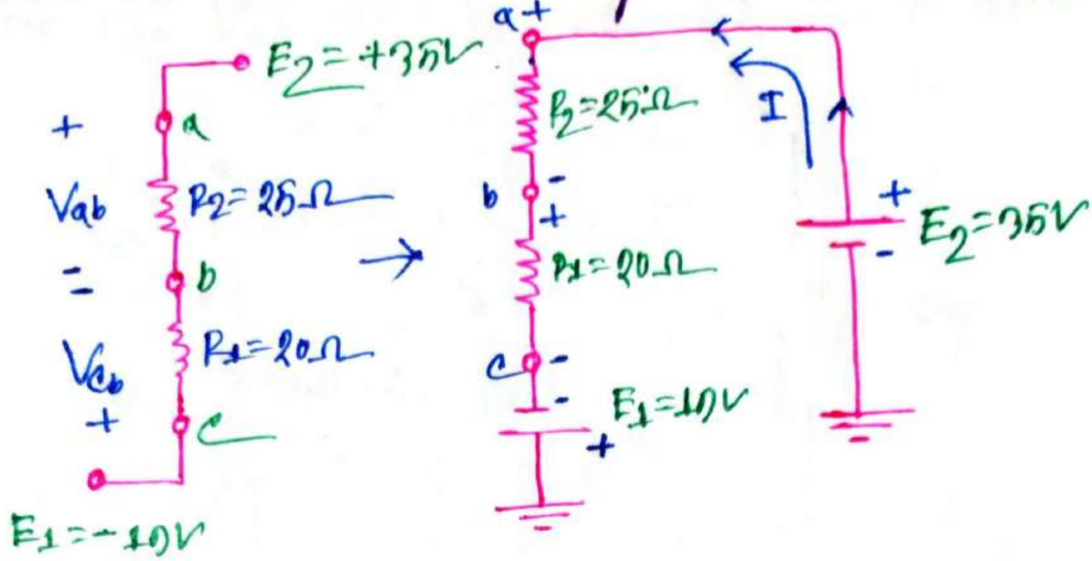
from 2nd loop:-

$8V + 10V - V_2 = 0$

$\therefore V_2 = 18V$ Ans:-

problem: 5.25:—

⇒ Determine the V_{ab}, V_{cb}, and V_c for the Network:



we have:—

$$E_2 = 35V \quad R_1 = 25\Omega$$
$$E_1 = 10V \quad R_2 = 20\Omega$$

we know:—

$$E_{net} = E_1 + E_2 = 35 + 10 = 54V$$

$$R_T = R_1 + R_2 = 25 + 20 = 45\Omega$$

$$\therefore I = \frac{E}{R} = \frac{54}{45} = 1.2A$$

$$V_{ab} = IR_2 = 1.2 \times 25 = 30V$$

$$V_{cb} = -IR_1 = 1.2 \times 20 = - \quad 24V$$

$$V_c = E_1 = -10V$$

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#Ch: 6 Parallel DC Circuit

DAY:

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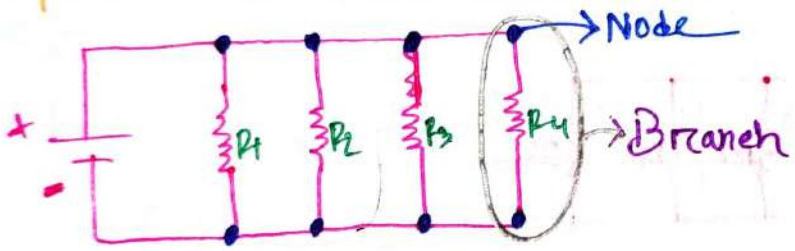
Parallel Circuit!—

⇒ For parallel Connection, the end of one circuit Component is Connected to two or more than two terminals of other components (like +ve end of battary is connected to end of R_1, R_2, R_3 & R_4).

(1) The Voltage of component in parallel remains same.

(2) The Current changes for each branch depending on the value of resistors.

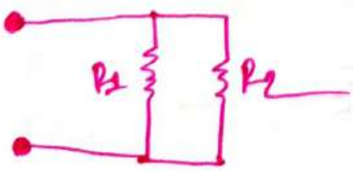
Note!— Opposite to a Series Circuit.



Parallel Resistance!—

$$\boxed{\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}}$$

Let, Say 2 resistor in parallel!—



$$\therefore \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_1+R_2}{R_1R_2}$$

$$\therefore \boxed{R_T = \left(\frac{R_1+R_2}{R_1R_2}\right)^{-1}}$$

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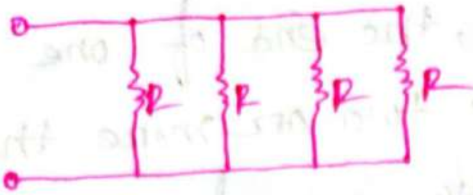
'n' equal resistor in parallel:—



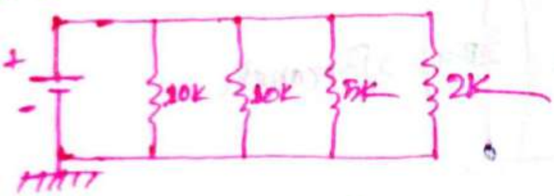
$$\therefore \frac{1}{R_T} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R}$$

$$= \frac{1}{R}(1+1+1+1)$$

$$= \frac{4}{R}$$

$$\boxed{\therefore R_T = \frac{R}{4}}$$

problem:—



⇒ Find the value of R_T

⇒ Is given:—

$R_1 = 10k\Omega$ | $R_3 = 5k\Omega$

$R_2 = 10k\Omega$ | $R_4 = 2k\Omega$

We know:—

$$R' = \frac{R}{2} = \frac{10k}{2} = 5k$$

$$R'' = \frac{5}{2} = 2.5k$$

$$R_T = \frac{R'' \times R_4}{R'' + R_4} = \frac{2 \times 2.5}{2.5+2} = 1.111k\Omega$$

Ans.

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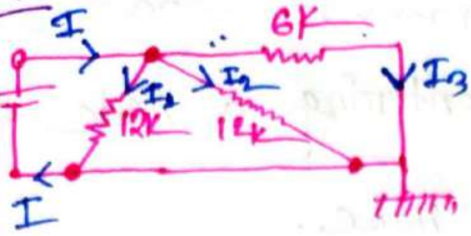
Problem:—



⇒ (i) Find R_T

(ii) Find I, I_1, I_2, I_3

⇒ In given:—

$E = 12V$

$R_1 = 12K\Omega$

$R_2 = 12K\Omega$

$R_3 = 6K\Omega$

(i)

we know—

$$R_T = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)^{-1}$$

$$= \left(\frac{1}{12K} + \frac{1}{12K} + \frac{1}{6K}\right)^{-1}$$

$$= 3K\Omega$$

(ii)

we know:—

$$I = \frac{E}{R_T} = \frac{12}{3\times10^3} = 4\times10^{-3}A = 4mA$$

$$I_1 = \frac{E}{R_1} = \frac{12}{12K} = 1mA$$

$$I_2 = \frac{E}{R_2} = \frac{12}{12K} = 1mA$$

$$I_3 = \frac{E}{R_3} = \frac{12}{6K} = 2mA$$

Ans:—

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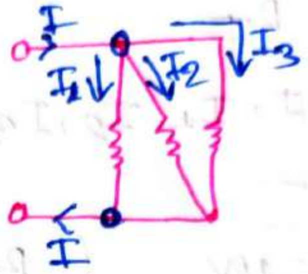
Kirchoff's Current law: (KCL)

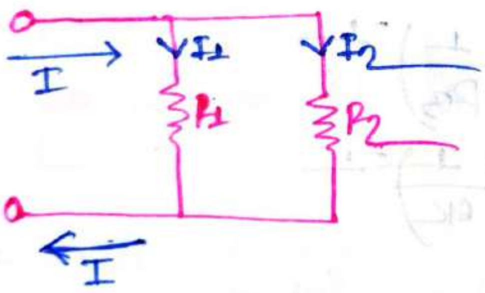
=> The ~~Same~~ Sum of Currents entering a node equals Sum of Currents exitting the node.

$$\therefore \boxed{I = I_1 + I_2 + I_3}$$



Current Divider rule: (CDR):-

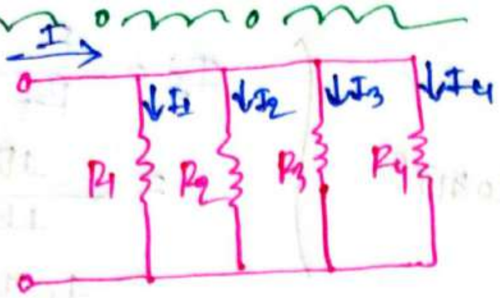
For 2 resistor:-



$$\therefore I_1 = \frac{R_2}{R_1 + R_2} \times I$$

$$I_2 = \frac{R_1}{R_1 + R_2} \times I$$

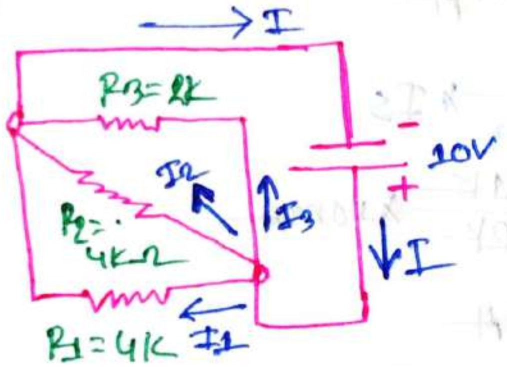
For Multiple resistor:—



$$\boxed{I_x = \frac{R_T}{R_x} \times I}$$

I_x = 'x'th resistor current.

problem:—



(i) Find the R_T

(ii) Find the branch current using CDR

(iii) Find the I_s

(iv) Verify KCL

⇒ Is given:—

$R_3 = 2K\Omega$, $R_2 = 4K\Omega$, $R_1 = 4K\Omega$

$E = 10\,V$

We know:—

(1)

$$R_T = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right)^{-1} = \left(\frac{1}{4K} + \frac{1}{4K} + \frac{1}{2K}\right)^{-1} = 1K$$

Ans:—

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(ii)

We know:-

$$I_1 = \frac{R_T}{R_1} \times I_s$$

$$I_1 = \frac{1k}{4k} \times 10 \times 10^{-3}$$

$$= 2.5mA$$

$$I_2 = \frac{R_T}{R_2} \times I_S$$

$$= \frac{1k}{4k} \times 10mA$$

$$= 2.5mA$$

$$I_3 = \frac{R_T}{R_3} \times I_S$$

$$= \frac{1k}{2k} \times 10mA$$

$$= 5mA$$

(iii)

$$I_S = \frac{E}{R_T}$$

$$= \frac{10}{1k}$$

$$= 10 \times 10^{-3} A$$

$$= 10mA$$

(iv)

The Value of total Entering Current $= I$
$= 10mA$

The Value of " Exiting Current $= I_1 + I_2 + I_3$
$= 2.5 + 2.5 + 5$
$= 10mA$

$\therefore I = I_1 + I_2 + I_3$

(Verified)

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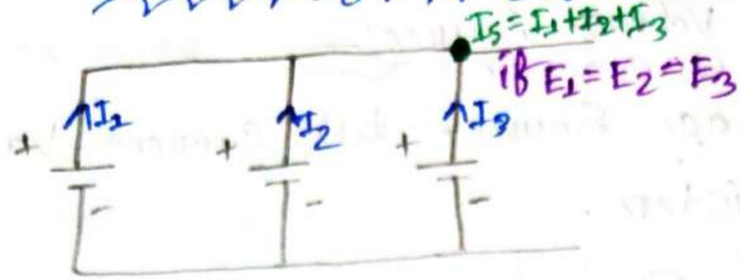
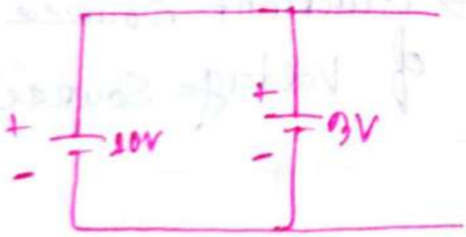
Voltage Source in Parallel:—



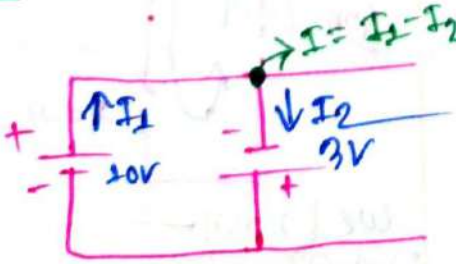
⇒ If Voltage Source are connected as above (all +ve terminal connected together and -ve terminal also connected together) Sources must be of similar value, hence voltage across will remain same.

Not recommended circuit:—



⇒ Not recommended in this type of circuit because in this case the low voltage battery will destroyed by the current from the high voltage battary.

but:—



In this case the total current decreases.

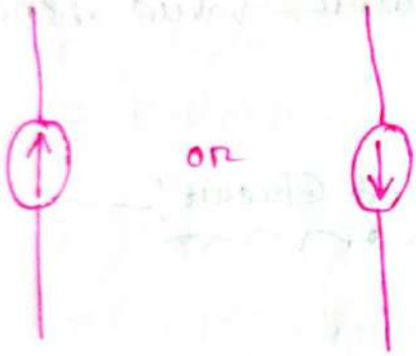
Current Source in Parallel:-

we know:- from Voltage Source:-

⇒ Constant voltage Source but Current Varies due to Connected resistor.

And the ~~Voltage~~ Current Source:-

⇒ Constant current Source but Voltage Varies due to Connected resistor.

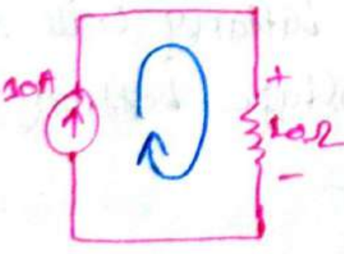


→ The arrow Sign indicate the direction of Current.

→ Current Source is a dual of Voltage Source.

Symbol of Current Source.

#problem:- 1



Q→ Find the Voltage

⇒ we know -

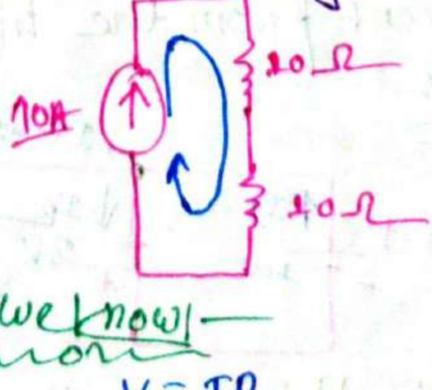
$V = IR$

$= 10A \times 10\Omega$

$= 100V$

#problem: 2

⇒ Find the Voltage?



we know:-

$V = IR$

$= 10(10+10)$

$= 200V$

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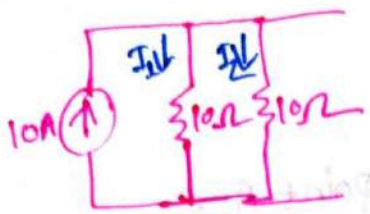
problem :- 3

⇒ Find the Voltage :- also Find I_1, I_2



⇒ We know -

$V = IR_T$

$= 10 \times 5$

$= 50V$

We know -

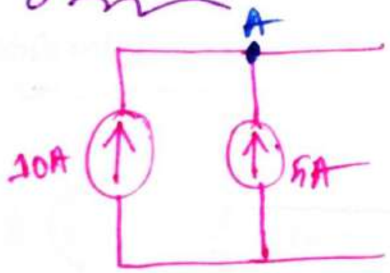
$R_T = \left(\frac{1}{10} + \frac{1}{10}\right)^{-1}$

$= 5\Omega$

$I_1 = \frac{R_T}{R_1} \times I_S = \frac{5}{10} \times 10 = 5A$

$I_2 = \frac{R_T}{R_2} \times I_S = \frac{5}{10} \times 10 = 5A$

problem : 4



⇒ Find the Value of current at point A

⇒ Seing

The direction of arrows are same side

So, $I = I_1 + I_2$ at point A

$= 10 + 5$

$= 15A$

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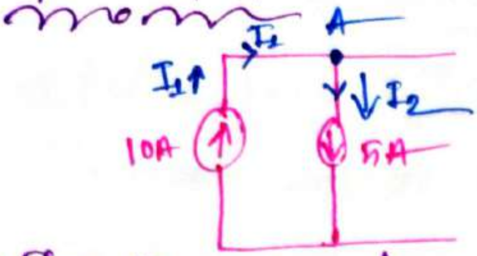
problem:- 5:-



⇒ Find the value of current at point A.

⇒ Since, The direction of arrows are opposite each other

So, $I = I_1 - I_2$

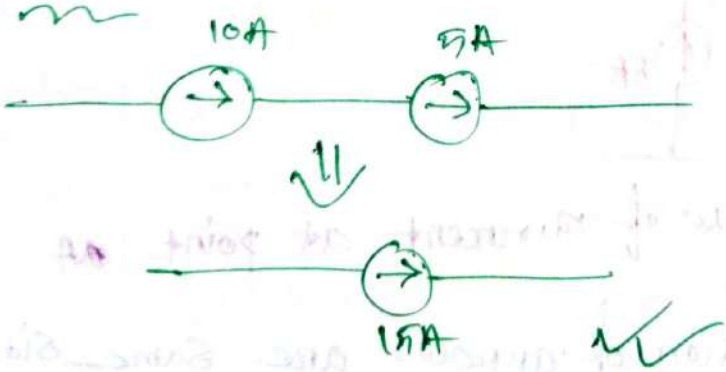
$= 10 - 5$

$= 5A$

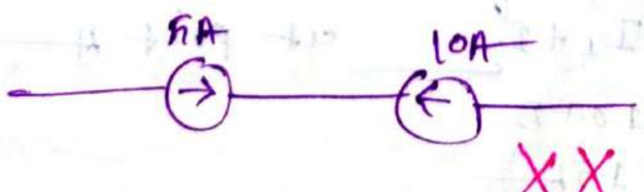
Note:-

⇒ It is not possible to connect two or more than two current source in series. (as current should be same in series)

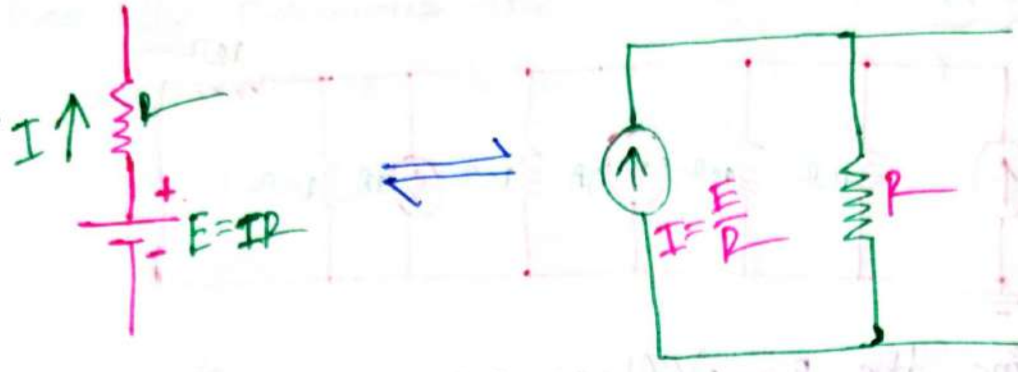
That means:-



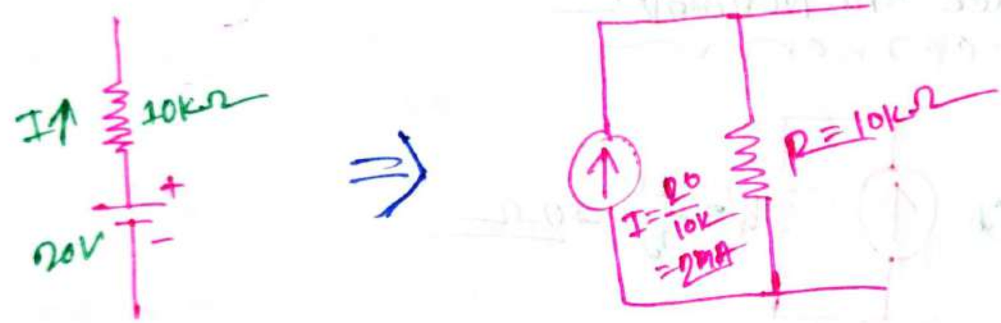
But:-



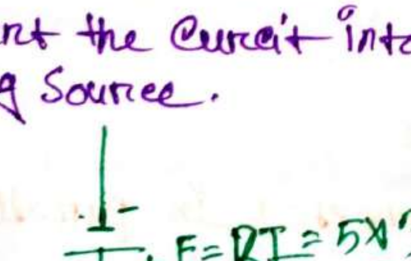
Source Transformation:-



Ex:-

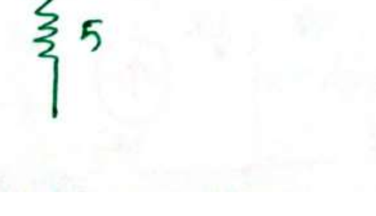


problem: 1:-



⇒ Convert the Circuit into a Voltag Source.

⇒



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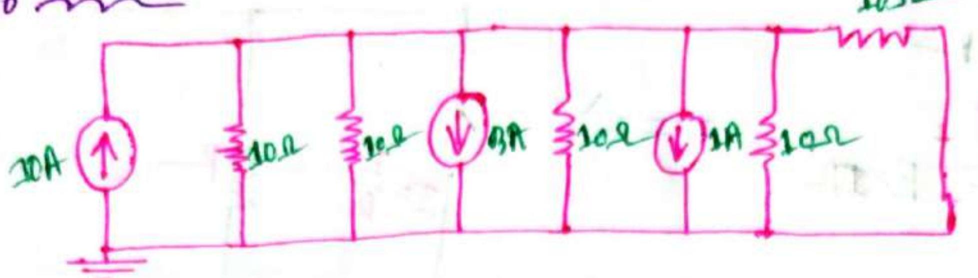
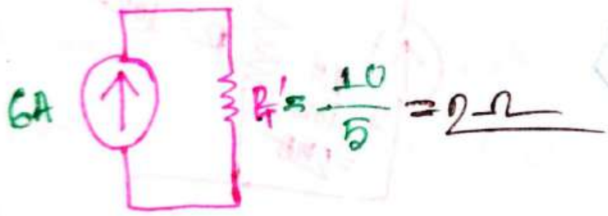
Problem Soln:

problem 1:



(i) Determine the R_T, Voltage across each Branch, Current for each resistor.

⇒ 1st reduce the Network:



We know:

$$R_T' = \frac{R}{5}\,\Omega$$ [Since every Resistor contain Same value]

$$= \frac{10}{5}\,\Omega$$

$$= 2\Omega$$

for voltage:

$$I = \frac{V}{R}$$

$$\therefore V = IR_T = 6 \times 2 = 12V$$

Hence every Resistor are connected in parallel. So, The Voltage are Same each Branch.

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For each resistor Current:-

⇒ Since, The resistors are Contained Same value, So current for each resistor will be Same.

$$\therefore V = I'R_1$$

$$\therefore I' = \frac{V}{R_1} = \frac{12}{10} = 1.2A$$

OR,

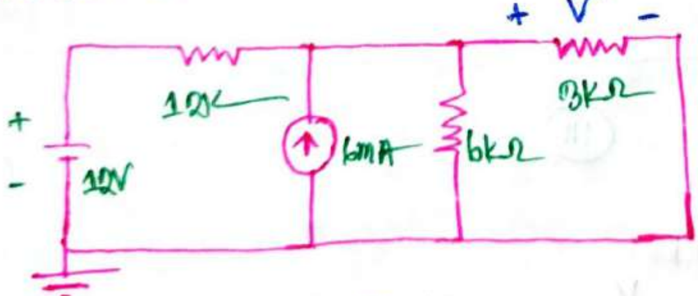
$$I' = \frac{R_T}{R_1} \times I_s$$

$$= \frac{2}{10} \times 6$$

$$= 1.2A$$

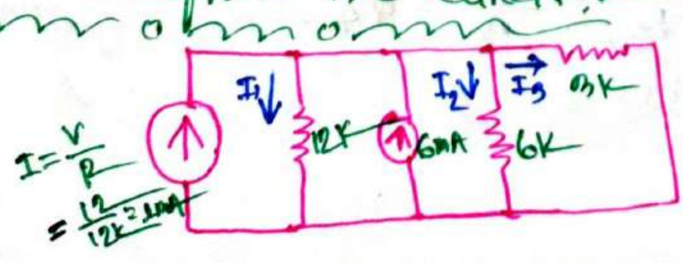
$$\therefore I_1 = I_2 = I_3 = I_4 = I_5 = 1.2A$$

Problem: 2



(1) Determine the total Current.
(2) Find V'
(3) Current through each resistor.

⇒ 1st transfrom the Circuit:-



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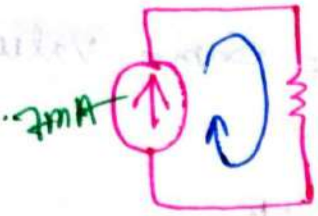
Now, Reduce the circuit:



$R' = \left(\frac{1}{12k} + \frac{1}{6k} + \frac{1}{12k}\right)^{-1}$
$= 1.714\,k\Omega$

(i)

The total Current $I = 6mA + 1mA$
$= 7mA$.

(ii)

Since, Every Resistor are Connected in parallel so the Voltage of every resistor will be equal.

$\therefore V' = IR_T$
$= 7m \times 1.714k$
$= 12V$

(iii)

We know

$V = IR$

$\therefore I_1 = \frac{V}{R_1} = \frac{12}{12} = 1A$

$I_2 = \frac{V}{R_2} = \frac{12}{6} = 2A$

$I_3 = \frac{V}{R_3} = \frac{12}{3} = 4A$

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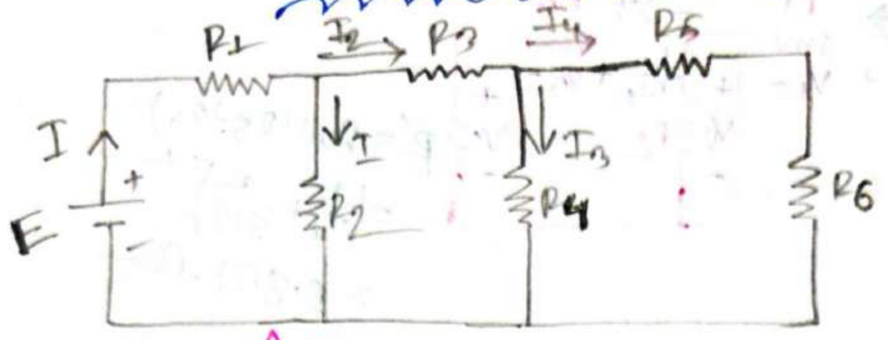
Series Parallel Network



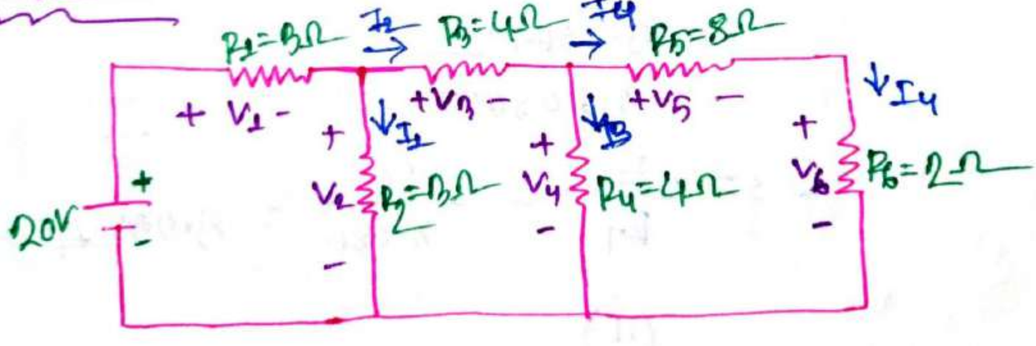
Ladder Network

⇒ To Solve the network/circuit, Reduce the ~~Current~~ Circuit from the other end of the Source find I_T and R_T then return.

$$R_T = R_1 + \left[R_2 \parallel \left[R_3 + \{ R_4 \parallel (R_5 + R_6) \} \right] \right]$$

$$\therefore I_s = \frac{E}{R_T}$$

Problems 1

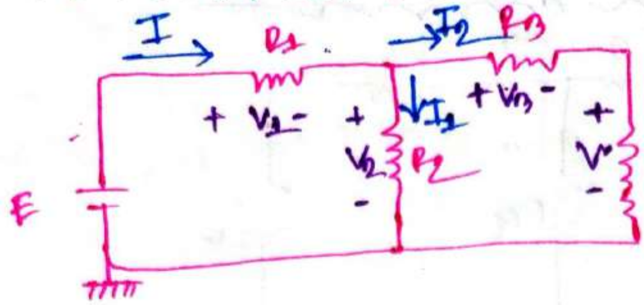


(1) Find R_T and I_s.

(2) Voltage across each Branch

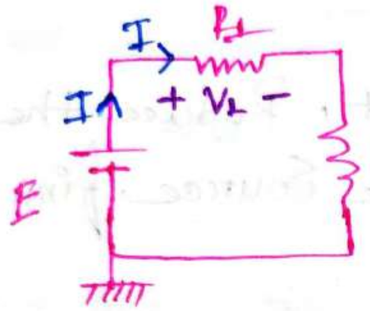
(3) Current for each resistor.

→ 1st reduce the network:-



$$R' = R_4 \parallel (R_5+R_6)$$
$$= \left(\frac{1}{4}+\frac{1}{8+2}\right)^{-1}$$
$$= 2.857\,\Omega$$

Again:-



$$R'' = R_2 \parallel (R_3+R')$$
$$= \left(\frac{1}{3}+\frac{1}{2.857+4}\right)^{-1}$$
$$= 2.086\,\Omega$$

(i)

We know:-

$$R_T = R_1 + R''$$
$$= 2.086 + 3$$
$$= 3.086 + 2$$
$$= 5.086\,\Omega$$

$$I_s = \frac{E}{R_T} = \frac{20}{5.086} = 3.932\,A$$

(ii)

We know:-

$$V_1 = I R_1 = 3.932 \times 3 = 11.796\,V$$

$$V_2 = I_1 R_2 = 2.734 \times 3 = 8.202\,V$$

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$V_3 = I_2 R_3$
$= 1.196 \times 4$
$= 4.78V$

$V_6 = I_4 R_6$
$= 0.341 \times 2$
$= 0.684V$

$V_4 = I_3 R_5$
$= 0.854 \times 4$
$= 3.416V$

$V_5 = I_4 R_5$
$= 0.341 \times 8$
$= 2.728V$

(iii)

We know,-

$I_1 = \frac{R''}{R_2} \times I$

$= \frac{2.086}{3} \times 3.932$

$= 2.734A$

$I_2 = \frac{R''}{R_3 + R''} \times I$

$= \frac{2.086}{4 + 2.857} \times 3.932$ OR

$= 1.196A$

From KCL:-

$I = I_1 + I_2$

$\therefore I_2 = I - I_1$

$= 3.932 - 2.734$

$= 1.198A$

$\therefore I_3 = \frac{R'}{R_4} \times I_2$

$= \frac{2.857}{4} \times 1.196A$

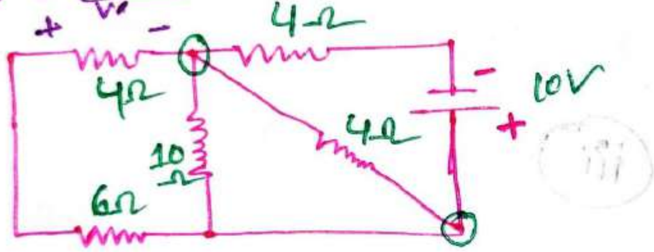
$= 0.854A$

$\therefore I_4 = I_2 - I_3$

$= 1.196 - 0.854$

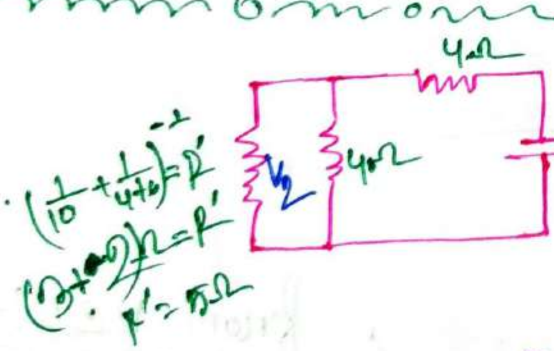
$= 0.341A$

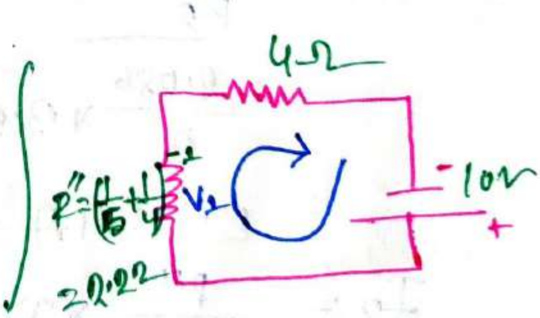
problems 2:-



⇒ Find V'?

⇒ 1st reduce the network:-





(1)

We know,

$R_T = R'' + 4 = 2.22 + 4 = 6.22\Omega$

$I_s = \frac{V}{R_T} = \frac{10}{6.22} = 1.6077$

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$$\therefore V_2 = \frac{R'' \times E}{R''+4} = \frac{22.2}{6.22} = 3.569V$$

Since v_1 and v_2 parrallel connected, so voltage of v_1 and v_2 are equal.

$$\therefore V' = \frac{4}{4+6} \times V_2$$

$$= \frac{4}{10} \times 3.569$$

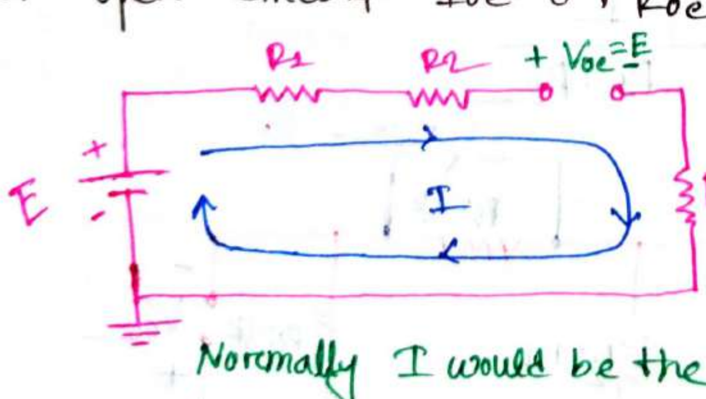
$$= 1.4276 \text{ Volt}.$$

Ans:-

Open Circuit:-

⇒ open circuit is when any of the connections are kept open in a closed circuit.

For open circuit $I_{oc} = 0$, $R_{oc} = \infty$



As $I = 0$ voltage Drops in R_1, R_2 and R_3 are $= 0$ hence open circuit Voltage $V_{oc} = E$

Normally I would be the total current here

If terminal A, B is open, the $I_{oc} = 0$, $R_{oc} = \infty$

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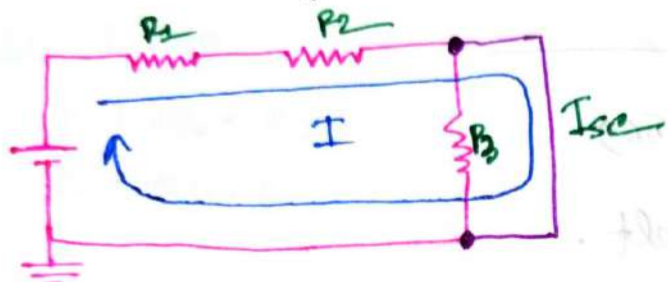
Short Circuit:-

⇒ It Occurs when a circuit component is shorted across.

⇒ As Current Always exist in the least resistive path, Short Current path always carry the current as $R_{sc}=0, V_{sc}=0$



Normally $I = \frac{E}{R_1+R_2+R_3}$

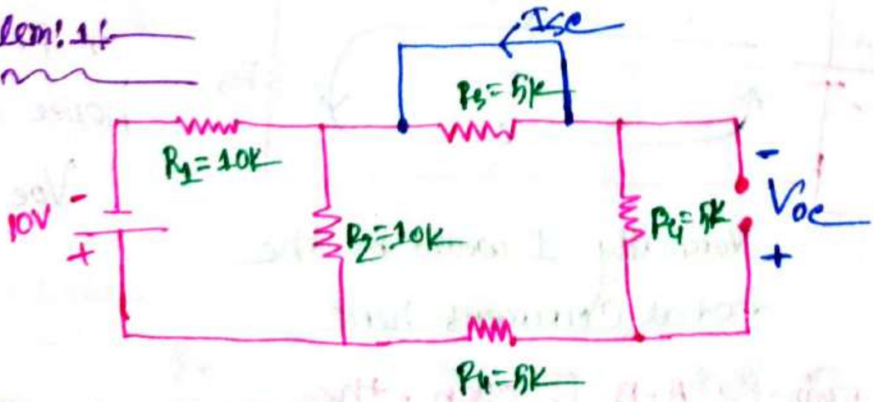
But:-

Due to R_3 being shorted Current I_{sc} will exist in the shorted path as $R_{sc}=0$. Hence, R_3 won't carry any current

So,

$$I = \frac{E}{R_1+R_2}$$

problem: 1:-



⇒ Determine the short circuited current I_{sc} and open circuit Voltage. V_{oc}.

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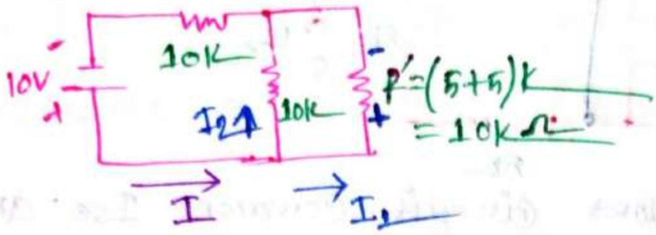
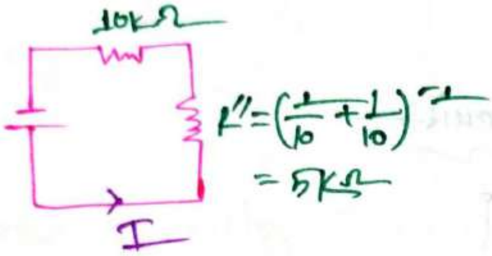
⇒ 1st reduce the circuit



Again:-



$\therefore R_T = 10k + 10k - 5k$
$= 20k - 5k$
$= 15k$

$\therefore I = \frac{E}{R_T} = \frac{10}{15k} = 0.667mA$

So, from CDR:-

$I_1 = \frac{R''}{R'} \times I = \frac{5K}{10K} \times 0.667mA$
$= 0.333mA$

Here, for thsin circuit:-

$I_1 = I_{sc} = 0.333mA$

and, Voltage V_{oc} is:-

$V_{oc} = I_1 R_4$
$= 0.333m \times 5K$
$= 1.667$ Volt.

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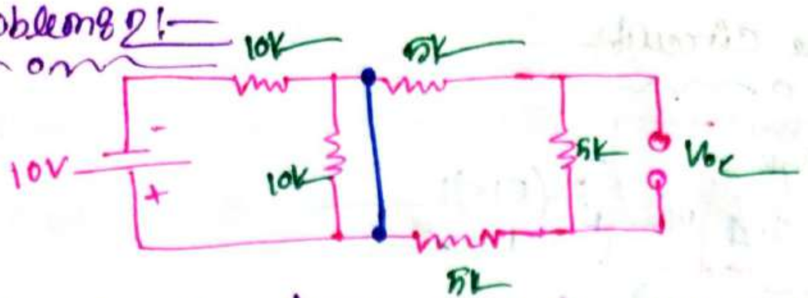
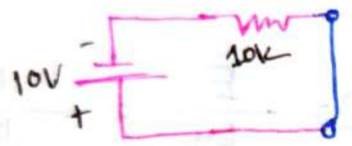
problem 21:-



⇒ Determine the short circuit current I_{sc} and Voltage of open circuit.

⇒ 1st reduce the Circuit:-



$\therefore \quad I_{sc} = \dfrac{10}{10K} = 1mA$.

ans Voltage of V_{oc} is:-

$V_5 = V_{oc} = I' R_5$

$= 0 \times 5$

$= 0$ Volt

Ans:-

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#Network Theorems:

Ch: 09

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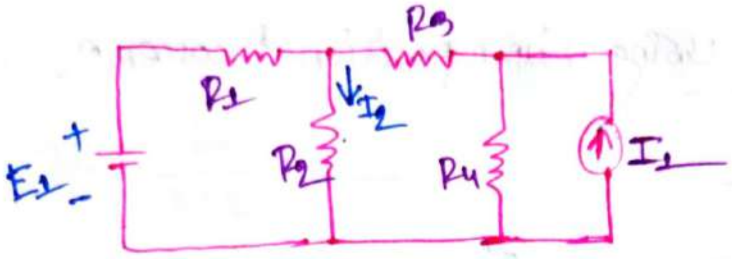
Superposition theorem:-

⇒ useful for solving a particular current or voltage with circuits of multiple source.



To Determine I_2 using Super position principle we have to Do,

(i) I_2 (For E_1 only) → Deactivating other Source.

(ii) I_2 (for I_1 only) → " " "

∴ $I_2 = I_2$ for E_1 + I_2 for I_1 (if the direction I_2 for E_1 & I_2 for I_1 are same as given I_2 directions)

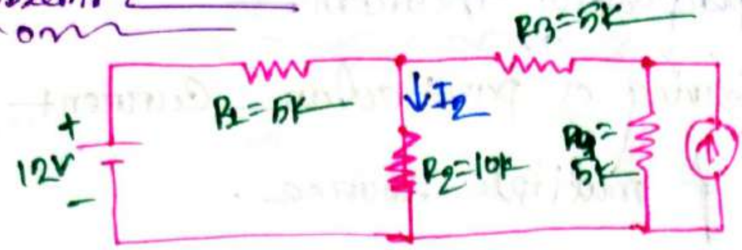
or, $I_2 = I_2$ for E_1 − I_2 for I_1 (If the Direction I_2 for E_1 is same as given I_2 Direction but I_2 for I_2 direction in opposite of given I_2 direction).

For Deactivating Source rule:-

① Voltage Source are short circuited.

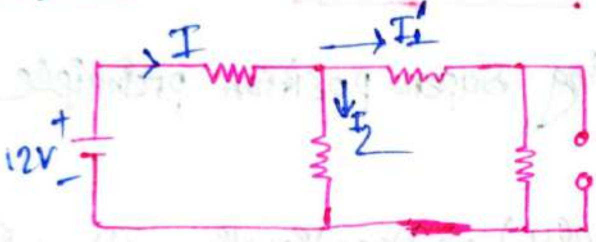
② Current Source are open circuited.

Problem: 2

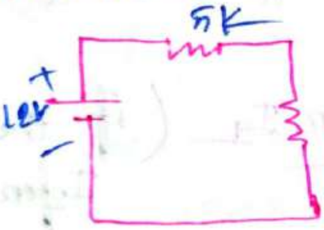


⇒ Determine I_2 using Super position theorem?

⇒ For E_1 only:-



Now Reduce the circuit:-

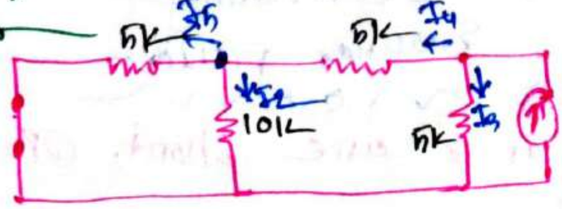


$$R' = \left(\frac{1}{10} + \frac{1}{5+5}\right)^{-1} = 5k\Omega$$

$$\therefore I = \frac{12}{10k} = 1.2\,mA$$

$$I_2 = \frac{R'}{R_2} \times I = \frac{5k}{10k} \times 1.2\,mA = 0.6\,mA$$

⇒ For I_1 only:-



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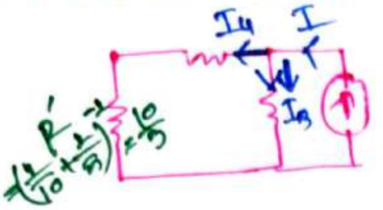
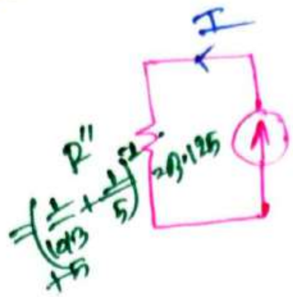
Now:- Reduce the Circuit:-





from CDR:-

$$I_4 = \frac{R_T}{R'+5} \times I$$

$$= \frac{3.125}{\frac{10}{3}+5} \times 6mA$$

$$= 2.25mA$$

$$\therefore \quad I_2 = \frac{R'}{R_2} \times I_4$$

$$= \frac{10/3}{10} \times 2.25mA$$

$$= 0.75mA$$

$$\therefore I_2 = I_2 \text{ for } E_1 + I_2 \text{ for } I_1$$

$$= 0.6mA + 0.75mA$$

$$= 1.35mA$$

Ans:-

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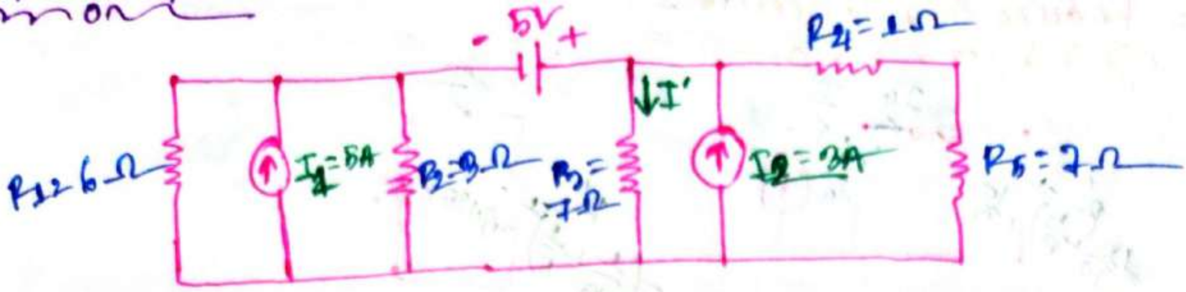
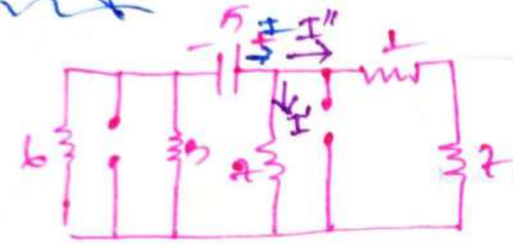
Problem:-



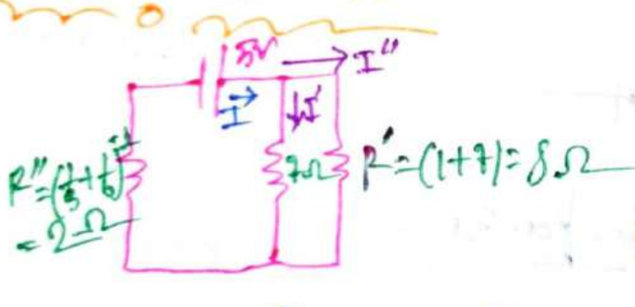
⇒ Determine the current I'

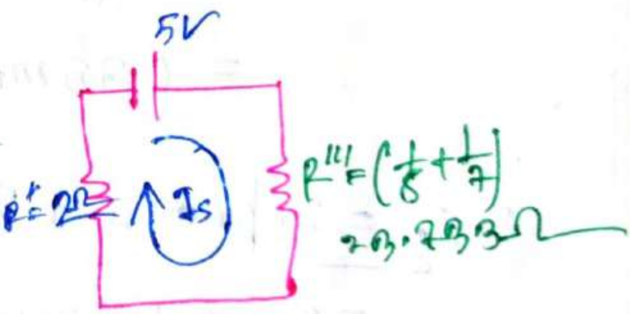
⇒ Using Super position:-

For E_1 only:-



Reduce the circuit:-





$\therefore I_S = \frac{E}{R_T} = \frac{5}{2+3.733} = 0.872\Omega$

$\therefore I' = \frac{R'''}{R_3} \times I_S$

$= \frac{3.733}{7} \times 0.872$

$\boxed{I' = 0.465A}$

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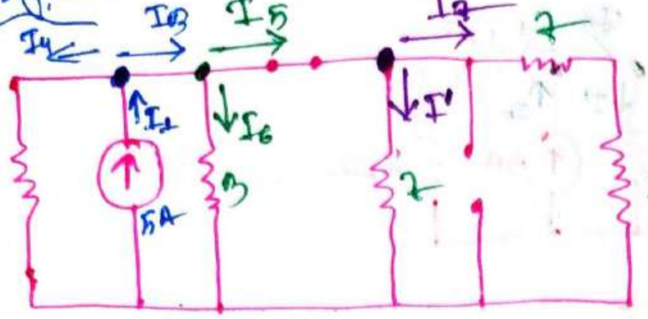
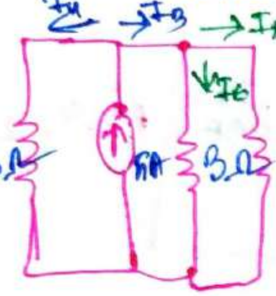
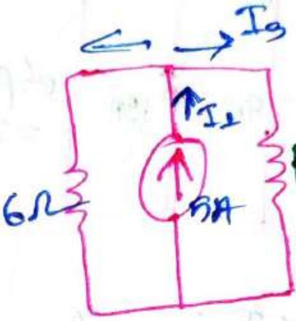
For I_1 Only:—



Reduce the circuit:—



$R' = \left(\frac{1}{8}+\frac{1}{7}\right)^{-1}$
$= 3.733\,\Omega$



$R'' = \left(\frac{1}{3}+\frac{1}{3.733}\right)^{-1}$
$= 1.663\,\Omega$

Now,

$$I_3 = \frac{R_T}{R''} \times I_1$$

$$= \frac{1.302}{1.663} \times 5$$

$$= 3.914A$$

$$\therefore R_T = \left(\frac{1}{6}+\frac{1}{1.663}\right)^{-1}$$
$$= 1.302\,\Omega$$

$$I_5 = \frac{R''}{R'} \times I_3$$

$$= \frac{1.663}{3.733} \times 3.914$$

$$= 1.743A$$

$$I' = \frac{R'}{R_3} \times I_5$$

$$= \frac{3.733}{7} \times 1.743$$

$$\boxed{I' = 0.929A}$$

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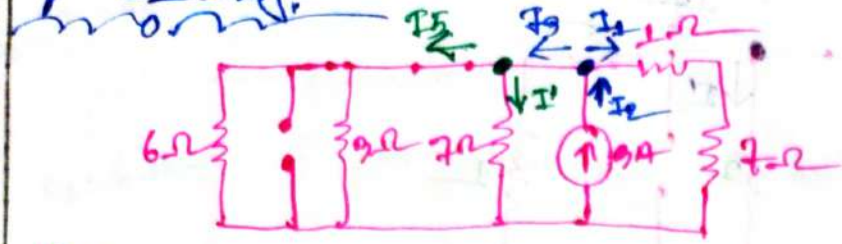
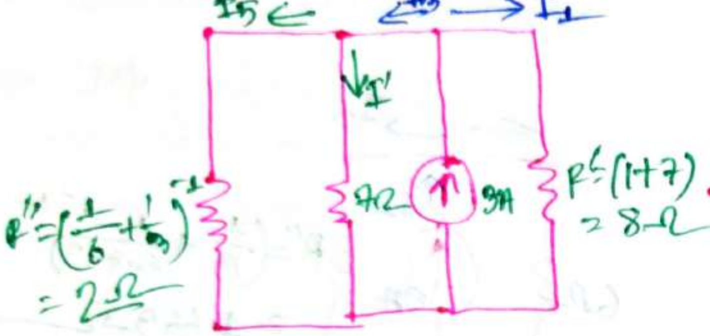
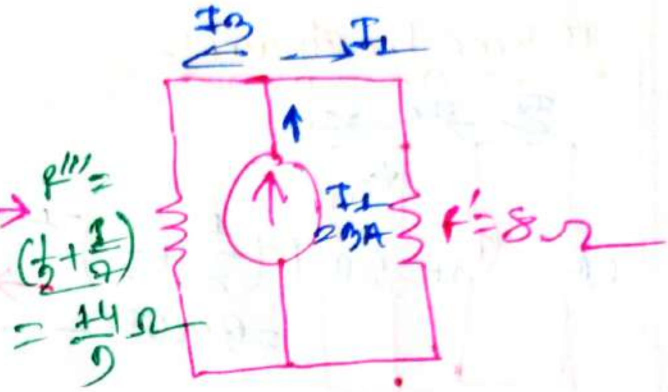
for I_2 only:



Reduce the circuit:





$$I_3 = \frac{R_T}{R'''} \times I_1$$

$$= \frac{1.302}{14/9} \times 3$$

$$= 2.511A$$

$$\therefore R_T = \left(\frac{1}{8} + \frac{1}{14/9}\right)^{-1}$$

$$= 1.302\Omega$$

$$I' = \frac{R'''}{R_3} \times I_3$$

$$= \frac{14/9}{7} \times 2.511A$$

$$\boxed{I' = 0.558A}$$

$\therefore I'_{net} =$ for $E_1(I')$ + for $I_1(I')$ + for $I_2(I')$

$= 0.465A + 0.929A + 0.558A$

$= 1.952A$

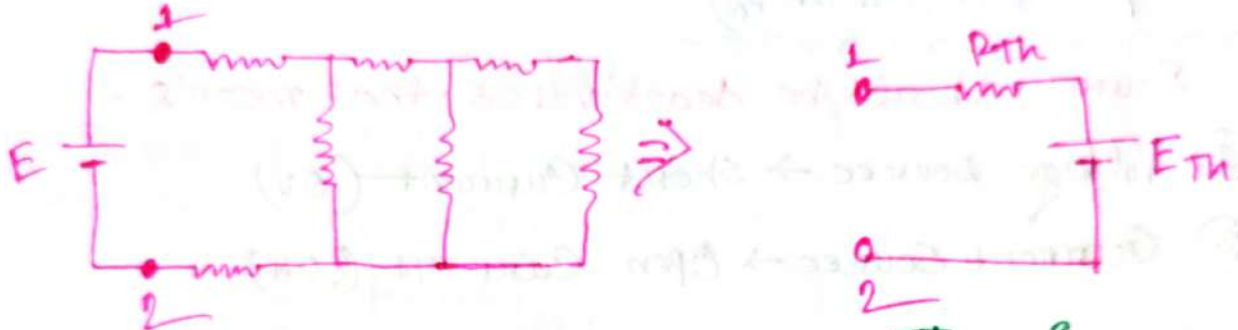
Ans:

TOPIC NAME : DAY : TIME : DATE : / /

THÉVENIN'S Theorem:—

⇒ Any two terminal circuit network can be reduced into —

(1) According to thevenins theorem → a Serie resistance and an emf (voltage) Source.



Thevenins Equivalent.

where,

R_{Th} = Thevenin's equivalent resistance.

E_{Th} = Thevenins equivalent Voltage.

Note:— For the two terminals present:—

1. The open Circuit Voltage = E_{Th}
2. The short Circuit Current = I_N

And:—

Total $\boxed{R_T = R_{Th} = R_N}$

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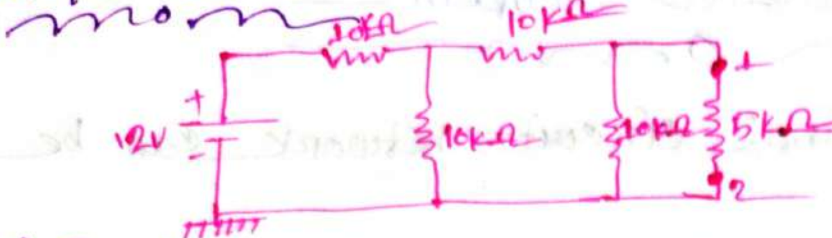
problem: 1:-



⇒ For the terminal 1 and 2 find the thevenins equivalent

Note:-

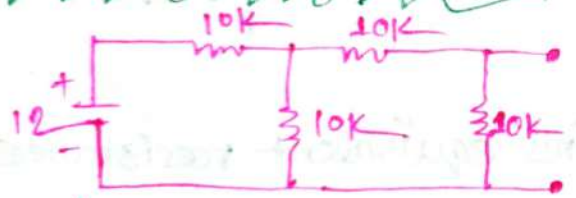
For find $R_T = (R_{Th}$ OR $R_N)$

⇒ Source should be deactivated that mean's .

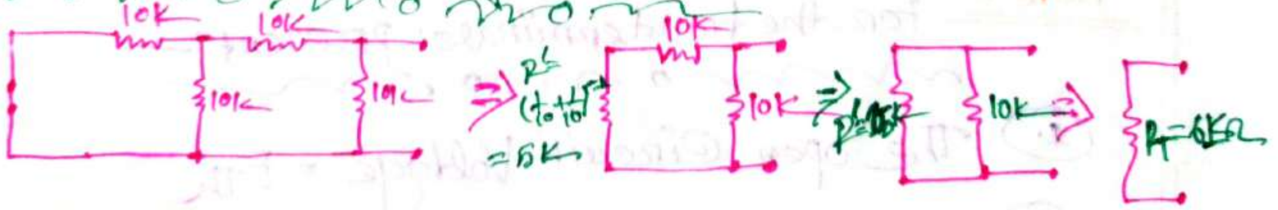
① Voltage source → short current (0V)

② Current source → Open current (0A)

⇒ Redraw the circuit:-

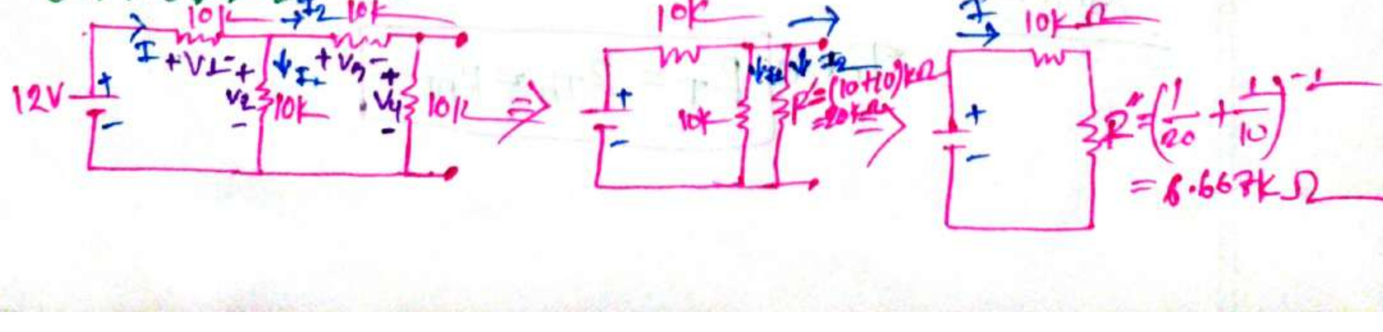


Now, for R_T, short the voltage source:-



$\therefore R_T = 6k\Omega$

Now:- for R_{Th}:-



$\therefore I = \frac{E}{R''+R} = \frac{12V}{(6.667+10)k} = 0.7199 mA$

$I_2 = \frac{R''}{R'} \times I = \frac{6.667k}{20k} \times 0.7199 mA$

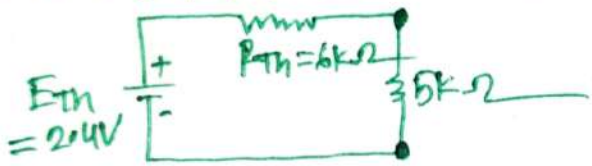
$= 0.24 mA$

$\therefore V_4 = E_{th} = V_{oc} = I_2 R_4$

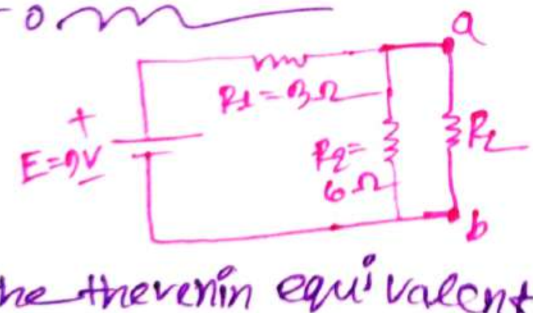
$= 0.24 mA \times 10k$

$= 2.4$ Volt.

∴ Thevenin Equivalent circuit

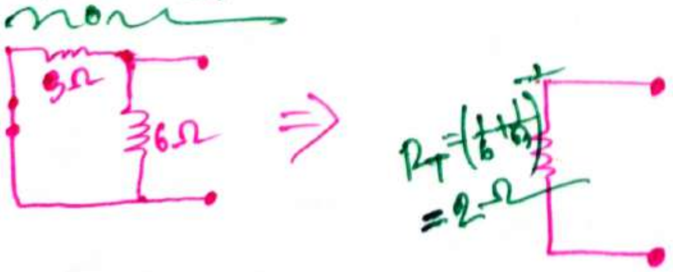


problem: D.6:



⇒ Find the thevenin equivalent circuit:

⇒ For R_{Th}:



$\therefore R_T = 2\Omega$

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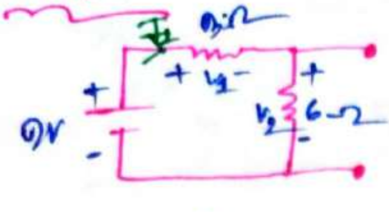
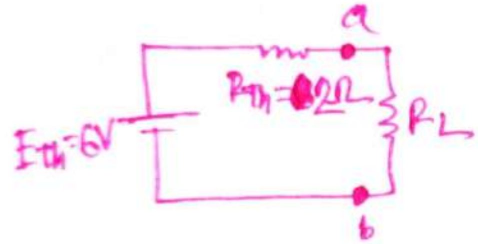
For Eth:—



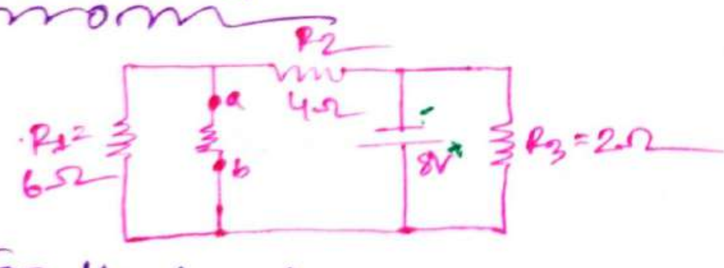
$$\therefore I_1 = \frac{E}{R_1+R_2} = \frac{9}{6+3} = 1A$$

$$\therefore V_2 = V_{oc} = E_{th} = IR_2 = 1 \times 6 = 6\,Volt$$

∴ The thevenen equivalent:—

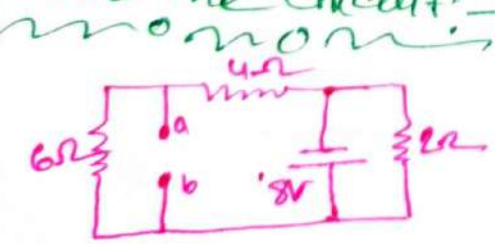


#Problem: 9.81—



⇒ For the terminal a and b Find the thevenen equivalent?

⇒ Redraw the circuit:—



For Rth:—

$$R_T = \left(\frac{1}{4}+\frac{1}{6}\right)^{-1} = 2.4Ω$$

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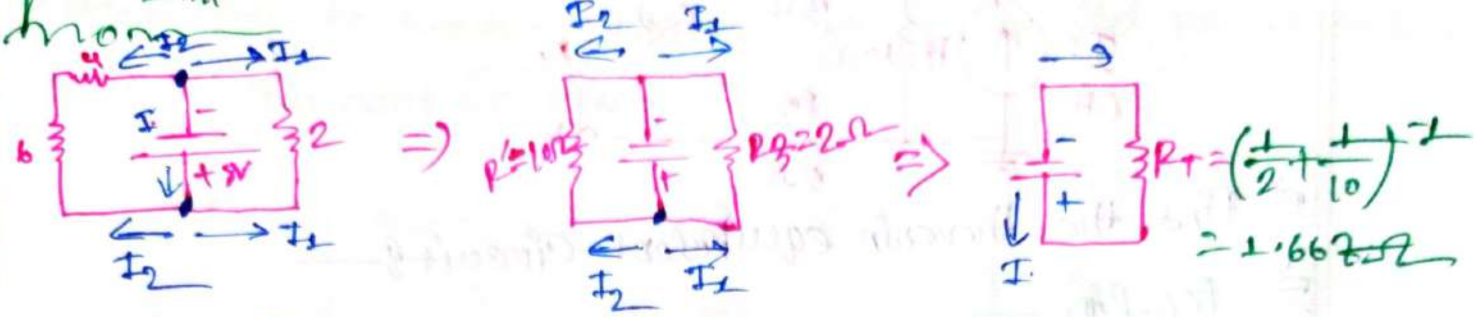
$\therefore R_{Th} = 2.4\,\Omega$

For E_{Th}:—

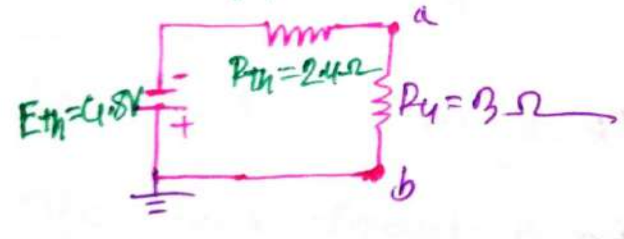


$$I = \frac{E}{R_T} = \frac{8}{1.667} = 4.8A$$

$$I_2 = \frac{R_T}{R'} \times I = \frac{1.667}{10} \times 4.8A = 0.80016A$$

$$V_1 = V_{oc} = I_2 R_1 = 0.80016 \times 6 = 4.8\,\text{Volt}$$

$\therefore$ Thevenin equivalent:—



(ii) Calculate the current passing through the R_4

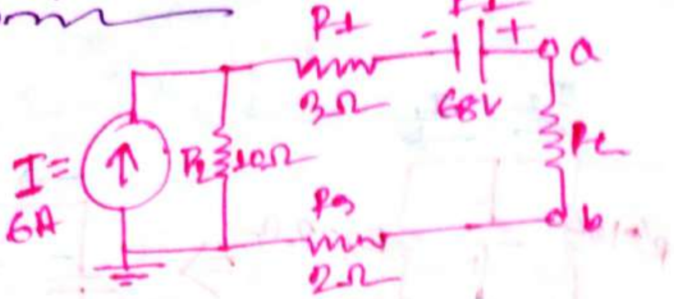
⇒ We have:—

$$E_{Th} = 4.8V$$
$$R_{Th} = 2.4V$$

$$\therefore I_4 = \frac{E_{Th}}{R_{Th}+R_4} = \frac{4.8}{2.4+3} = 0.889A$$

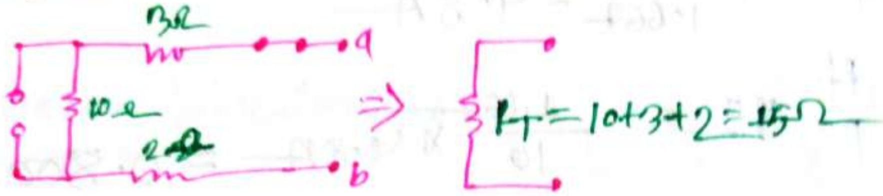
Ans:—

problems Q.13.A:—



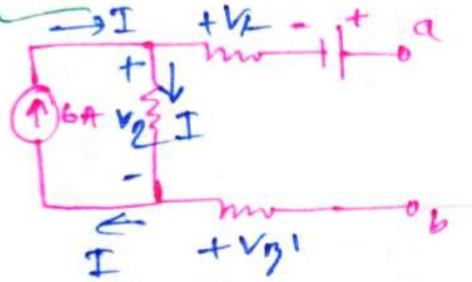
⇒ Find the thevenin equivalent circuit:—

⇒ For R_{Th}:—



$\therefore R_{Th} = 15\,\Omega$

⇒ For E_{Th}:—



$\therefore V_2 = IR_2 = 6\times 10 = 60\,\Omega$

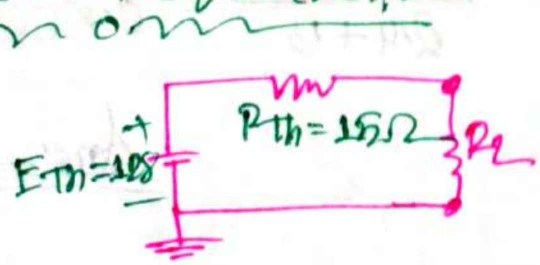
Appling KVL:—

$E_{Th} = V_2 + E_1$

$= (60+68)V$

$= 128V$

$\therefore$ Thevenin equivelent:—



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#Northon's Theorem:—

⇒ According to Northon's theorem → A parallel resistance with a current Source.

Northon's equivalent

R_N = Northon's equivalent resistance

I_N = Northon's equivalent current.

$$\boxed{\text{Total} \cdot R_T = R_N = R_{Th}}$$

☒ For the two terminal presents:—

① The Short circuit current = I_N

② The Open circuit Voltage = E_{th}

☒ For R_T (R_{th} OR R_N):—

⇒ Sources Should be deactivated thatmeans

(i) Voltage ~~source~~ Source → Short circuit (0V)

(ii) Current Source → open circuit (0A)

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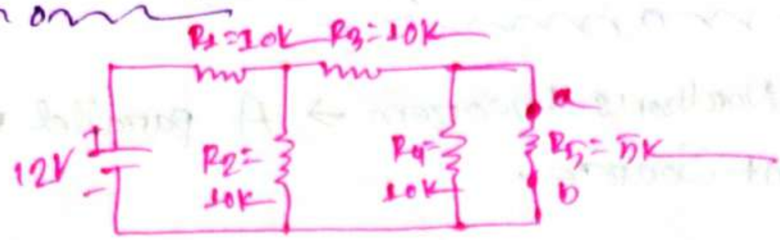
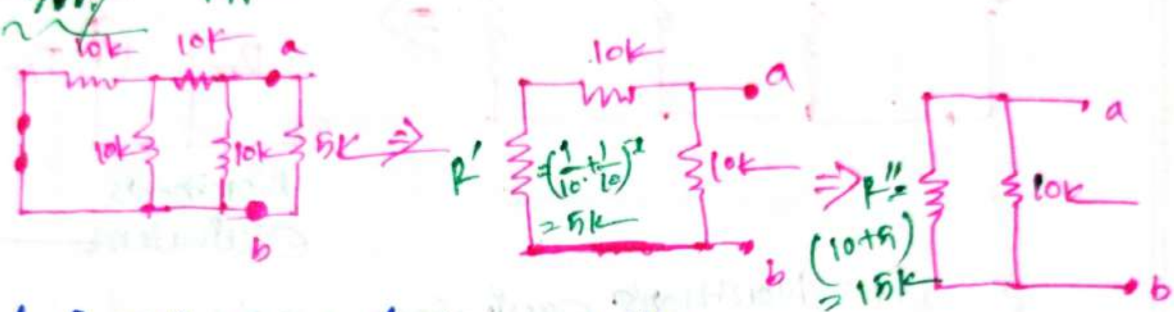
problem: 1 :—



⇒ For terminal a and b Find the Norton's equivalent?

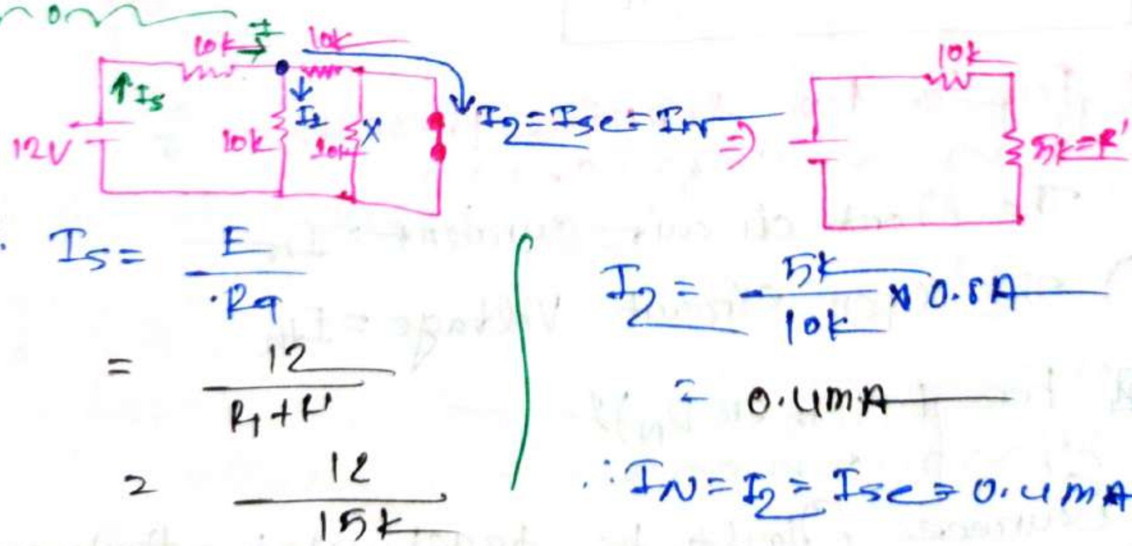
⇒ For ~~RN~~ RN



$\therefore R_T = R_N = R_{Th} = \left(\frac{1}{15k} + \frac{1}{10k}\right)^{-1}$

$= 6K$

⇒ For I_N:—



$\therefore I_S = \frac{E}{R_q}$

$= \frac{12}{R_1 + R'}$

$= \frac{12}{15k}$

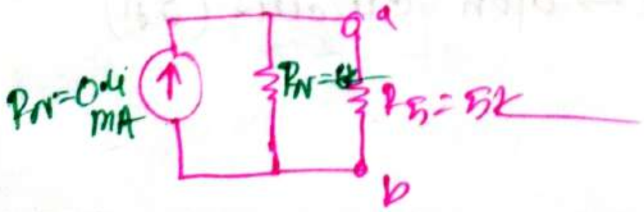
$= 0.8\ mA$

$I_2 = \frac{5k}{10k} \times 0.8A$

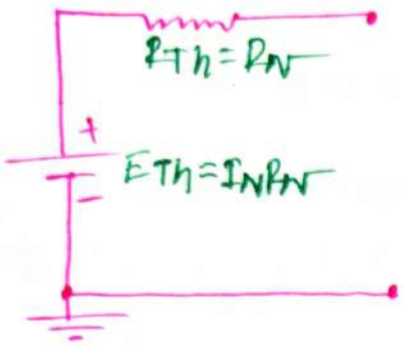
$= 0.4 mA$

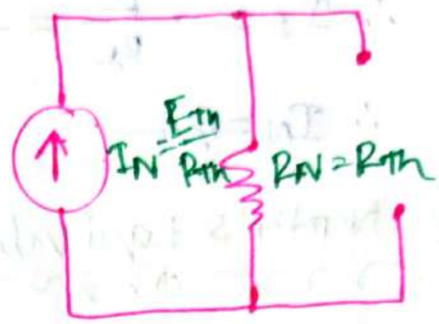
$\therefore I_N = I_2 = I_{sc} = 0.4 mA$

∴ Norton equivalent:—

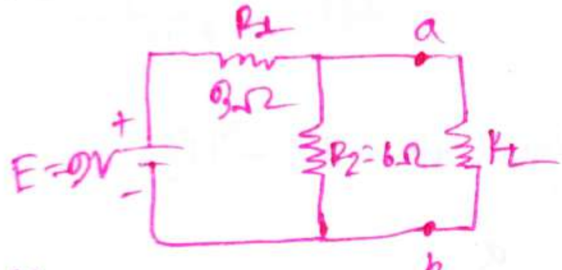


Conversion Between Thevenin and Norton's Equivalent circuit



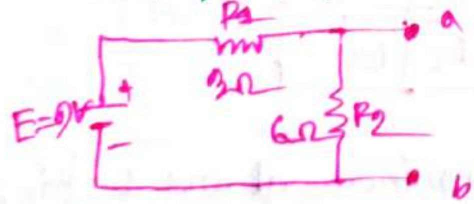


Problem: Dn!

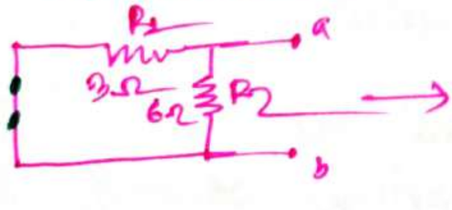


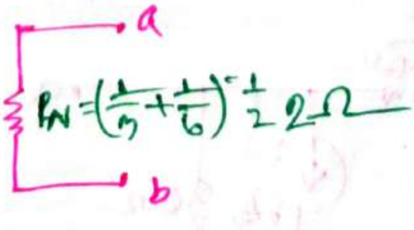
⇒ Find the Norton equivalent circuit at point a, b.

⇒ 1st draw the new circuit



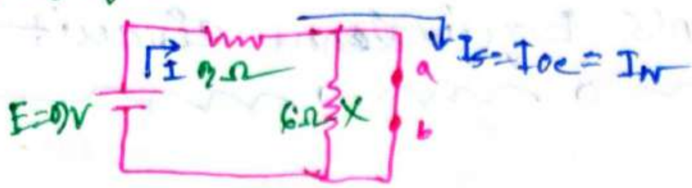
⇒ Now, for R_N:—





$\therefore R_N = 2\Omega$

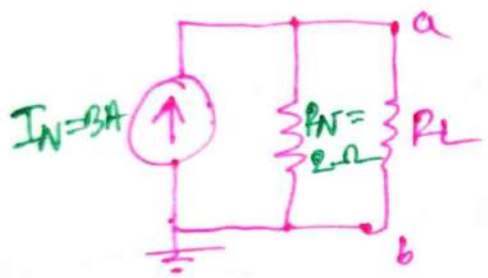
Now for I_N:—



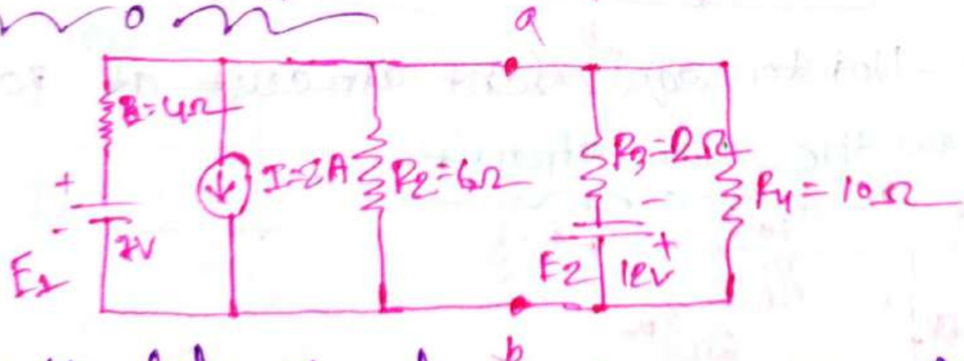
$\therefore I_S = \frac{E}{R_1} = \frac{9}{3} = 3A$

$\therefore I_N = 3A$

So, Norton's Equivalent:—

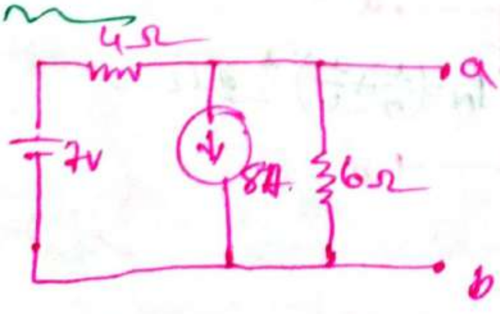


problem: 9.13:—



⇒ For the left side of terminal a and b find the Northon equivalent circuit.

⇒ 1st step:—



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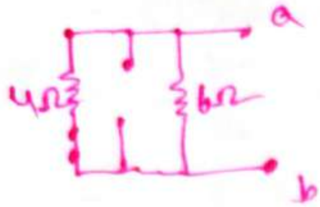
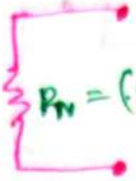
for R_N:—



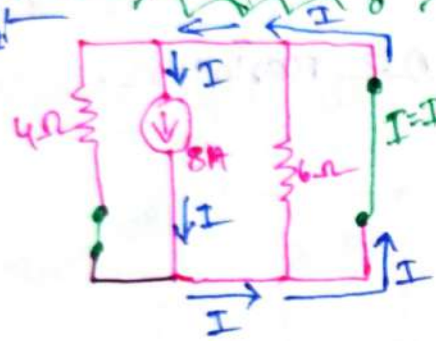


$R_N = \left(\frac{1}{4}+\frac{1}{6}\right)^{-1} = 2.4\,\Omega$

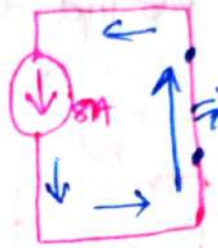
$\therefore R_N = 2.4\,\Omega$

for I_N:— (using Super position theorem):—

I only:—



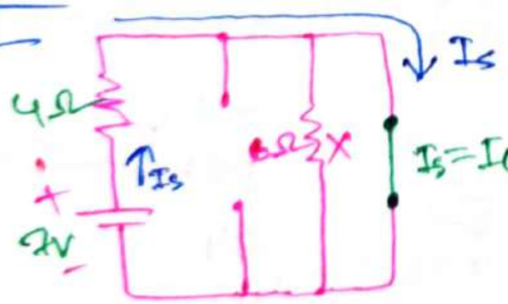
$\Rightarrow$



$I = I_{os} = I_N''$ [Due to Short circuit]

$\therefore I_N'' = 8A$ Anticlockwise.

E_1 only:—

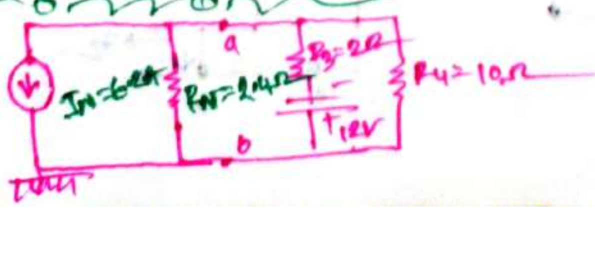


$\therefore I_s = \frac{E}{R_1} = \frac{7}{4} = 1.75A$

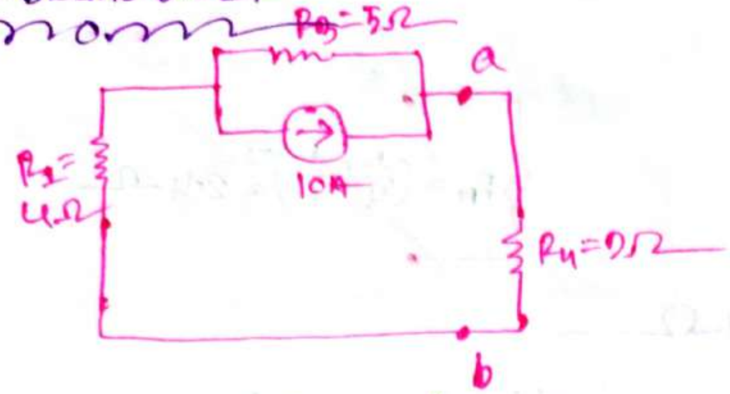
$\therefore I_N''' = 1.75A$ clockwise.

So, $I_N = I_N'' - I_N''' = (8-1.75)A = 6.2A$

So, Norton equivalent:—



problems 9.12:-

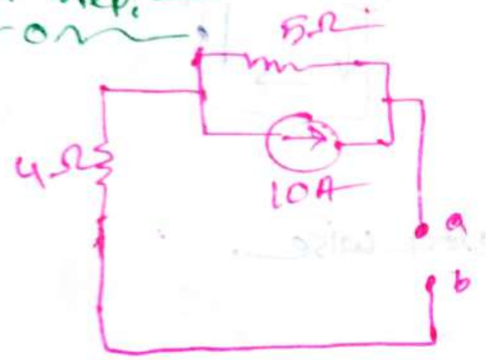


(i) For the terminal a & b Find the Norton equivalent

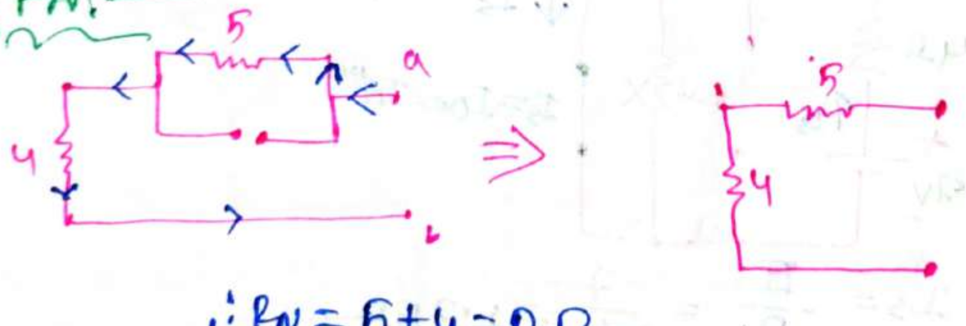
(ii) Current passing through 9Ω resistor.

⇒ (i)

1st step:-

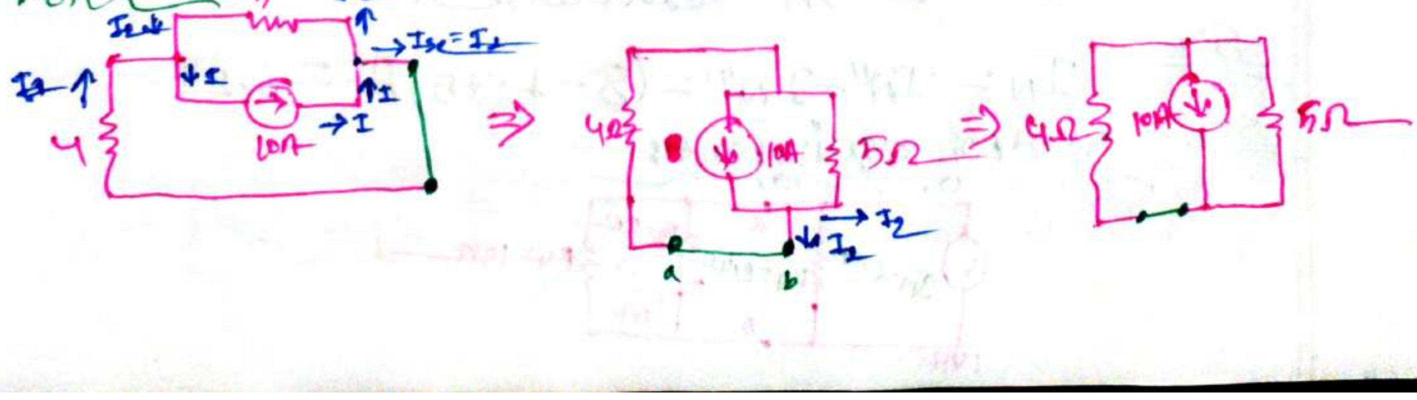


For R_N:-



$\therefore R_N = 5 + 4 = 9\Omega$

For I_N:-



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$\therefore R_T = \left(\frac{1}{5} + \frac{1}{4}\right)^{-1}$

$= 2.222\Omega$

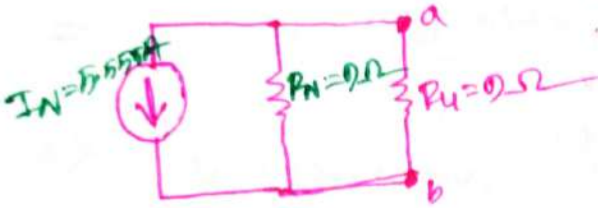
$I_1 = \frac{R_T}{R_1} \times I$

$= \frac{2.222}{4} \times 10$

$= 5.555A$

$\therefore I_1 = I_{oc} = I_N = 5.5.55A$

So, Norton Equivalent:-



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$\therefore R_T = \left(\frac{1}{R_N} + \frac{1}{R_4}\right)^{-1} = \left(\frac{1}{9} + \frac{1}{9}\right)^{-1} = 4.5\Omega$

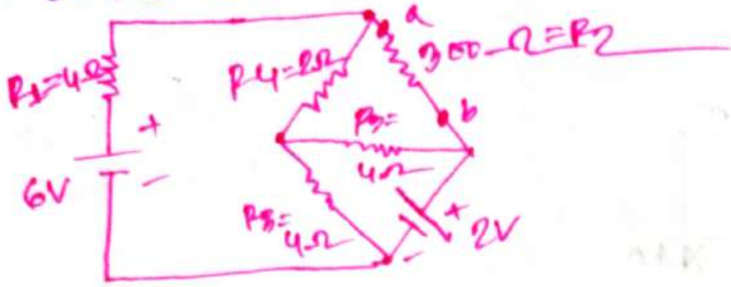
$\therefore I_{R_4} = \frac{R_T}{R_4} \times I_N$

$= \frac{4.5}{9} \times 5.555$

$= 2.777A$

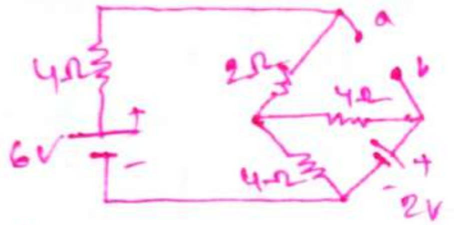
Ans:-

problem: 23:—

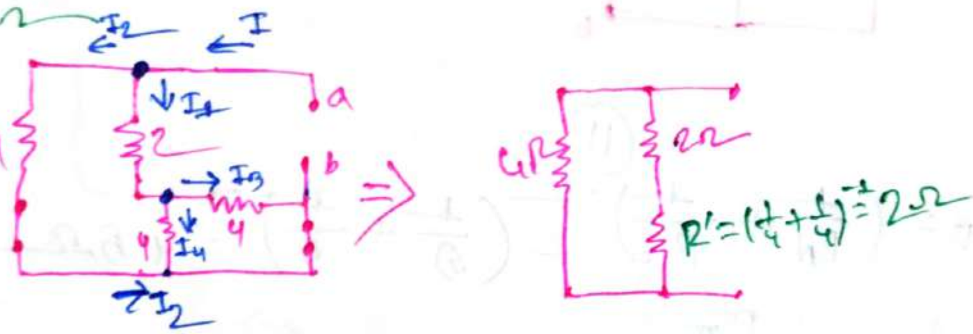


⇒ For the terminal a and b find the Norton equivalent.

⇒ 1st step:—



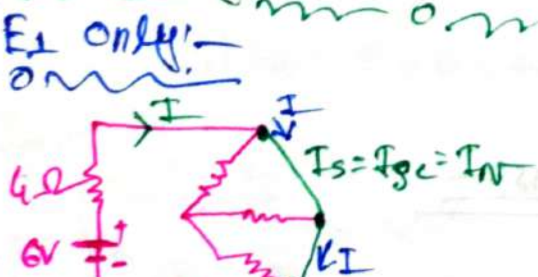
⇒ for R_N:—



$$\therefore R_T = \left(\frac{1}{4}+\frac{1}{4}\right)^{-1} = 2\,\Omega = R_N$$

⇒ for I_N: Using Superposition theorem:—

For E_1 only:—



$$I_s = I_{sc} = I_N' = \frac{E}{R_T} = \frac{6}{4} = 1.5A \text{ clockwise.}$$

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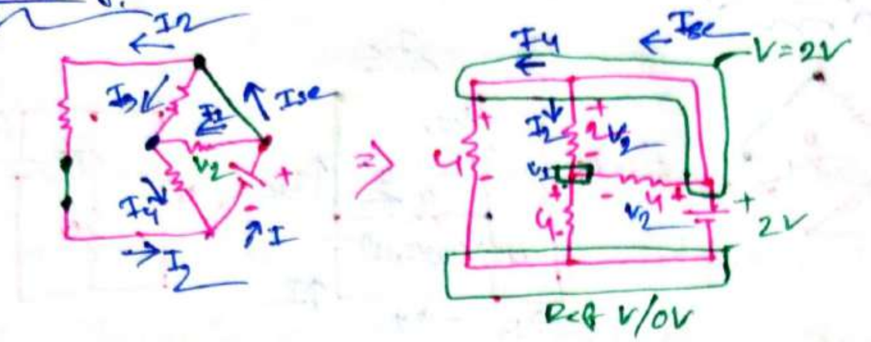
For E_2 only:—



$$\therefore I_4 = \frac{2-0}{4} = 0.5A$$

$$V_1 \Rightarrow \frac{V_1-0}{4} + \frac{V_1-2}{2} + \frac{V_1-2}{4} = 0$$

$$\Rightarrow \frac{V_1+2(V_1-2)+V_1-2}{4} = 0$$

$$\Rightarrow V_1+2V_1-4+V_1-2=0$$

$$\Rightarrow 4V_1 = 6$$

$$\therefore V_1 = \frac{6}{4} = 1.5 \text{ Volt}$$

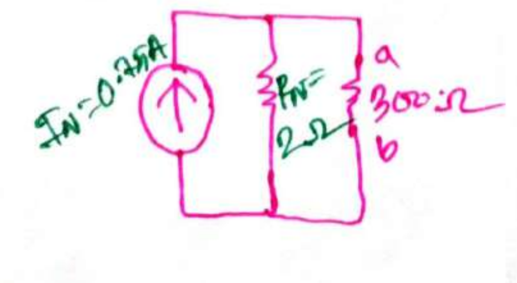
So, $V_2 = (2-1.5) = 0.5$ Volt

$$\therefore I_2 = \frac{V_2}{R_4} = \frac{0.5}{2} = 0.25A$$

$I_N'' = \therefore I_{sc} = I_4 + I_2 = (0.5+0.25) = 0.75A$ Anticlockwise

$$\therefore I_N = I_N' - I_N''$$
$$= (1.5 - 0.75)$$
$$= 0.75A$$

So, the Norton Equivalent:—



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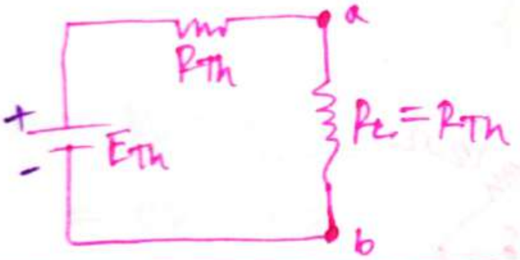
Maximum Power Transfer Theorem:

For thevenin:

$\Rightarrow$ A load will receive/consume/absorb maximum power from a network when its resistance is exactly equal to the Thevenin resistance for the network applied to the load.

$$\boxed{R_L = R_{Th}}$$



$$\therefore \boxed{I_L = \frac{E_{Th}}{R_{Th}+R_L} = \frac{E_{th}}{R_{Th}+R_{Th}} = \frac{E_{th}}{2R_{th}}}$$

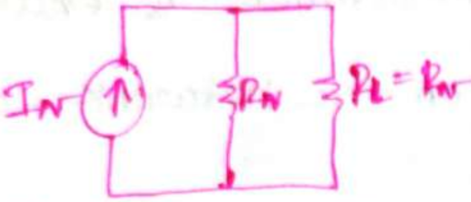
$$\therefore \text{power } P_{L_{max}} = I_L^2 R_L = \frac{(E_{th})^2}{(2R_{th})^2} \times R_{th} = \frac{E_{th}^2 \times R_{th}}{4R_{th}^2}$$

$$\therefore \boxed{P_{L_{max}} = \frac{E_{th}^2}{4R_{th}}}$$

For Norton:—

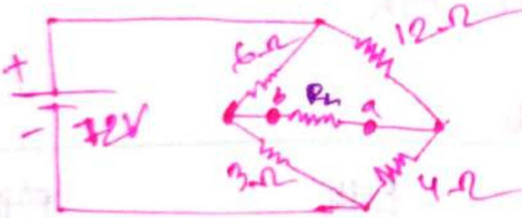
⇒ Maximum power will be delivered to the load when

$$\boxed{R_L = R_N}$$



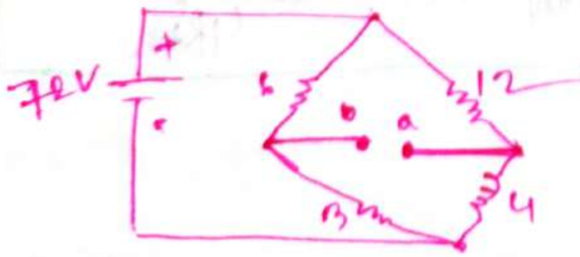
$$\boxed{\therefore P_{Lmax} = \frac{I_N^2 R_N}{4}}$$

problem: 0.6.1:—



⇒ Find the value of R_L for maximum power to R_L and determine the maximum power to R_L

⇒ 1st Step:

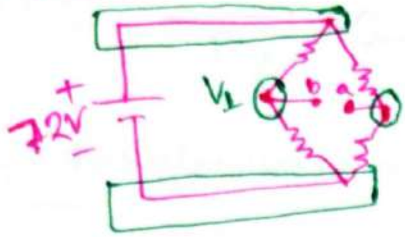


For R_{TH}:—

$R'' = \left(\frac{1}{12}+\frac{1}{4}\right)^{-1}$
$= 3\Omega$

$\therefore R_{Th} = 3+2 = 5\Omega$

For E_{th}:—



$$\frac{V_1-72}{6}+\frac{V_2-0}{3}=0$$

$$\frac{V_1-72+2V_1}{6}=0$$

$$3V_1-72=0$$

$$V_1 = 24\text{ Volt}$$

$$\frac{V_2-72}{12}+\frac{V_2-0}{4}=0$$

$$\frac{V_2-72+3V_2}{12}=0$$

$$\therefore 4V_2 = 72$$

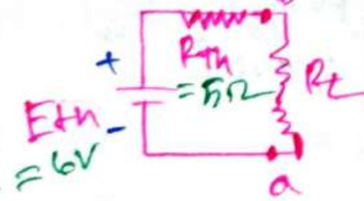
$$V_2 = 18\text{ Volt}$$

$$\therefore E_{th} = V_1 - V_2$$

$$= 24 - 18$$

$$= 6\text{ Volt}$$

$\therefore$ thevenin equivalent:—



$\therefore R_L = R_{Th} = 5\Omega$ for maximum power.

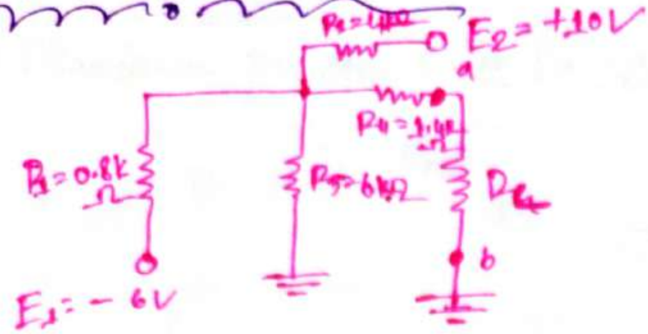
$$\therefore P_{Lmax} = \frac{E_{th}^2}{4R_{th}}$$

$$= \frac{6^2}{4\times5}$$

$$= 1.8W$$

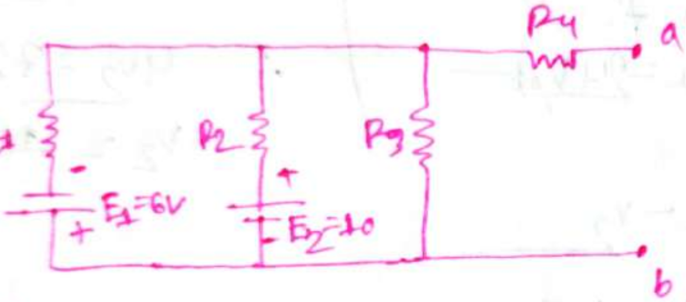
Ans.

Problem-Q.5.2:

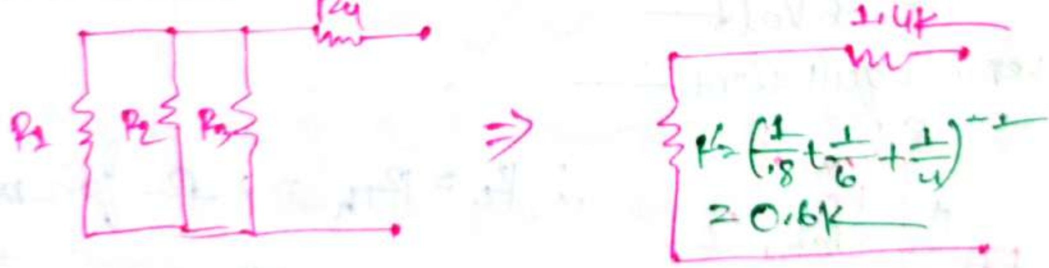


⇒ Find the value of R_L for maximum power to R_L and Determine the maximum power to R_L

⇒ 1st Step:



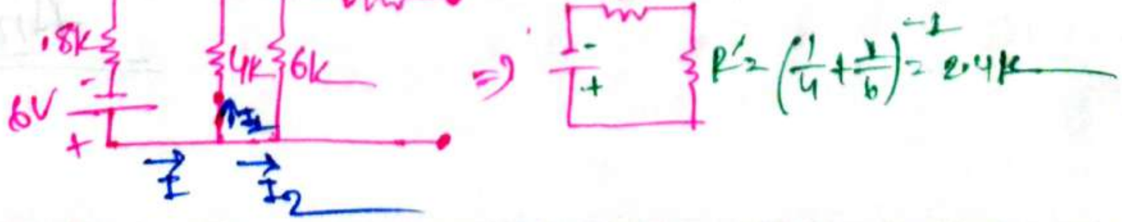
⇒ for Rth:



$\therefore R_{Th} = 0.6k + 1.4k$
$= 2k\Omega$

⇒ for Eth: using Super position principle:

E_1 only:

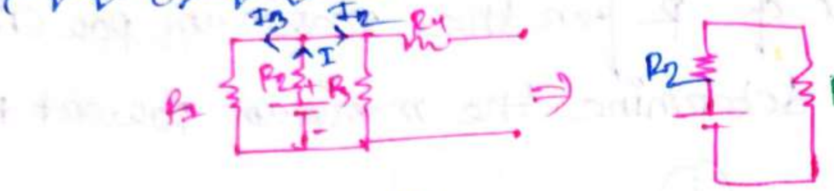


$$I = \frac{E}{R' + R_1} = \frac{6V}{(2.4 + .8)k} = 1.875mA$$

$$I_2 = \frac{R'}{R_3} \times I = \frac{2.4k}{6k} \times 1.875mA = 0.75mA$$

$\therefore I_2' =$ Anticlockwise $= 0.75mA$

For E_2 only:



$$R' = \left(\frac{1}{.8} + \frac{1}{6}\right)^{-1} = 0.705K$$

$$\therefore I = \frac{E}{R' + R_2} = \frac{10}{(0.705 + 4)k} = 2.125mA$$

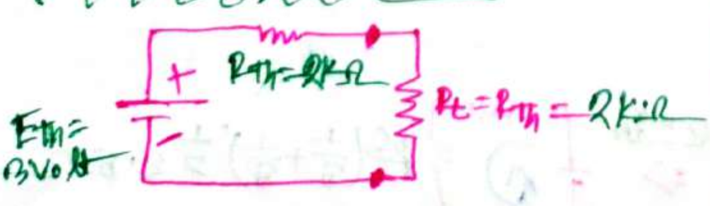
$$I_2'' = \frac{R'}{R_3} \times I = \frac{0.705k}{6k} \times 2.125mA = 0.249mA$$

$I_2 =$ Clockwise $= 0.249mA$

$\therefore$ effective $I_2 = (I' - I'') = (0.75 - 0.249)\,mA = 0.5mA$

$\therefore$ Voltage $V_3 = V_{oc} = R_{th} = I_2 \times R_3 = 0.5m \times 6K = 3$ Volt.

$\therefore$ Thevinen Equivalent:



$R_L = R_{th} = 2k\Omega$

$$\therefore P_{Lmax} = \frac{E_{th}^2}{4R_{th}} = \frac{3^2}{(4 \times 2)k} = 1.125mW$$

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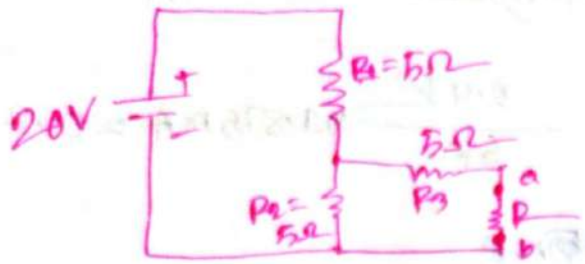
Problem: Q.5.21 :—

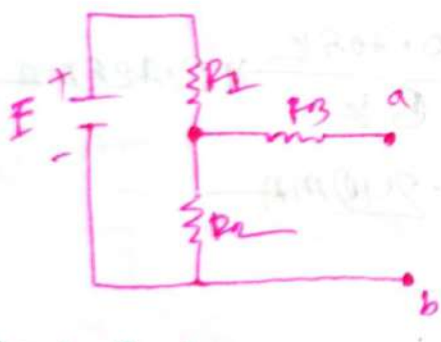


(i) Find the Norton equivalent of terminal a and b.

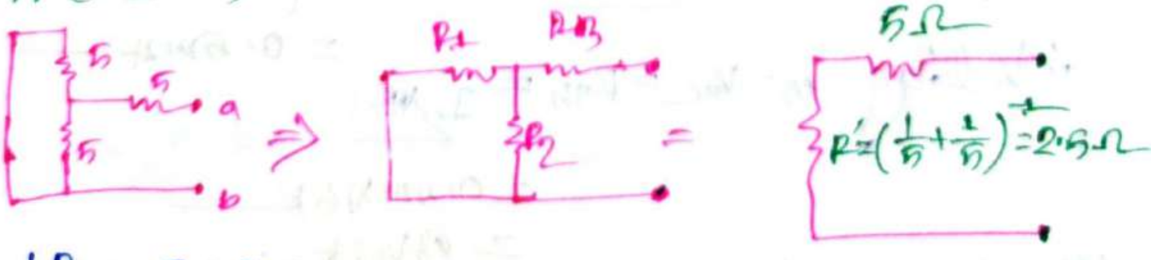
(ii) Find the value of R for the maximum power to R and determine the maximum power to R

(i)

1st Step :—

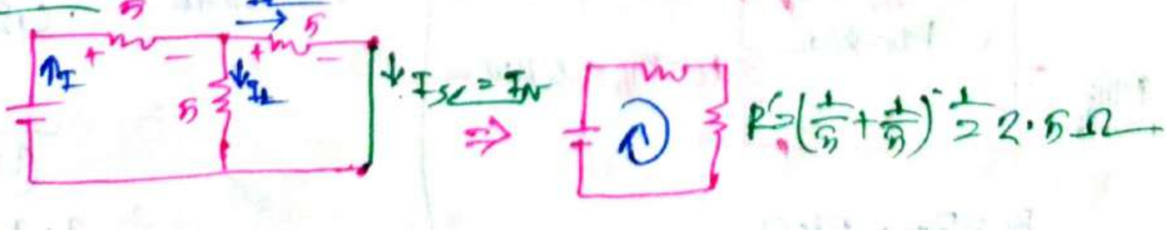


For $R_{th} = R_N$:—



$\therefore R_N = 5 + 2.5 = 7.5\Omega$

For I_N :—



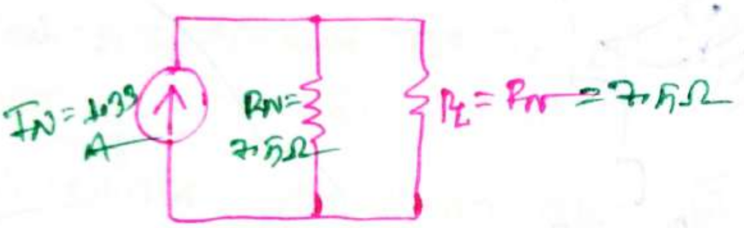
$\therefore R_T = 2.5 + 5 = 7.5\,\Omega$

$\therefore I_s = \dfrac{E}{R_T} = \dfrac{20}{7.5} = 2.667\,A$

$\therefore I_2 = \dfrac{R'}{R_3} = \dfrac{2.5}{5} \times 2.667 = 1.333\,A$

$\therefore I_2 = I_{sc} = I_N = 1.333\,A$

$\therefore$ Norton equivalent:—



for maximum power

$\therefore R_L = R_N = 7.5\,\Omega$

$\therefore P_{L\,max} = \dfrac{I_N^2 R_N}{4}$

$= \dfrac{(1.333)^2 \times 7.5}{4}$

$= 3.331\,W$

Ans—

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Methods of Analysis

Ch. 08

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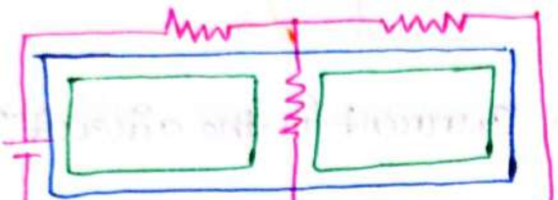
#KVL for Mesh/Loop Analysis:—

⇒ লুপ হচ্ছে একটি circuit এর close path।



⇒ এখানে দেখা যাচ্ছে, উক্ত circuit এর মোট লুপ হচ্ছে ৩টি

⇒ Mesh হচ্ছে circuit এর যে লুপটি যা, তার নিজের ভিতরে আর কোনো লুপ ধারণ করে না।

∴ এখানে মেশ/Mesh হচ্ছে ২টি।

Note:—

⇒ All meshes are loops but not all loops are meshes

For mesh Analysis:—

1st step:— Detect all meshes.

2nd step:— Assume Current for each meshes.

3rd step:— Assume that, either all current in clockwise or all current in anticlockwise.

4th Step:— Now, label the current as $I_1, I_2, \ldots$ etc

5th step:— Write equation for each mesh. Num of eq = Num of meshes

6th step:— Then Solve the equation

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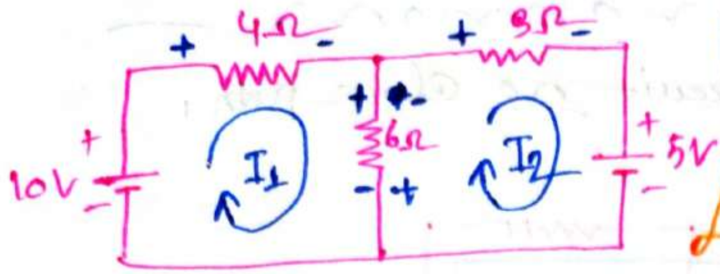
Problem:—



Note:—

⇒ When we are writing mesh 1's equation $I_1 > I_2$

⇒ When we are writing mesh 2's equation $I_2 > I_1$

⇒ Determine the mesh current in the circuit?

⇒ for I_1:—

$$-10 + I_1 4 + (I_1 - I_2)6 = 0$$

$$-10 + 4I_1 + 6I_1 - 6I_2 = 0$$

$$10I_1 - 6I_2 = 10 \quad \text{—— (I)}$$

⇒ for I_2:—

$$3I_2 + 5 + 6(I_2 - I_1) = 0$$

$$3I_2 + 5 + 6I_2 - 6I_1 = 0$$

$$\therefore -6I_1 + 9I_2 = -5 \quad \text{—— (II)}$$

(I) × 6 + (II) × 10 ⇒

$$60I_1 - 36I_2 = 60$$

$$-60I_1 + 90I_2 = -50$$

$$54I_2 = 10$$

$$\therefore I_2 = 0.185A$$

$$\therefore I_1 = \frac{10 + I_2 \times 6}{10} = \frac{10 + 6 \times 0.185}{10} = 1.111A$$

$\therefore I_1 = 1.11A$

$I_2 = 0.185A$

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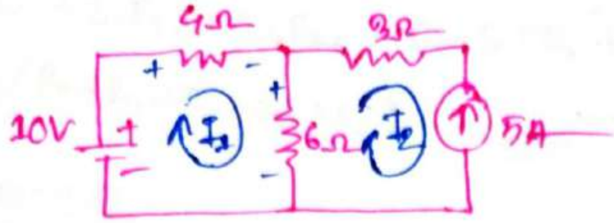
Problem: 2



⇒ Determine the mesh current:–

⇒ Here,
$I_2 = -5A$ because 5A current Direction exact opposite to our assuming current.

Now, for I_1:–

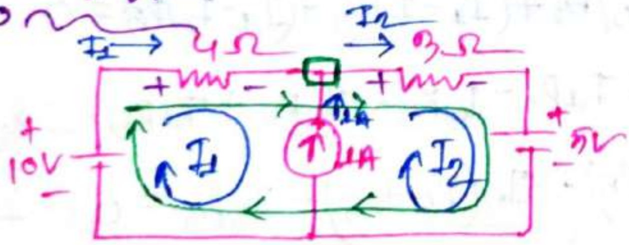
$$-10 + 4I_1 + 6(I_1 - (-5A)) = 0$$

$$4I_1 + 6I_1 = 10 - 30$$

$$10I_1 = -20$$

$$\therefore I_1 = -2A$$

Problem: 3:–



① Determine the mesh Current.

⇒ Supermesh:–

$$-10 + 4I_1 + 3I_2 + 5 = 0$$

$$4I_1 + 3I_2 = 5 \quad \text{—— (i)}$$

Using KCL.

$$-I_1 + I_2 = 1A \quad \text{—— (ii)}$$

Note:– Supermesh তৈরি করার পর আমরা যেকোনো একটি eq পাবো। অন্য eq এর জন্য আমাদের আগে বের করতে হবে কোন Current Source এর জন্য Supermesh টি তৈরি হলো। সেই current source থেকে নির্গত current টি যে Node এ যাচ্ছে সেই Node এ KCL Apply করতে হবে।

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(i) + (ii)×4 ⇒

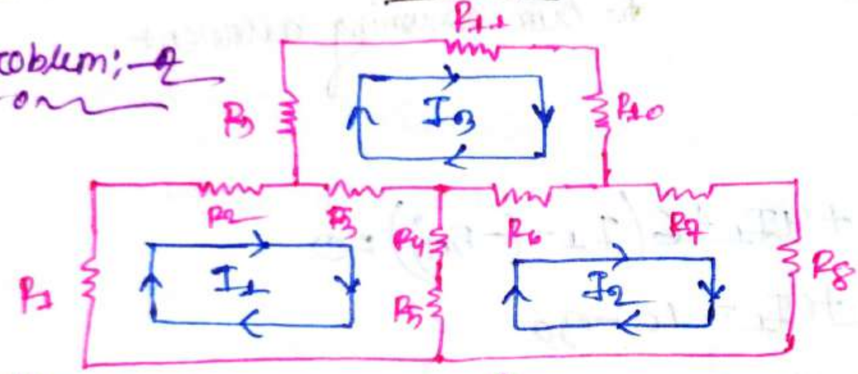
$$4I_1+3I_2=5$$
$$-4I_1+4I_2=4$$
$$7I_2=9$$

$\therefore I_2 = 1.285A$

$\therefore I_1 = 0.285A$

Ans:

problem: 2



⇒ Write the loop equation for the following circuit

⇒ For I_1 :

$$I_1R_1+I_1R_2+(I_1-I_3)R_3+(I_1-I_2)R_4+(I_1-I_2)R_5=0$$

$$\Rightarrow I_1R_1+I_1R_2+I_1R_3+I_1R_4-I_3R_3-I_2R_4+I_1R_5-I_2R_5=0$$

$$\Rightarrow I_1(R_1+R_2+R_3+R_4+R_5)-I_2(R_4+R_5)-I_3R_3=0 \quad \text{--- (i)}$$

⇒ For I_2 :

$$I_2R_7+I_2R_8+(I_2-I_1)R_5+(I_2-I_1)R_4+(I_2-I_3)R_6=0$$

$$I_2R_7+I_2R_8+I_2R_5-I_1R_5+I_2R_4-I_1R_4+I_2R_6-I_3R_6=0$$

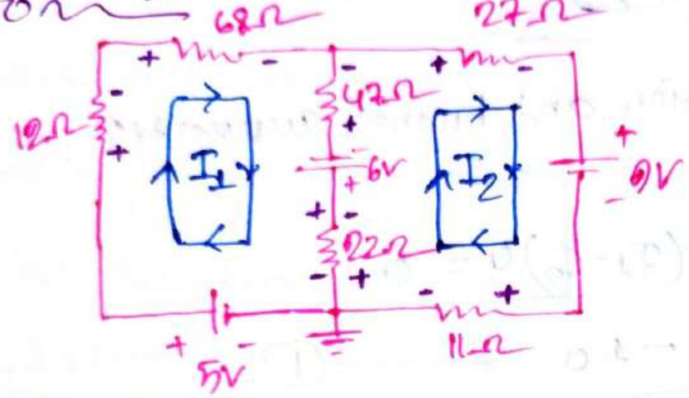
$$I_2(R_4+R_5+R_6+R_7+R_8)-I_1(R_4+R_5)-I_3R_6=0 \quad \text{--- (ii)}$$

For I_3:—

$$I_3R_9 + I_3R_{10} + I_3R_{11} + (I_3 - I_2)R_6 + (I_3 - I_1)R_3 = 0$$

$$I_3(R_3+R_6+R_9+R_{10}+R_{11}) - I_2R_6 - I_1R_3 = 0 \quad \text{——— (iii)}$$

problem: 3



Find the current I_1 and I_2

For I_1:—

$$-5+12I_1+68I_1+(I_1-I_2)47-6+(I_1-I_2)22=0$$

$$-5+12I_1+68I_1+47I_1-47I_2-6+22I_1-22I_2=0$$

$$I_1(12+68+47+22) - I_2(47+22) = 11$$

$$149I_1 - 69I_2 = 11 \quad \text{——— (i)}$$

For I_2:—

$$27I_2+9+11I_2+(I_2-I_1)22+6+(I_2-I_1)47=0$$

$$I_2(27+11+22+47) - I_1(22+47) = -15$$

$$-69I_1+107I_2=-15 \quad \text{——— (ii)}$$

Now Solve the equation:—

$\therefore I_1 = 0.0126A$

$I_2 = -0.1310A$

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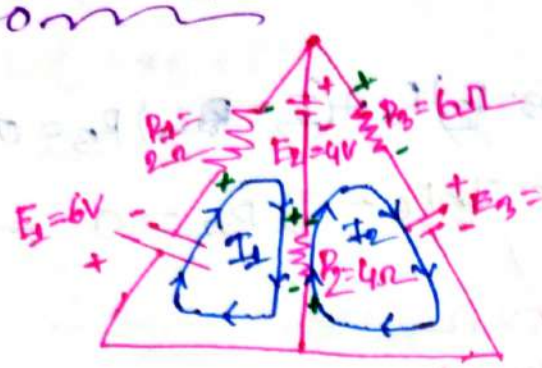
Problems 8.13:-



⇒ Find the mesh equation and branch currents?

⇒ For I_1:-

$$6 + 2I_1 + 4 + (I_1 - I_2)4 = 0$$

$$6I_1 - 4I_2 = -10 \quad \text{———— (i)}$$

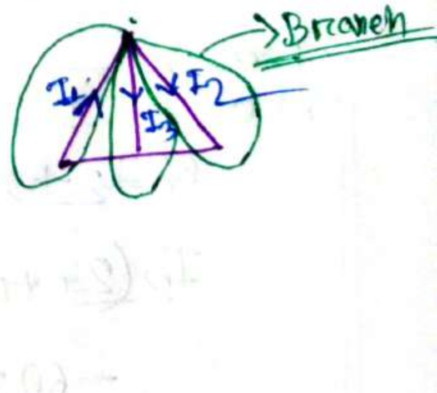
⇒ For I_2:-

$$3 + 4(I_2 - I_1) - 4 + 6I_2 = 0$$

$$-4I_1 + 10I_2 = 1 \quad \text{———— (ii)}$$

(i)×4 + (ii)×6 ⇒

$$24I_1 - 16I_2 = -40$$
$$-24I_1 + 60I_2 = 6$$



$$44I_2 = -34$$

$$\therefore I_2 = -0.772A$$

$$\therefore I_1 = -2.18A$$

$$\therefore I_3 = (I_1 - I_2) = (-2.18 + 0.772)$$
$$= -1.4072A$$

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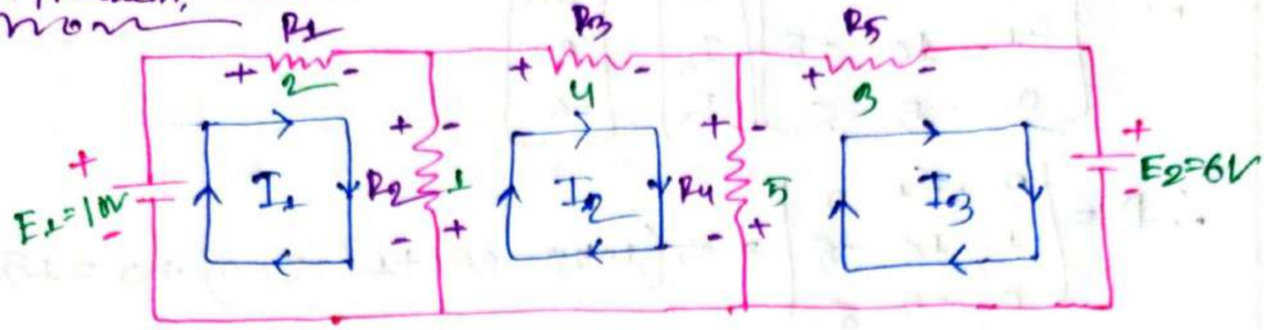
Problem:—



(a) Write the mesh equation for the each loop of the networks.

(b) Solve for the loop current.

(c) Find the current of each branch.

⇒ For I_1:— (I)

$$-10 + 2I_1 + (I_1 - I_2) = 0$$

$$3I_1 - I_2 = 10 \quad \text{——— (I)}$$

⇒ For I_2:—

$$(I_2 - I_1) + 4I_2 + (I_2 - I_3)5 = 0$$

$$-I_1 + 10I_2 - 5I_3 = 0 \quad \text{——— (II)}$$

⇒ For I_3:—

$$(I_3 - I_2)5 + 3I_3 + 6 = 0$$

$$8I_3 - 5I_2 = -6 \quad \text{——— (III)}$$

We have:— (II)

$$3I_1 - I_2 = 10V \quad \text{——— (I)}$$

$$-I_1 + 10I_2 - 5I_3 = 0V \quad \text{——— (II)}$$

$$-5I_2 + 8I_3 = -6V \quad \text{——— (III)}$$

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$$\therefore \begin{bmatrix} 3 & -1 & 0 \\ -1 & 10 & -5 \\ 0 & -5 & 8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ -6 \end{bmatrix}$$

$$\therefore D = \begin{vmatrix} 3 & -1 & 0 \\ -1 & 10 & -5 \\ 0 & -5 & 8 \end{vmatrix} = 3(10\times8-25)+1(-8-0)+0 = 157$$

$$D_1 = \begin{vmatrix} 10 & -1 & 0 \\ 0 & 10 & -5 \\ -6 & -5 & 8 \end{vmatrix} = 10(80-25)+1(0-30)+0 = 520$$

$$D_2 = \begin{vmatrix} 3 & 10 & 0 \\ -1 & 0 & -5 \\ 0 & -6 & 8 \end{vmatrix} = 3(0-30)-10(-8+0)+0 = -10$$

$$D_3 = \begin{vmatrix} 3 & -1 & 10 \\ -1 & 10 & 0 \\ 0 & -5 & -6 \end{vmatrix} = 3(-60)+1(6-0)+10(5-0) = -124$$

$$\therefore I_1 = \frac{D_1}{D} = \frac{520}{157} = 3.31A$$

$$I_2 = \frac{D_2}{D} = \frac{-10}{157} = -0.0636A$$

$$I_3 = \frac{D_3}{D} = \frac{-124}{157} = -0.7898A$$

(III)

We have:
now

$I_1 = 3.31A$

$I_2 = -0.0636A$

$I_3 = -0.7898A$

$\therefore I_4 = (I_1 - I_2)$

$= (3.31 + 0.0636A)$

$= 3.403A$

$I_5 = I_2 - I_3$

$= (-0.0636 + 0.7898)$

$= 0.7262A$

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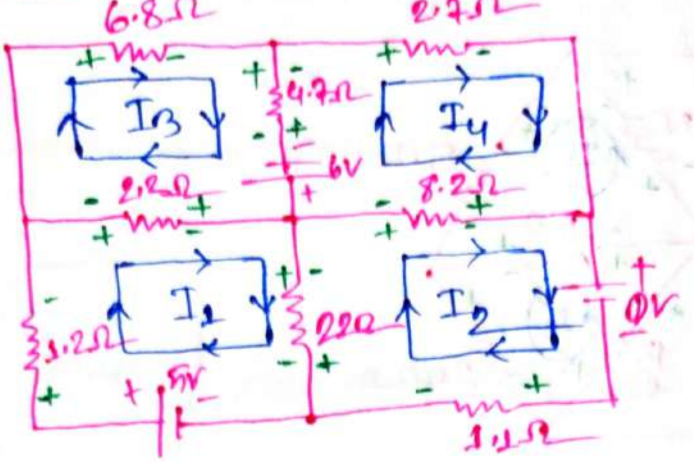
problem: 4:—



⇒ write the mesh equation for the every branch:—

⇒ For I_1:—

$$-5+1.2I_1+(I_1-I_3)2.2+(I_1-I_2)22=0$$

$$I_1(1.2+2.2+22)-2.2I_3-22I_2=0+5$$

$$25.4I_1-22I_2-2.2I_3=5 \quad \text{—— (I)}$$

⇒ For I_2:—

$$9+1.1I_2+(I_2-I_1)22+8.2(I_2-I_4)=0$$

$$I_2(1.1+22+8.2)-22I_1-8.2I_4=-9$$

$$-22I_1+31.3I_2-8.2I_4=-9 \quad \text{—— (II)}$$

⇒ For I_3:—

$$6.8I_3+4.7(I_3-I_4)-6+2.2(I_3-I_1)=0$$

$$-2.2I_1+13.7I_3-4.7I_4=6 \quad \text{—— (III)}$$

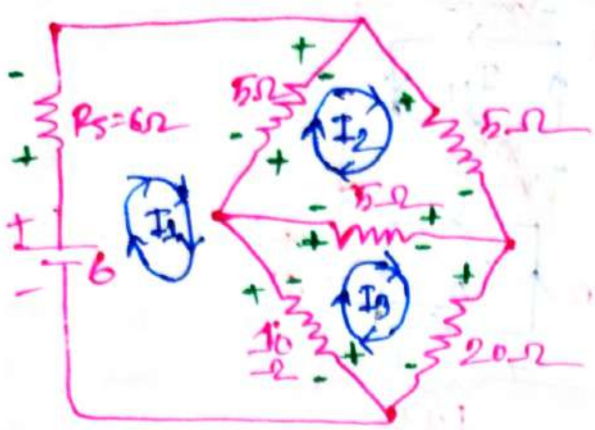
⇒ For I_4:—

$$2.7I_4+8.2(I_4-I_2)+6+4.7(I_4-I_3)=0$$

$$-8.2I_2-4.7I_3+15.6I_4=-6 \quad \text{—— (IV)}$$

Ans:

problems 5:—



=> write the Loop equation:—

=> For I_1:—

$$-6 + 6I_1 + 5(I_1 - I_2) + 10(I_1 - I_3) = 0$$

$$I_1(6+5+10) - 5I_2 - 10I_3 = 6$$

$$21I_1 - 5I_2 - 10I_3 = 6 \quad \text{——— (I)}$$

=> For I_2:—

$$5I_2 + 5(I_2 - I_3) + 5(I_2 - I_1) = 0$$

$$-5I_1 + 15I_2 - 5I_3 = 0 \quad \text{——— (II)}$$

=> for I_3:—

$$20I_3 + 10(I_3 - I_1) + 5(I_3 - I_2) = 0$$

$$-10I_1 - 5I_2 + 35I_3 = 0 \quad \text{——— (III)}$$

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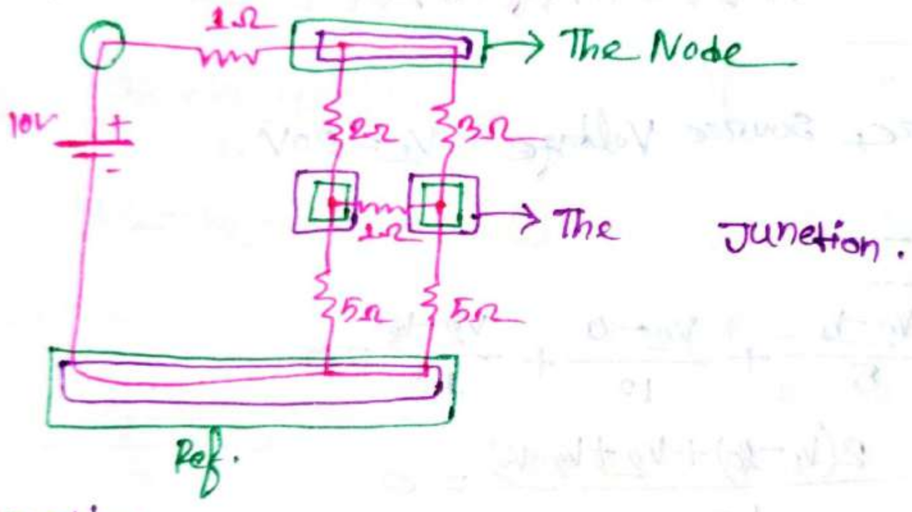
#Nodal Analysis:—

Branch:—

⇒ Any element of a circuit

NODE:—

⇒ Meeting point of two or more brances.



Junction:—

⇒ Meeting point of three or more branch

For Node Analaysis:—

Step 1:— Identify each node with labelling.

Step 2:— Write equation for each node. Using KCL.

Step:3:— Solve the equation to find the unknown node Voltage.

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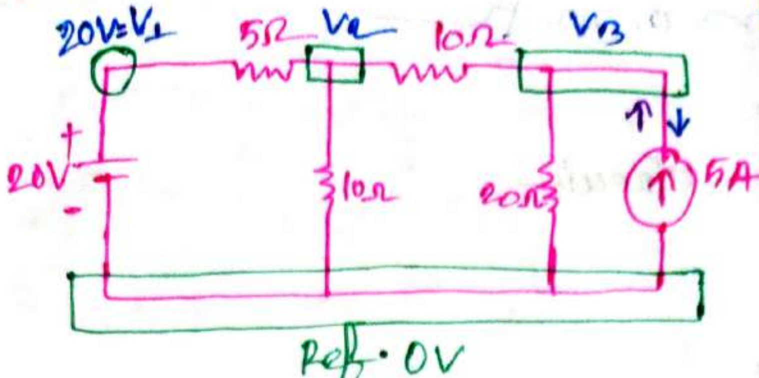
problem: 1:



Note:

1. Assumption: Current leaves from node.
2. Leaving Current is positive
3. Entering Current in negative
4. Node Voltage is highest in his own node equation.

⇒ Find the V_2, V_1, V_3 using node analysis:

⇒ For V_1:

here, source Voltage $= V_1 = 20V$

⇒ For V_2:

$$\Rightarrow \frac{V_2 - V_1}{5} + \frac{V_2 - 0}{10} + \frac{V_2 - V_3}{10} = 0$$

$$\Rightarrow \frac{2(V_2 - 20) + V_2 + V_2 - V_3}{10} = 0$$

$$\therefore 2V_2 + V_2 + V_2 - V_3 = 40$$

$$4V_2 - V_3 = 40 \quad \text{—— (I)}$$

⇒ For V_3:

$$\frac{V_3 - V_2}{10} + \frac{V_3 - 0}{20} - 5 = 0$$

$$\Rightarrow \frac{2(V_3 - V_2) + V_3 - 100}{20} = 0$$

$$\Rightarrow 2V_3 + V_3 - 2V_2 = 100$$

$$\Rightarrow -2V_2 + 3V_3 = 100 \quad \text{—— (II)}$$

Solve the equation:

$V_2 = 22$

$V_3 = 48$

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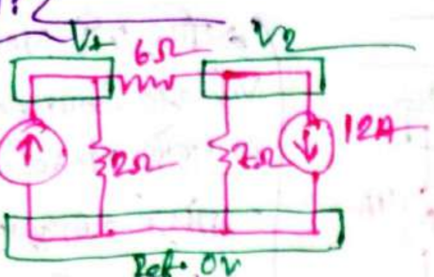
Problem: 2



⇒ Find the Node Voltages:—

⇒ For V_1:—

$$-3 + \frac{V_1 - V_2}{6} + \frac{V_1 - 0}{2} = 0$$

$$\Rightarrow \frac{-18 + V_1 - V_2 + 3V_1}{6} = 0$$

$$\therefore 4V_1 - V_2 = 18 \quad \text{——} \quad (i)$$

⇒ For V_2:—

$$\frac{V_2 - V_1}{6} + \frac{V_2 - 0}{7} + 12 = 0$$

$$\Rightarrow \frac{7(V_2 - V_1) + 6V_2 + 504}{42} = 0$$

$$\Rightarrow 7V_2 - 7V_1 + 6V_2 = -504$$

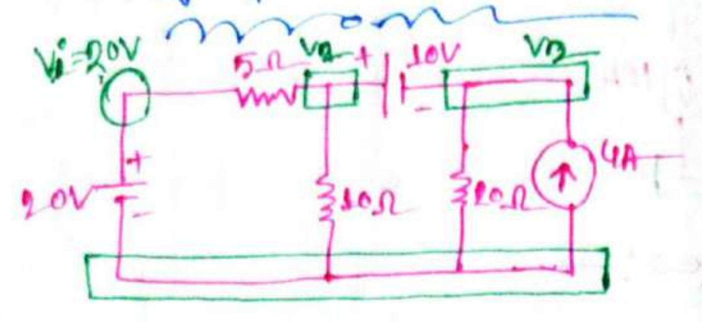
$$\Rightarrow -7V_1 + 13V_2 = -504 \quad \text{——} \quad (ii)$$

⇒ Now, Solve the equation:—

$$\therefore V_1 = -6 \text{ Volt}$$

$$V_2 = -42 \text{ Volt.}$$

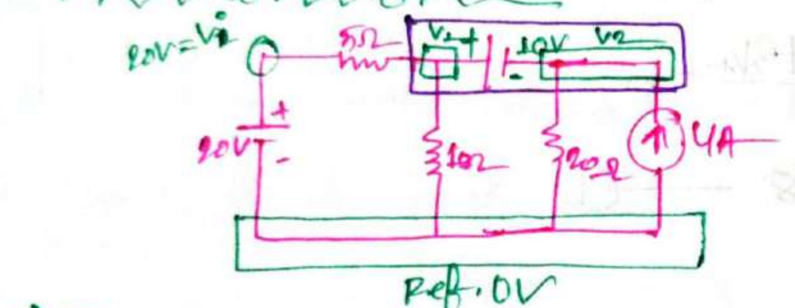
Special Case:



যদি দুটি Node এর মাঝখানে Voltage Source থাকে তাহলে ঐ দুটি Node + Voltage Source সম্পূর্ণ অংশকে একটি Node হিসাবে Consider করতে হবে। যার নাম হচ্ছে SuperNode.

⇒ Determine the Node Voltage:

⇒ Draw the Super Node:



SuperNode এর ক্ষেত্রে আমরা কেবল মাত্র একটি eq পাবো so, অন্য Eq এর জন্য আমরা Super-Node এর Voltage Source এ যেখানে আছে তার +ve প্রান্তের সাথে যে Node Voltage আছে তা থেকে -ve প্রান্তের Node Voltage বাদ দিয়ে Voltage Source এর value টি লিখতে হবে।

⇒ For SuperNode:

$$\frac{V_1 - V^0}{5} + \frac{V_1 - 0}{10} + \frac{V_2 - 0}{20} - 4 = 0$$

$$\Rightarrow \frac{4(V_1 - 20) + 2V_1 + V_2 - 80}{20} = 0$$

$$\Rightarrow 6V_1 + V_2 = 160 \quad \text{——— (I)}$$

⇒ Now: For 2nd eq:

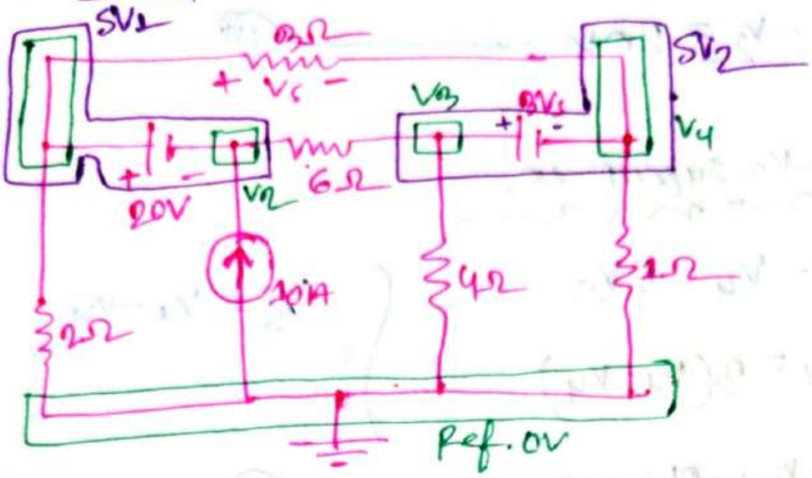
$$V_1 - V_2 = 10 \quad \text{——— (II)}$$

Solve the equation:

$$\therefore V_1 = 24.28V$$
$$V_2 = 14.28V$$

TOPIC NAME : ______ DAY : ______ TIME : DATE : / /

problems 3:-



⇒ Find the Node Voltage:-

⇒ For Super Node 1:-

$$\frac{V_1-0}{2}+\frac{V_1-V_4}{3}+\frac{V_2-V_3}{6}-10=0$$

$$\Rightarrow \frac{3V_1+2(V_1-V_4)+V_2-V_3-60}{6}=0$$

$$\Rightarrow 5V_1+V_2-V_3-2V_4=60 \quad \text{——— (I)}$$

For Super Node 2:-

$$\frac{V_4-V_1}{3}+\frac{V_4-0}{1}+\frac{V_3-V_2}{6}+\frac{V_3-0}{4}=0$$

$$\Rightarrow \frac{2(V_4-V_1)+6V_4+V_3-V_2+1.5V_3}{6}=0$$

$$\Rightarrow -2V_1=V_2+2.5V_3+8V_4=0 \quad \text{——— (II)}$$

TOPIC NAME : DAY : TIME : DATE : / /

For V_1 and V_2 Supernode:—

$$V_1 - V_2 = 20V \quad \text{——— (iii)}$$

For V_2 and V_3 supernode:—

$$V_3 - V_4 = 3V_S \qquad V_S = V_1 - V_4$$

$$\therefore V_3 - V_4 = 3(V_1 - V_4)$$

$$3V_1 - V_3 - 2V_4 = 0 \quad \text{——— (iv)}$$

From the equation:—

$$V_1 - V_2 = 20V$$

$$\therefore V_1 = 20 + V_2$$

Put the value of V_1 in eq (iv), (i), (ii) =)

$$6V_2 - V_3 - 2V_4 = -40 \quad \text{——— (v)}$$

$$-3V_2 + 2.5V_3 + 8V_4 = 40 \quad \text{——— (vi)}$$

$$3V_2 - V_3 - 2V_4 = -60 \quad \text{——— (vii)}$$

Now solve the equation:—

$$\therefore V_2 = 6.667V$$

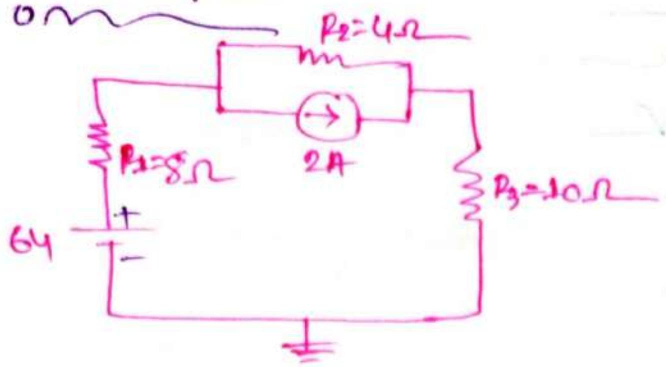
$$V_3 = 173.333V$$

$$V_4 = -46.667$$

$$V_1 = 20 + V_2$$

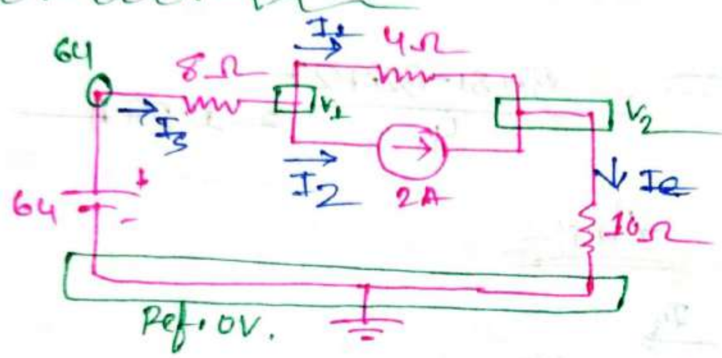
$$= 26.667V$$

problem:- 8.20:-



⇒ Applying Nodal ~~anylsi~~ analysis in the network

⇒ 1st redraw the circuit:-



For V_1:-

$$\frac{V_1-64}{8}+\frac{V_1-V_2}{4}+2=0$$

$$\Rightarrow \frac{V_1-64+2V_1-2V_2+16}{8}=0$$

$$\Rightarrow 3V_1-2V_2=48 \quad \text{———} (I)$$

For V_2:-

$$\frac{V_2-0}{10}+\frac{V_2-V_1}{4}-2=0$$

$$\Rightarrow \frac{2V_2+5(V_2-V_1)-40}{20}=0$$

$$\Rightarrow 2V_2+5V_2-5V_1=40$$

$$\Rightarrow -5V_1+7V_2=40 \quad \text{———} (II)$$

Solve two equation:—

$\therefore V_1 = 37.8181V$

$V_2 = 32.727V$

For I:—

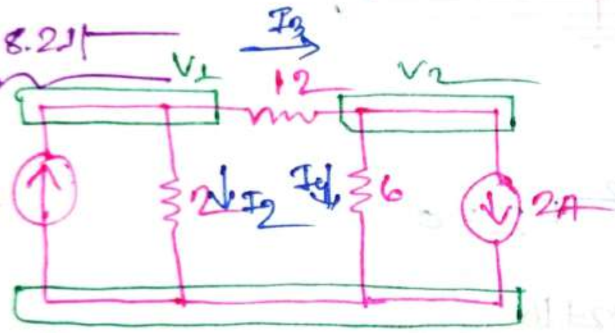
$$I_8 = \frac{64 - 37.8181}{8} = 3.272 \ A = I_e$$

For I_1:—

$$I_1 = \frac{V_1 - V_2}{4} = \frac{37.81 - 32.727}{4} = 1.27A$$

Ans:—

problem: 8.21



⇒ Determine the Nodal Voltage.

⇒ For V_1:—

$$\frac{V_1 - 0}{2} + \frac{V_1 - V_2}{12} - 4 = 0$$

$$\Rightarrow 6V_1 + V_1 - V_2 = 48$$

$$\Rightarrow 7V_1 - V_2 = 48 \quad \text{—— (i)}$$

⇒ For V_2:—

$$\frac{V_2 - V_1}{12} + \frac{V_2}{6} + 2 = 0$$

$$V_2 - V_1 + 2V_2 = -24$$

$$-V_1 + 3V_2 = -24 \quad \text{—— (ii)}$$

Solve the equation:—

$\therefore V_1 = 6V$

$V_2 = -6V$

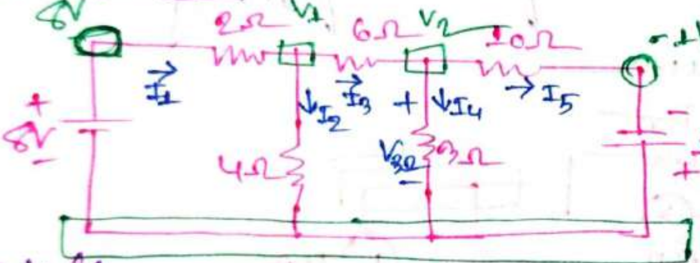
$\therefore I_3 = \dfrac{V_1 - V_2}{12} = \dfrac{(6+6)}{12} = 1A$

$I_2 = \dfrac{V_1}{2} = \dfrac{6}{2} = 3A$

$I_4 = \dfrac{V_2}{6} = \dfrac{-6}{6} = -1A$

Ans:—

problem 8.24:—



⇒ Find the voltage across the 3Ω Resistor:—

⇒ For V_1:—

$$\frac{V_1 - 8}{2} + \frac{V_1 - 0}{4} + \frac{V_1 - V_2}{6} = 0$$

$$\Rightarrow \frac{6V_1 - 48 + 3V_1 + 2V_1 - 2V_2}{12} = 0$$

$$\Rightarrow 11V_1 - 2V_2 = 48 \text{ ——— (I)}$$

⇒ For V_2:—

$$\frac{V_2 - V_1}{6} + \frac{V_2 - (-1)}{10} + \frac{V_2 - 0}{3} = 0$$

$$\Rightarrow \frac{5V_2 - 5V_1 + 3V_2 + 3 + 10V_2}{30} = 0$$

$$\Rightarrow -5V_1 + 18V_2 = -3 \text{ ——— (II)}$$

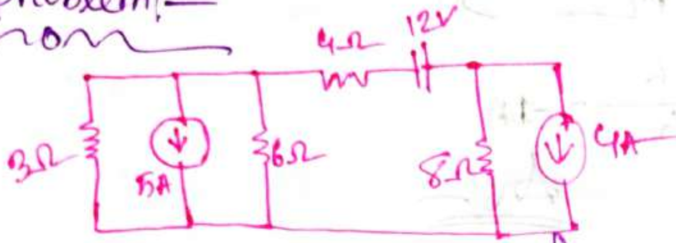
⇒ Solve the equation:—

$V_1 = 4.56V$

$V_2 = 1.101V$

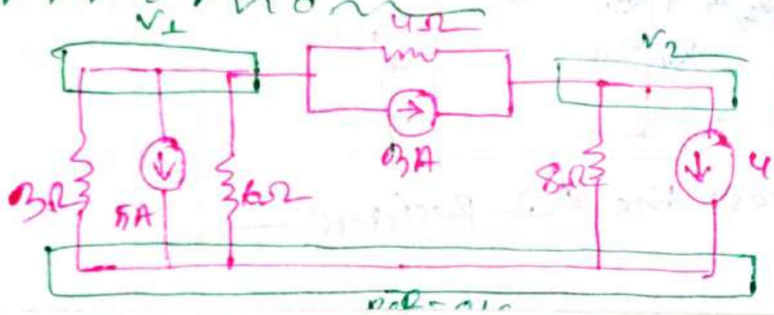
$\therefore V_{3\Omega} = V_2 - 0 = 1.101V$

problem:—



⇒ Written the nodal equations for the following circuit

⇒ Re Draw the circuit:—



For V_1:—

$$\frac{V_1}{3}+5+\frac{V_1}{6}+\frac{V_1-V_2}{4}+3=0$$

$$\Rightarrow \frac{4V_1+60+2V_1+3V_1-3V_2+36}{12}=0$$

$$\Rightarrow 9V_1-3V_2=-96 \quad \cdots\cdots (1)$$

For V_2:—

$$\frac{V_2-V_1}{4}+\frac{V_2}{8}+4-3=0$$

$$\Rightarrow \frac{2V_2-2V_1+8+V_2}{8}=0$$

$$\Rightarrow -2V_1+3V_2=-8 \quad \cdots\cdots (11)$$

Capacitors

Ch: 10:

Capacitor:—

⇒ A Capacitor Consists of two conducting plates seperated by an insulator (or dielectric)

Capacitance:—

⇒ Capacitance is a mesure the ability of a capacitor to store charge on it's plates.

Releationship among the Applied Voltage (E=V), the charge (Q) and The Capacitance C:—

⇒ Capacitance is the ratio of the charge (Q) in one plate of a capacitor to the difference Voltage between the two plates.

$$\therefore \boxed{C = \frac{Q}{V}}$$

C = Capacitance of capacitor (F)
Q = Charge (C)
V = Voltage (V)

SI Unit CV^{-1}

Common Unit Name is Farad (F)

Problem: 3:-

⇒ Find the Capacitance of a parallel plate capacitor if 1200μC of charge are deposited on its plates when 10V are applied across the plates.

⇒ In given:-

$Q = 1200\mu C = 1200 \times 10^{-6} C$

$V = 10\,V$

We know:-

$$C = \frac{Q}{V} = \frac{1200 \times 10^{-6}}{10} = 1.2 \times 10^{-6} F$$

$$= 120\,\mu F$$

Ans:-

problem: 4:-

⇒ How much charge is deposited on the plates of a 0.15μF Capacitor if 45V are applied across the Capacitor.

⇒ In given:-

$C = 0.15\mu F$

$V = 45V$

We know:-

$$C = \frac{Q}{V}$$

$$\therefore Q = CV = 0.15\mu \times 45 = 6.75\mu F$$

Ans:-

Capacitance Based on physical Dimension:—

$$\boxed{C = \epsilon \times \frac{A}{d}}$$

A = Surface Area of each plate

d = Distance between plate.

ϵ = permittivity

$$\boxed{C = \epsilon_0 \times \epsilon \frac{A}{d}}$$

$$\boxed{\epsilon_0 = 8.85 \times 10^{-12}}$$

ϵ_0 = relative permittivity.

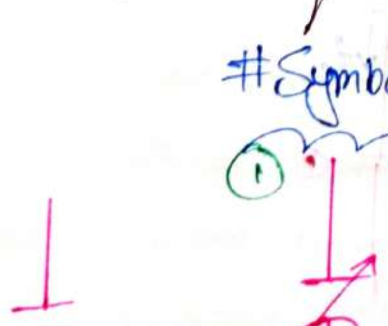
$$\boxed{\begin{aligned} C &= \epsilon_r C_0 \\ C_0 &= \epsilon_0 \frac{A}{d} \end{aligned}}$$

C_0 = Capacitance Consider at air.

C = Capacitance for any other dielectric material.

Symbol of Capacitance:—

(ii) (i)



Fixed Variable.

Transients in Capacitive Network

We know:–

$$q = CV$$

$\therefore$ $i = \frac{dq}{dt}$ with respect to time.

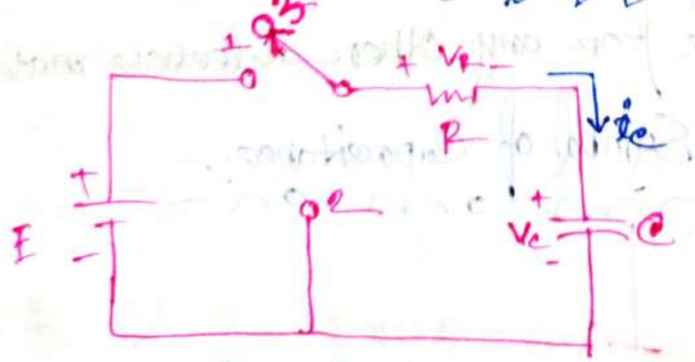
$$i_c = \frac{dq}{dt} = \frac{dV_c}{dt} \times C$$

$$V_c = \frac{1}{C}\int_{t_0}^{t} i_c\,dt + V_c(t_0)$$

$\therefore$ Energy storage by a Capacitor

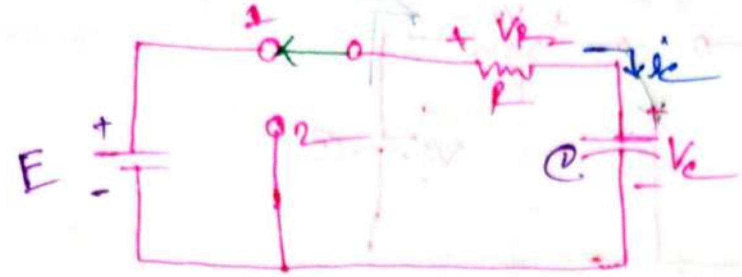
$$\boxed{W_c = \frac{1}{2} C V_c^2 \ (J)}$$

Charging and Discharging in a Capacitor:



Stored phase.

Charging Phase:-



Charging Phase:-

Applying KVL:-

$$E = V_R + V_c$$
$$= Ri_c + V_c$$
$$= RC\frac{dV_c}{dt} + V_c \quad \text{.......}(1)$$

From eq (1):- :

	Unit
$V_c = E(1 - e^{-\frac{t}{\tau}})$	Volt, V
$i_c = \frac{E}{R}e^{-\frac{t}{\tau}}$	Ampere, A
$V_R = Ee^{-t/\tau}$	Volt, V
$\tau = RC$	Second, s

τ = time Constant.

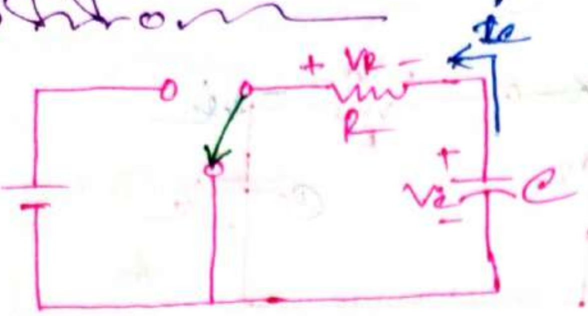
V_c = Voltage Across Capacitance

i_c = Current Across Capacitance

V_R = Voltage Across Resistor

$T = 5\tau$ = Total charging time.

Discharging phase:—



Applying KVL:—

$V_R = V_c \quad Ri_c = E$

$\therefore E = RC \cdot \frac{dV_c}{dt} \cdots\cdots$ (11)

From eq (11):—

$$\boxed{\begin{array}{l} V_c = Ee^{-\frac{t}{\tau}} \\ i_c = -\frac{E}{R}e^{-\frac{t}{\tau}} \\ V_R = -Ee^{-\frac{t}{\tau}} \\ \tau = RC \end{array}}$$

V_c = Voltage of Conductor/Voltage across capacitance

i_c = Current Across Capacitence

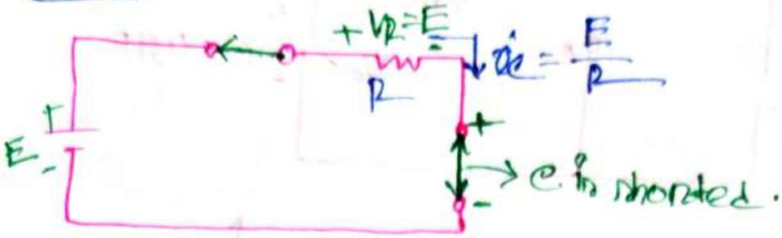
V_R = Voltage of Resistor/Voltage across Resistance.

τ = time constant

$t = 5\tau$ = Total time/Total Discharging time.

At the instant of switch is closed:—

$\Rightarrow$ At time $t=0$



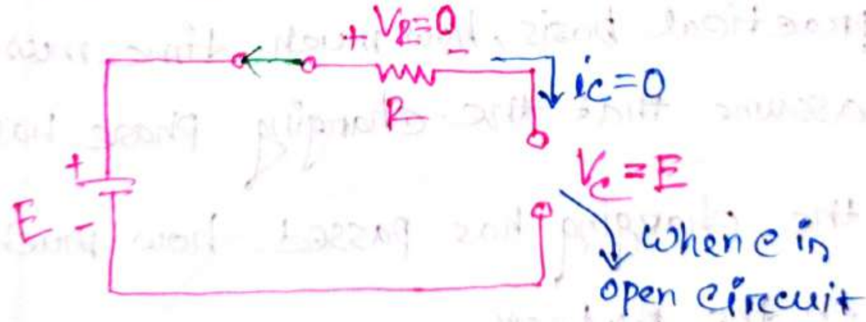
$$\boxed{V_c=0}$$

$$\boxed{i_c=\frac{E}{R}}$$

$$\boxed{V_R=E}$$

Stedy-State Condition:—

$\Rightarrow$ At time $t=\infty$:-

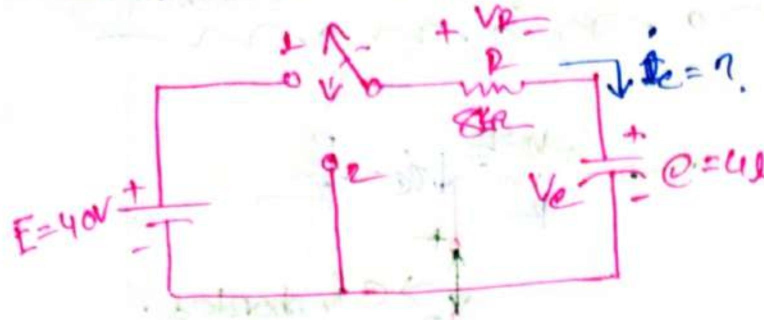


$$\boxed{\therefore V_c=E}$$

$$\boxed{V_R=0}$$

$$\boxed{i_c=0}$$

problems 10.6:-



(a) Find the mathematical expression for the transient behavoir of V_c, i_c and V_R if switch in close at $t = 0$ s

(b) plot the waveform of V_c versus the time constant of the network.

(c) plot the wavefrom of i_c and V_R versus the time constant of the network.

(d) what in the value of V_c at time $t = 20$ms?

(e) On a practical basis, how much time must pass before we can assume that the charging phase has passed.

(f) When the charging has passed how much charge is seeting on the plates?

⇒ is given:-

(i)

$R = 8k\Omega$
$V_c = ?$
$C = 4\mu F$
$E = 40V$

we know:-

$\tau = RC$
$= 8k \times 4\mu = 32ms$

We know:—

(a)

For charging:—

$$V_C = E(1 - e^{-\frac{t}{\tau}})$$

$$= 40(1 - e^{-t/32ms})\,V$$

$$i_C = \frac{E}{R} e^{-\frac{t}{\tau}}$$

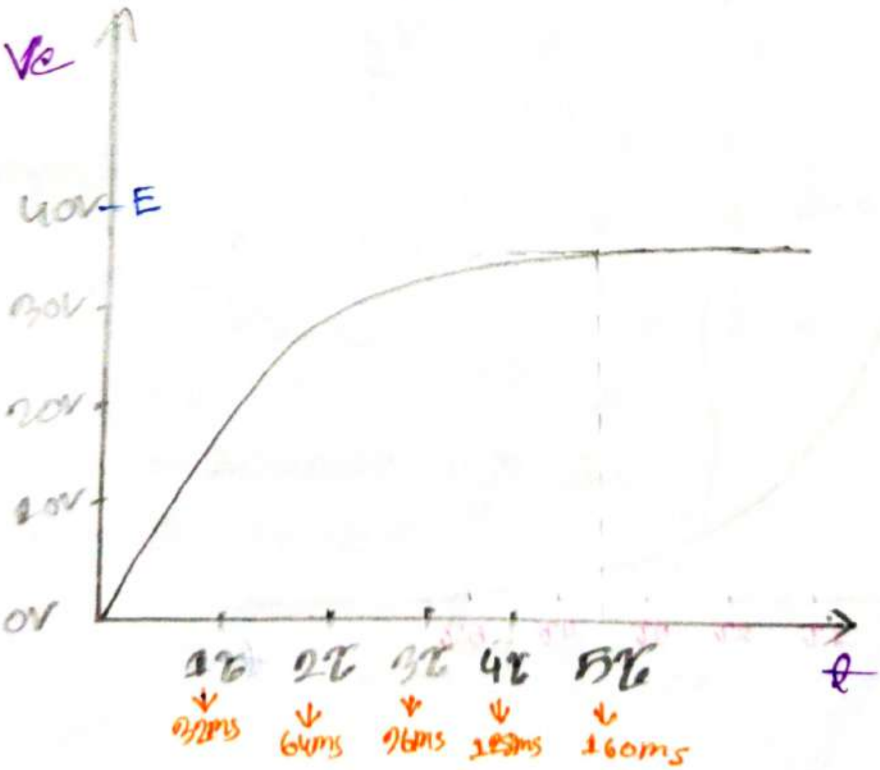
$$= \frac{40}{8K} e^{-\frac{t}{32ms}}$$

$$= 5m e^{-t/32m}\,A$$

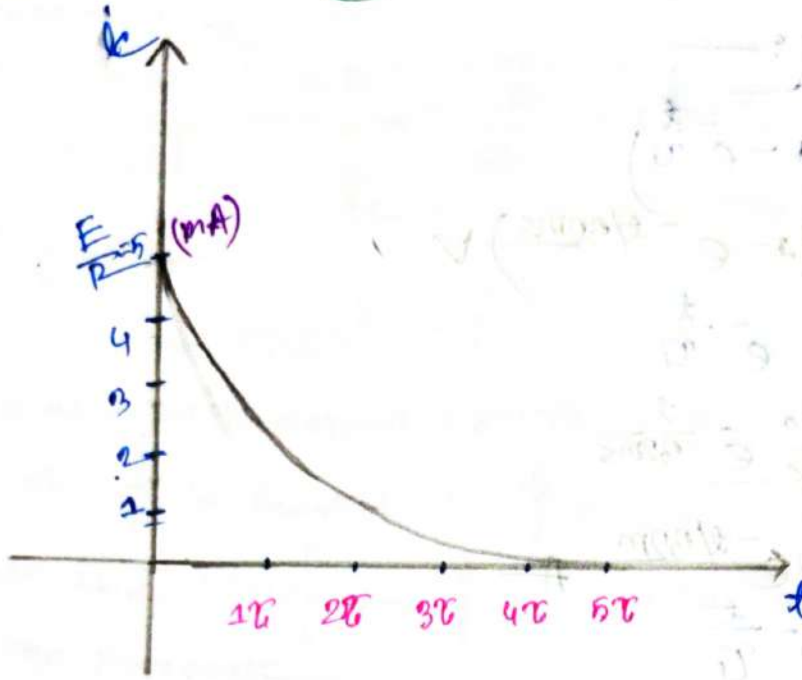
$$V_R = E e^{-\frac{t}{\tau}}$$

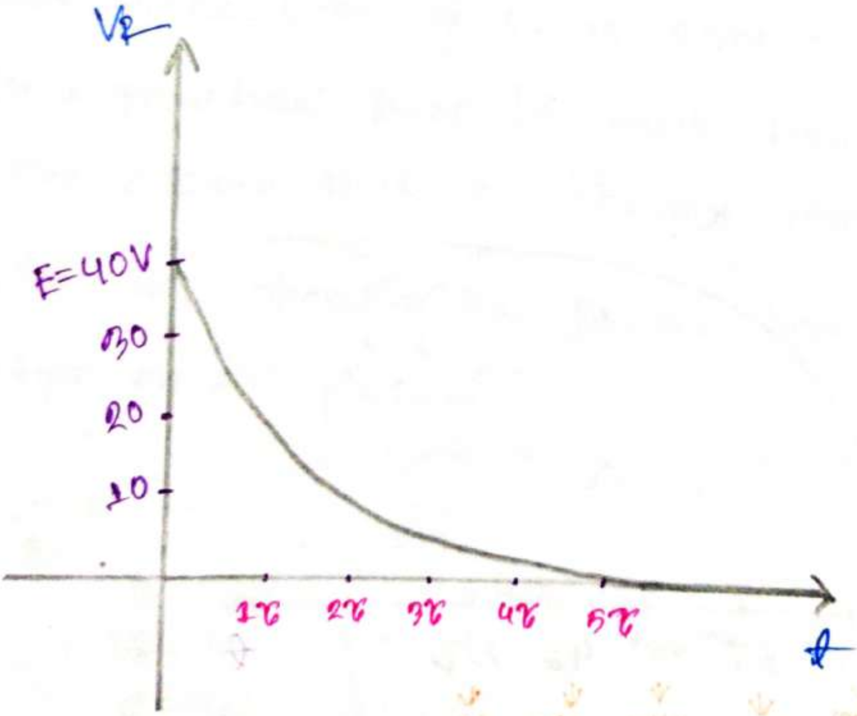
$$= 40 \times e^{-t/32m}\,V$$

(b)



(c)





(d)

In given:—

$t = 20ms$, $E = 40V$

We know:—

$$V_c = E\left(1 - e^{-\frac{t}{\tau}}\right)$$

$$= 40\left(1 - e^{-\frac{20m}{32m}}\right)$$

$$= 18.589 \text{ Volt}.$$

$$\tau = RC$$
$$= 8k \times 4\mu$$
$$= 32ms$$

(e)

We know:—

Total changing time $t = 5\tau$

$$= 5 \times 32$$
$$= 160ms$$
$$= 0.16S$$

(F)

We know:—

$$C = \frac{Q}{V_{cmax}}$$

$$\therefore Q = V_c \times C$$

$$= 4\mu \times 40$$

$$= 160\mu C$$

$t_{max} = 5\tau$

$$V_{cmax} = E\left(1 - e^{-\frac{t_{max}}{\tau}}\right)$$

$$= E\left(1 - e^{-\frac{5\tau}{\tau}}\right)$$

$$= 40(1 - e^{-5})$$

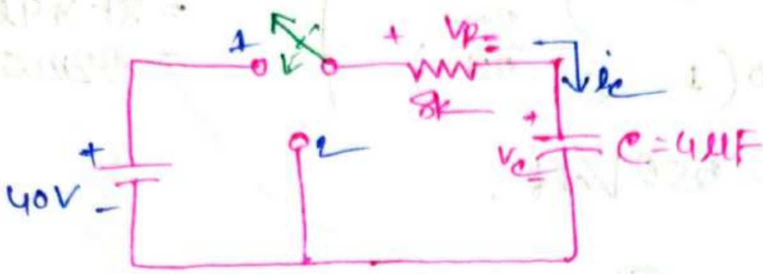
$$= 39.73$$

$$\approx 40V$$

Ans:—

problems 10.7:

⇒ Plot the waveforms of V_c and I_c resulting from Switching between contacts 1 and 2 in the following figure every 5 time constant.



⇒ For Charging:

is given:

$$E=40V;\ R=8k;\ C=4\mu F$$

We know:

$$\tau=RC=8k\times4\mu=32ms$$

$$\therefore V_c=E(1-e^{-t/\tau})$$

$$=40(1-e^{-t/32m})\ Volt$$

$$i_c=\frac{E}{R}e^{-t/\tau}$$

$$=40/8k\times e^{-t/32m}$$

$$=5m e^{-t/32m}\ A$$

$$V_R=Ee^{-t/\tau}$$

$$=40\times e^{-t/32m}\ Volt$$

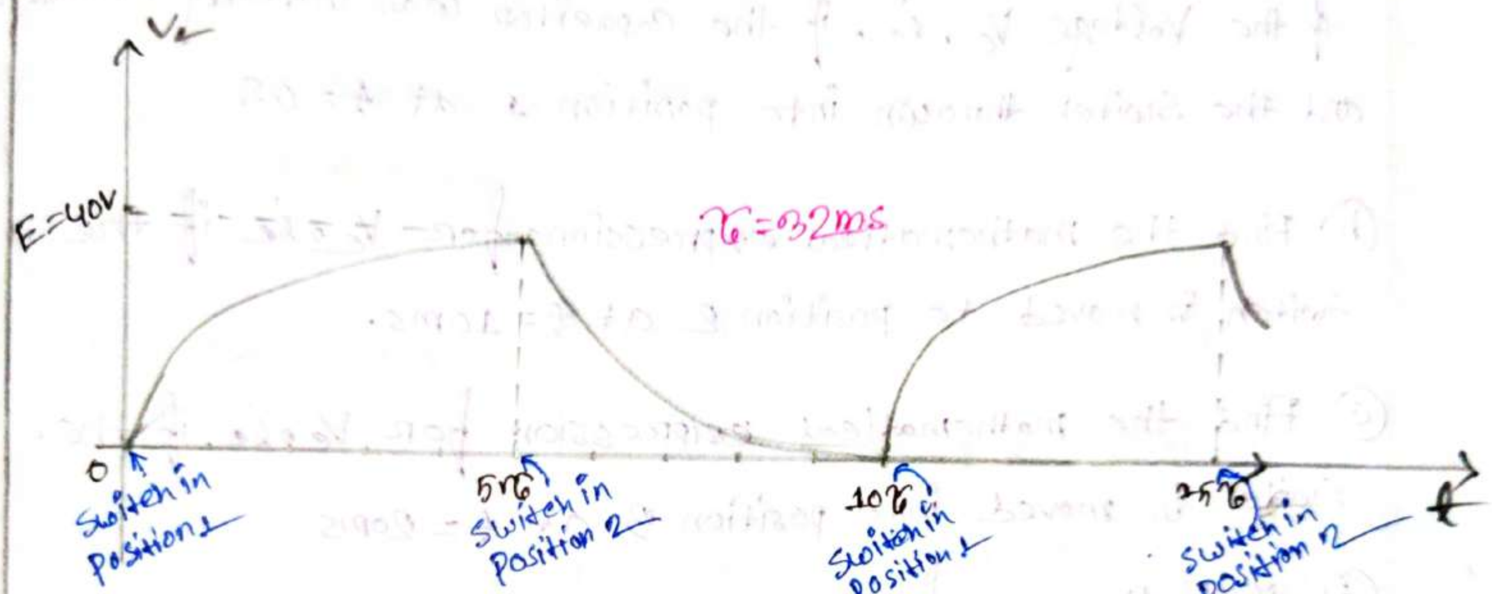
⇒ For Discharging:-

$$V_C = Ee^{-t/\tau} = 40e^{-t/32m} \text{ Volt}$$

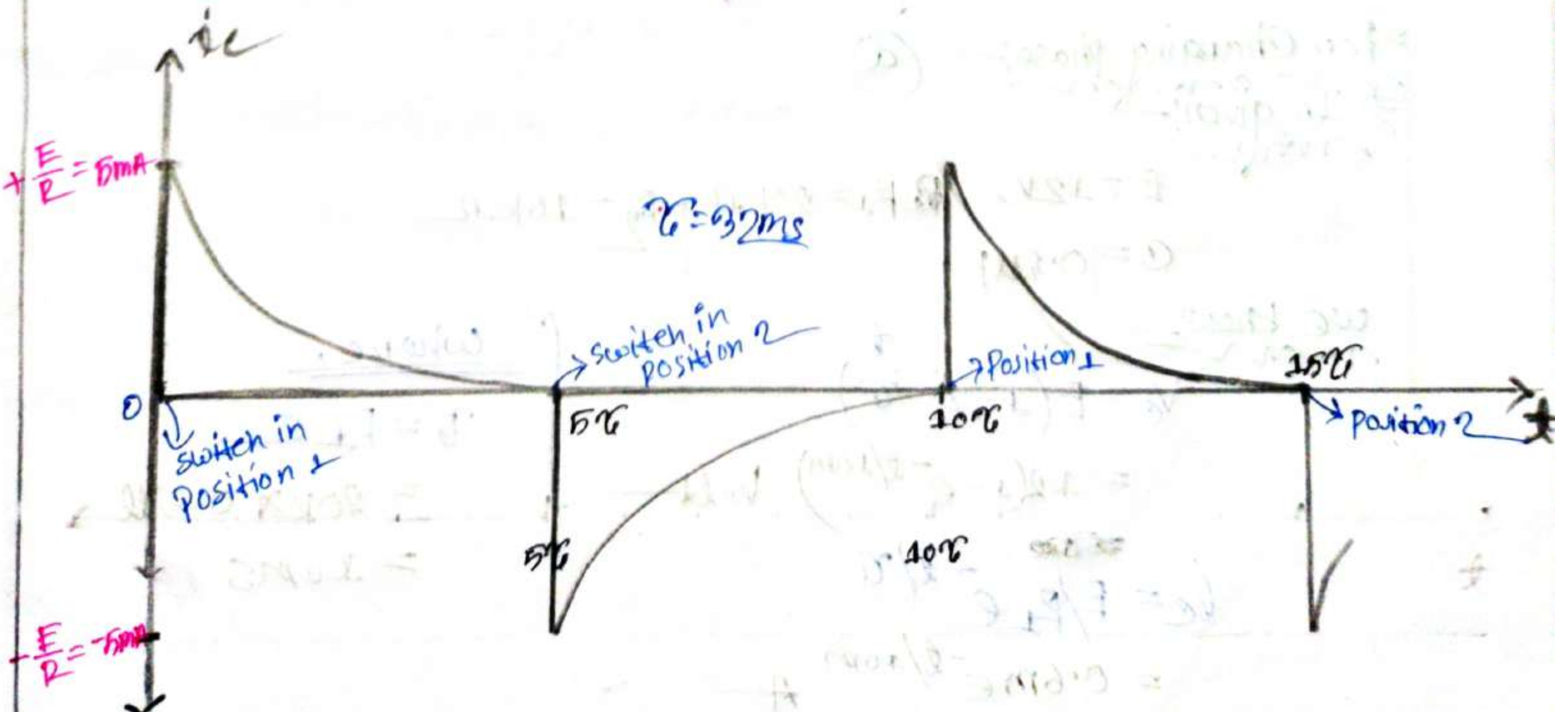
$$i_c = -\frac{E}{R}e^{-t/\tau} = -5me^{-t/32m} \text{ A}$$

$$V_R = -Ee^{-t/\tau} = -40e^{-t/32m} \text{ Volt}$$

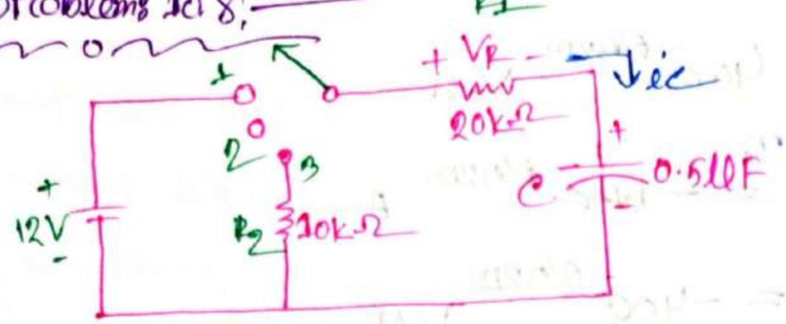
graph for V_C:-



graph for i_c:-



problems 10 8:-



(a) Find the mathematical expression for the transient of the voltage V_C, i_C, if the capacitor was initially uncharged and the switch thrown into position 1 at $t = 0S$

(b) Find the mathematical expressions for V_C, i_C if the switch is moved to position 2 at $t = 10ms$.

(c) Find the mathematical expression for V_C, i_C if the switch is moved into position 3 at $t = 20ms$

(d) Plot the waveforms obtained in parts (a)-(c) on the same time axis using the defined polarities.

= For Charging phase:- (a)

⇒ In given:-

$E = 12V$, $R_1 = 20k\Omega$, $R_2 = 10k\Omega$

$C = 0.5\mu F$

we know:-

$$V_C = E\left(1 - e^{-\frac{t}{\tau}}\right)$$

$$= 12\left(1 - e^{-t/10m}\right) \text{ Volt}$$

where,

$$\tau = R_1 C$$

$$= 20k \times 0.5\mu$$

$$= 10ms$$

$$i_C = E/R_1 \, e^{-t/\tau}$$

$$= 0.6m e^{-t/10m} \text{ A}$$

(b)

For Storage phase:—

$$V_c = E(1 - e^{-\frac{t}{\tau}})$$

$$\boxed{V_c = 40(1 - e^{-10ms/10ms})}$$

$$V_c = 7.585\,V.$$

$$i_c = \frac{E}{R} e^{-\frac{t}{\tau}}$$

$$\boxed{i_c = 0.6m\, e^{-\frac{10ms}{10ms}}}$$

$$= 2.20 \times 10^{-4} A$$

(d)

For Discharging phase.

$$V_c = E e^{-\frac{t}{\tau'}}$$

$$\tau' = (R_1 + R_2) C$$

$$= 30k \times 0.5\mu$$

$$= 15ms$$

$$V_c = 12 e^{-\frac{20ms}{15ms}}$$

$$\boxed{V_c = 12 e^{-t/15ms}}$$

$$i_c = -E/R_T \times e^{-\frac{t}{\tau'}}$$

$$\boxed{i_c = -0.4m\, e^{-t/15ms}}$$

⇒ if $C = 0.05\mu F$ then calculate all the question Again.

⇒ is given:—

$$E = 12V \qquad R_1 = 20k\Omega$$
$$C = 0.05\mu F \qquad R_2 = 10k\Omega$$

(a)

For charging phase:—

$$V_c = E(1 - e^{-\frac{t}{\tau}})$$

$$\therefore V_c = 12(1 - e^{-t/1m})\ V$$

$$\tau = R_1 C = 20k \times 0.05\mu = 1ms$$

$$i_c = \frac{E}{R_1}(e^{-t/\tau})$$

$$= \frac{12}{20k} \times (e^{-t/1m})$$

$$= 0.6m\, e^{-t/1m}$$

(b)

For Storage phase:—

The 10 ms time in equal to 10τ. that means the capacitor is fully charged. At this movement the value of V_c and i_c is fixed.

$$V_c = 12V$$
$$i_c = 0A$$

(c)

For Discharging phase:-

$$V_c = E e^{-t/\tau'}$$
$$= 12 \times e^{-t/1.5ms}$$

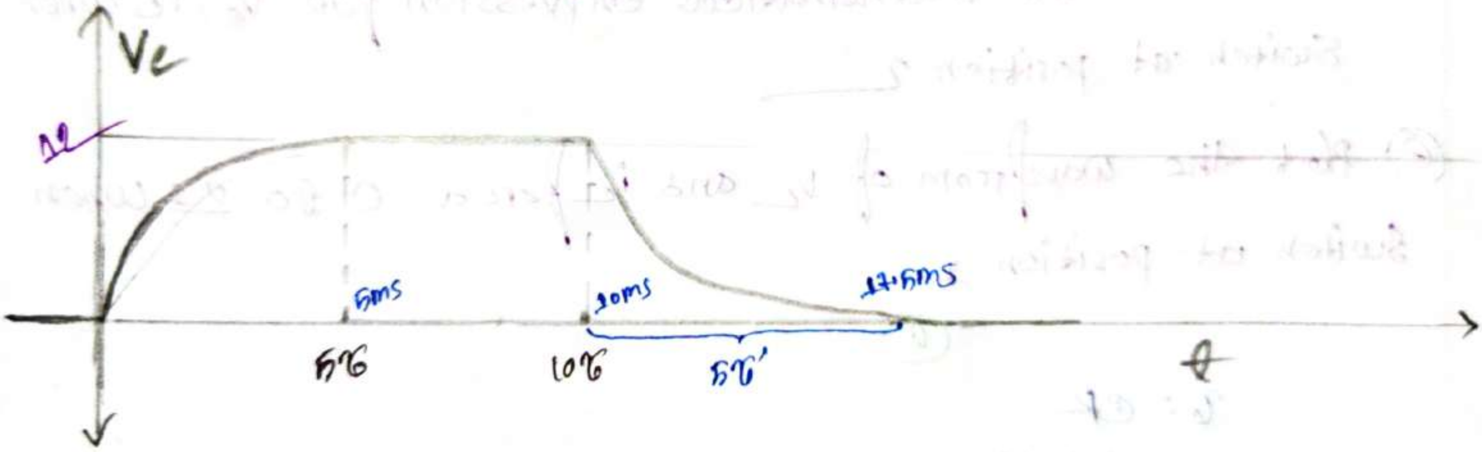
$$i_c = \frac{-E}{R_1+R_2} \times e^{-t/\tau'}$$
$$= \frac{-12}{30k} e^{-t/1.5ms}$$
$$= -0.4 e^{-t/1.5ms}$$

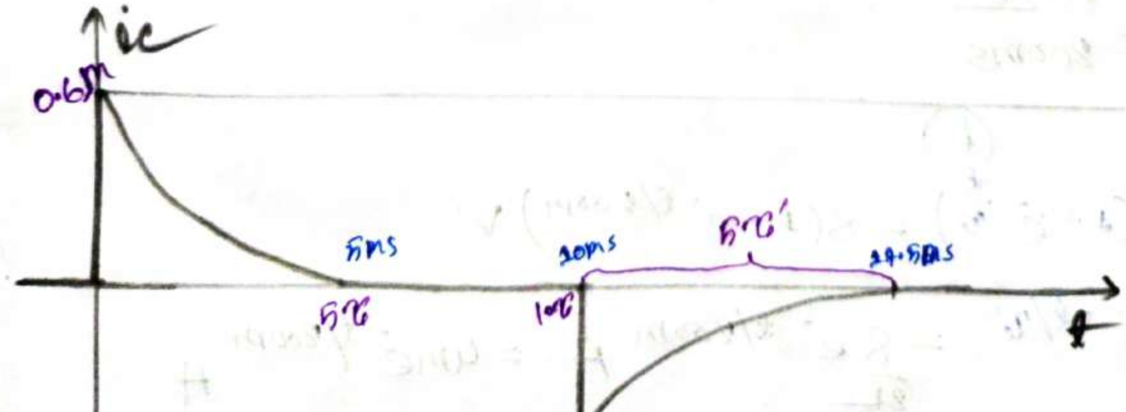
$$\tau' = (R_1+R_2)C$$
$$= 30k \times 0.05\mu$$
$$= 1.5ms$$

$$5\tau' = 5 \times 1.5$$
$$= 7.5ms$$

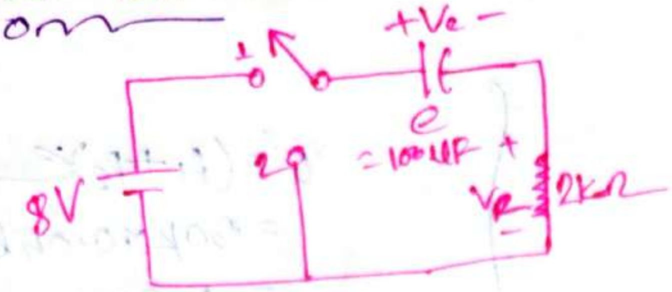
(d)

For V_L graph:-





Problem: 27:—



(a) Determine the time constant.

(b) Find the mathematical expression for V_c, i_c after switch at position 1

(c) Determine the magnitude of V_c, i_c instant the switch in thrown into position 2 at $t = 1s$

(d) Determine the mathematical expression for V_c, i_c when switch at position 2

(e) Plot the wavefrom of V_c and i_c for a 0 to 2s when switch at position 1.

(a)

$$\tau = CR$$
$$= 100\mu \times 2k$$
$$= 0.2s$$
$$= 200ms$$

(b)

$$V_c = E(1 - e^{-\frac{t}{\tau}}) = 8(1 - e^{-t/200m})\,V$$

$$i_c = \frac{E}{R} e^{-t/\tau} = \frac{8}{2k} e^{-t/200m}\,A = 4m e^{-t/200m}\,A$$

(c)

$$V_c = E(1 - e^{-t/\tau})$$
$$= 8(1 - e^{-1/0.2})$$
$$= 7.96V$$

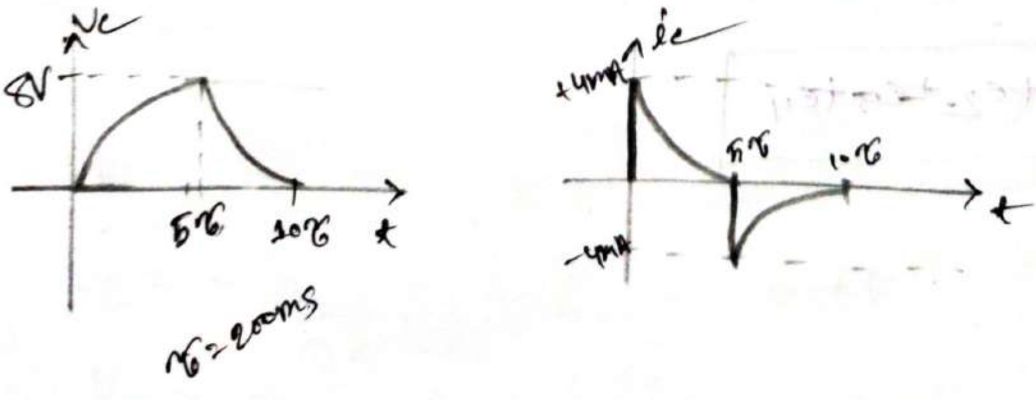
$\tau = 200ms$
$t = 1s$

$$i_c = \frac{E}{R} e^{-t/\tau}$$
$$= 0.004e^{-1/0.2}$$
$$= 2.695 \times 10^{-5} A$$

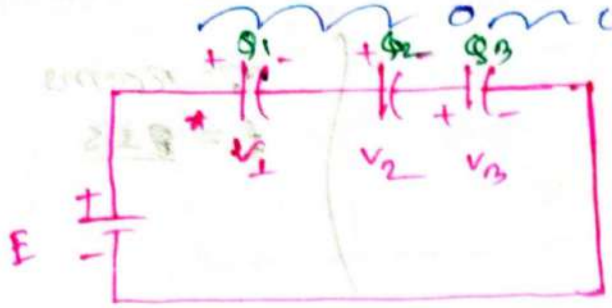
(d)

$$V_c = Ee^{-t/\tau}$$
$$= 8e^{-t/0.2} V$$
$$i_c = -\frac{E}{R} e^{-t/\tau}$$
$$= -\frac{8}{2K} e^{-t/0.2}$$
$$= 0.004e^{-t/0.2}$$
$$= 4m e^{-t/200m} A$$

(d)

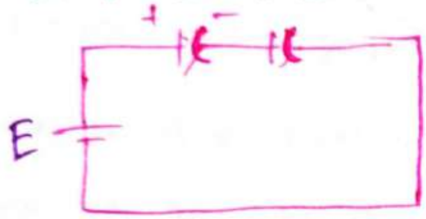


Capacitors In Series:—



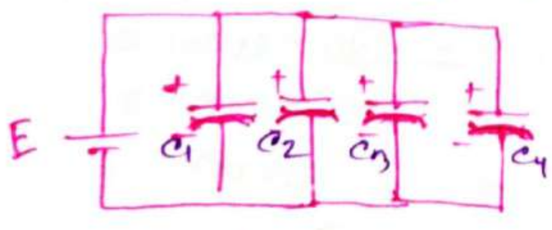
$$\boxed{\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

For 2 Capacitor:—



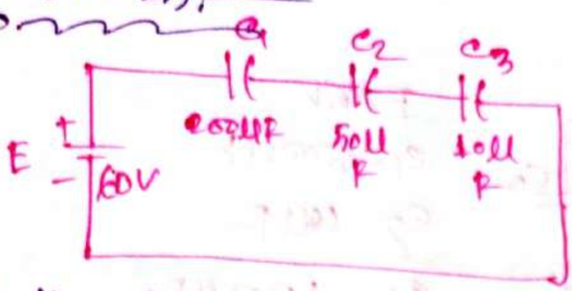
$$\boxed{C_T = \frac{C_1 C_2}{C_1 + C_2}}$$

Capacitors in Parallel:—



$$\boxed{C_T = C_1 + C_2 + C_3 + C_4}$$

problem: 10.1A :-



(a) Find the total Capacitance

(b) Determine the charge on each plate.

(c) Find the Voltage Across the capacitor

(a)

we know:-

$$C_T = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)^{-1}$$

$$= \left(\frac{1}{200\mu} + \frac{1}{50\mu} + \frac{1}{10\mu}\right)^{-1}$$

$$= 8\mu F$$

(b)

we know:-

$$C = \frac{Q}{V}$$

$$\therefore Q_T = C_T E$$

$$= 8\mu \times 60$$

$$= 480\mu F$$

$$Q_T = Q_1 = Q_2 = Q_3$$

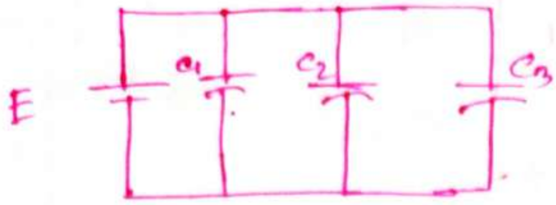
(c)

$$V_1 = \frac{Q_1}{C_1} = \frac{480\mu}{200\mu} = 2.4V$$

$$V_2 = \frac{Q_2}{C_2} = \frac{480}{50} = 9.6V$$

$$V_3 = \frac{Q_3}{C_3} = \frac{480}{10} = 48V$$

problem: 10.16:—



$E = 48V$
$C_1 = 800\mu F$
$C_2 = 60\mu F$
$C_3 = 1200\mu F$

(a) Find the total Capacitance.
(b) Determine the charge of each plate.
(c) Find the total charge.

(a)

We know:—

$C_T = C_1 + C_2 + C_3$
$= 1200 + 60 + 800$
$= 2060\mu F$

(b)

We know:—

$Q_1 = C_1 E = 800\mu \times 48 = 38.4mC$
$Q_2 = C_2 E = 60\mu \times 48 = 2.88mC$
$Q_3 = C_3 E = 1200\mu \times 48 = 57.6mC$

(c)

We know:—

$Q_T = Q_1 + Q_2 + Q_3$
$= 38.4 + 2.88 + 57.6$
$= 98.88\ mC$

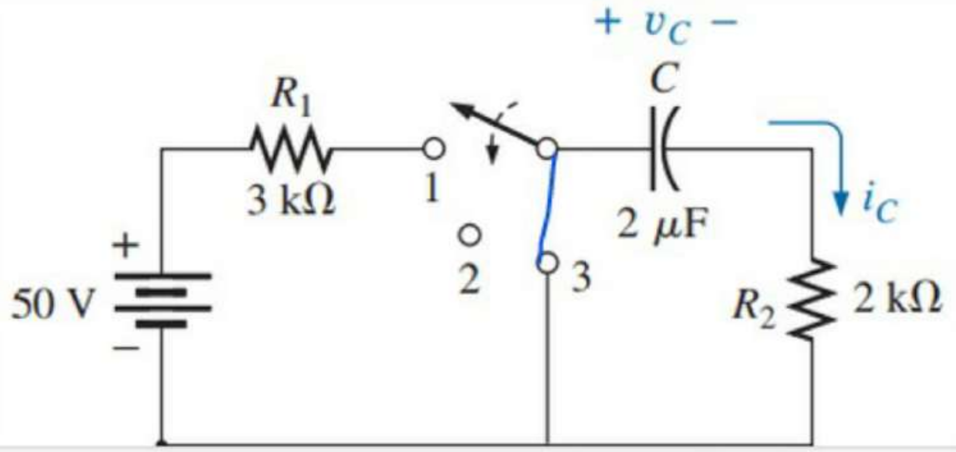
28. For the network in Fig. 10.91, composed of standard values:

a. Write the mathematical expressions for the voltages v_C, and v_{R_1} and the current i_C after the switch is thrown into position 1.

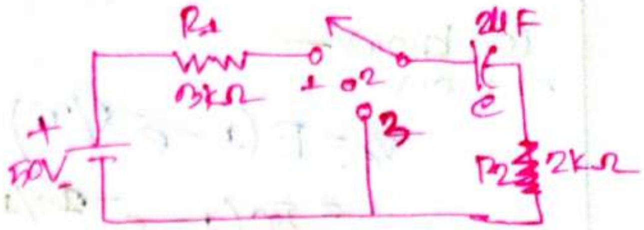
b. Find the values of v_C, v_{R_1}, and i_C when the switch is moved to position 2 at $t = 100$ ms.

c. Write the mathematical expressions for the voltages v_C and v_{R_2} and the current i_C if the switch is moved to position 3 at $t = 200$ ms.

d. Plot the waveforms of v_C, v_{R_2}, and i_C for the time period extending from 0 to 300 ms.



Problem: 28:—



(a)

In given:—

$$E_1 = 50V$$
$$R_1 = 3k\Omega$$
$$R_2 = 2k\Omega$$
$$C = 2\mu F$$

we know:—

$$\tau = (R_1 + R_2)C$$
$$= (3+2)k \times 2\mu$$
$$= 10ms$$

$$\therefore V_C = E(1 - e^{-t/\tau})$$
$$= 50(1 - e^{-t/10ms})\ \text{Volt}$$

$$i_C = \frac{E}{R_T}(e^{-\frac{t}{\tau}})$$
$$= \frac{50}{5k} e^{-t/10ms}$$
$$= 10m e^{-t/10ms}\ A$$

$$V_{R_1} = E(e^{-t/\tau})$$
$$= (e^{-t/10ms}) \times 30$$
$$= 30 e^{-t/10ms}$$

$$V_1 = IR_1$$
$$= 10m \times 3k$$
$$= 30V$$

$$V_{R_2} = V_2(e^{-t/\tau})$$
$$= 20 e^{-t/10ms}$$

$$V_2 = IR_2$$
$$= 10m \times 2k$$
$$= 20V$$

(b)

We know:—

$$V_c = E(1 - e^{-t/\tau})$$
$$= 50(1 - e^{-100/10})$$
$$= 49.99V = E$$

$V_{R_1} = 0V$

$i_c = 0A$

$\tau = 10ms$

$5\tau = 100ms$

$t = 100ms$ that means The Capacitor is fully charged.

(c)

We know:—

$$V_c = E(e^{-t/\tau'})$$
$$= 50e^{-t/4ms}\ V$$

$$i_c = -\frac{E}{R_2} \times e^{-t/\tau'}$$
$$= -\frac{50}{2k} \times e^{-t/4ms}$$
$$= 25m e^{-t/4ms}\ A$$

$$V_{R_2} = -Ee^{-t/\tau'}$$
$$= -50e^{-t/4ms}$$

$\tau' = CR_2$

$= 2\mu \times 2k$

$= 4ms$

Ans

#Inductor

Ch: 11

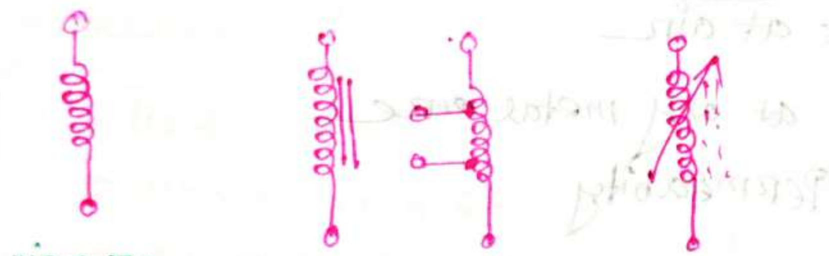
Inductor:-

⇒ An inductor is a coil of conducting wire.

Inductance:-

⇒ Inductance is the property for which an inductor opposes the change of current flowing through it.

Types of Inductors:-



air-core Ferromagnetic Core Tapped Variable.

Inductance Based on physical Dimension:-

$$\boxed{L = \frac{\mu N^2 A}{l}}$$

	Unit
L = Inductance	Henries (H)
μ = permeability	Wb/A · m
N = Number of turns	
A = Area	m^2
l = length	m

$$\boxed{L = \frac{\mu_0 \mu_R N^2 A}{l}}$$

μ_R = permeabity relative

$\boxed{\mu_0 = 4\pi \times 10^{-7}}$ (wb/A·m)

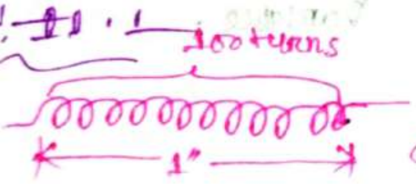
$\boxed{L = \mu_R L_0}$ where $\boxed{L_0 = \frac{\mu_0 N^2 A}{l}}$

L_0 = inductance at air

L = inductance at any metal core

μ_R = relative permeabity

problem: 1.1



$d = \frac{1}{4}$"

(a) Find the inductance.

(b) Find the inductance if a metalic core with $\mu_R = 2000$ in insert in the coil.

⇒ in given:—

(a)

$$l = 1 \text{ in} = 0.0254 m$$

$$d = \frac{1}{4} in = \frac{1}{4} \times 0.0254 = 6.35 \times 10^{-3} m$$

$$\therefore r = \frac{d}{2} = 3.175 \times 10^{-3} m$$

$$\therefore A = \pi r^2 = 3.1416 \times (3.175 \times 10^{-3})^2$$

$$= 3.166 \times 10^{-5} m^2$$

$\mu_0 = 4\pi\times10^{-7}$

$N = 100$

we know:—

$$L_0 = \frac{\mu_0 N^2 A}{l}$$

$$= \frac{4\pi\times10^{-7}\times(100)^2\times 3.166\times10^{-5}}{0.0254}$$

$$= 1.566\times10^{-5}\,H$$

(b)

we know:—

$$\boxed{L = \mu_r\times L_0}$$

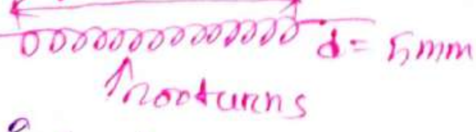
$$= 2000\times1.566\times10^{-5}$$

$$= 31.32\times10^{-3}\,H$$

$$= 31.32\,mH$$

Ans:—

Problems: Q.P 504:—



(a) Find the inductance.

⇒ In given:—

$N = 200$

$l = 100\times10^{-3}\,m$

$d = 5\times10^{-3}\,m$

$\therefore r = 2.5\times10^{-3}$

$\mu_0 = 4\pi\times10^{-7}$

$\therefore A = \pi(2.5\times10^{-3})^2 = 1.963\times10^{-5}\,m^2$

We know—

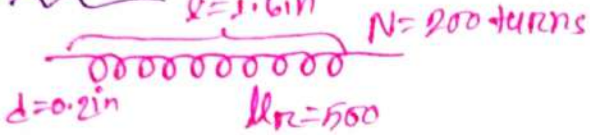
$$L = \frac{AN^2\mu_0}{l}$$

$$= \frac{1.963\times10^{-5}\times(200)^2\times 4\pi\times10^{-7}}{100\times10^{-3}}$$

$$= 0.86\times10^{-6}\,H$$

$$= 0.86\,\mu H$$

problem: 3:—



(a) Find the inductance.

⇒ In given:—

$l = 1.6\text{ in} = 1.6\times0.0254 = 0.04064\,m$

$d = 0.2\text{ in} = 0.2\times0.0254 = 5.08\times10^{-3}\,m$

$r = 2.54\times10^{-3}\,m$

$\therefore A = \pi r^2 = \pi\times(2.54\times10^{-3})^2 = 2.026\times10^{-5}\,m^2$

$\mu_0 = 4\pi\times10^{-7}$

$\mu_r = 500$

$N = 200.$

We know—

$$L = \frac{\mu_r\mu_0 N^2 A}{l}$$

$$= \frac{500\times4\pi\times10^{-7}\times(200)^2\times2.026\times10^{-5}}{0.04064}$$

$$= 12.52\times10^{-3}\,H$$

$$= 12.52\,mH$$

Transients in inductive Network

⇒ If current in allowed to pass through an inductor, it is found that the voltage across the inductor is directly proportional to the rate change of the current

$$V_L = L\frac{di_L}{dt}$$

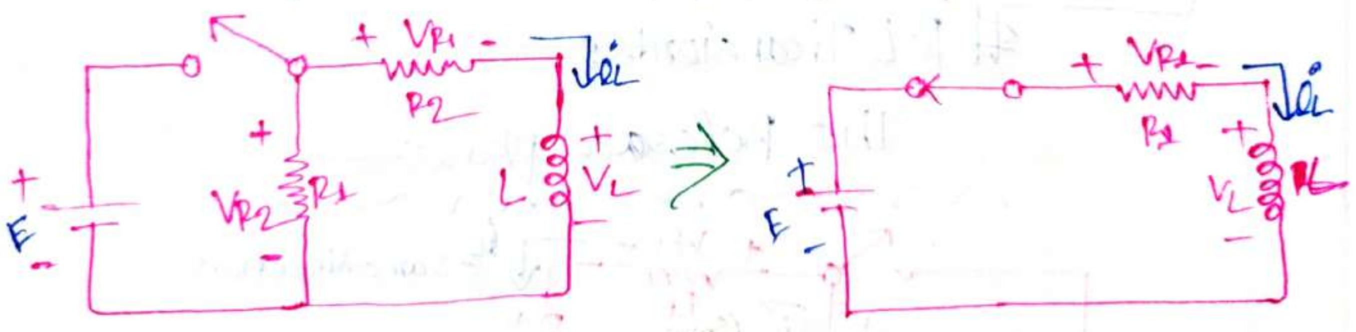
$$i_L = \frac{1}{L}\int_0^t V_L\,dt + i_L(t_0)$$

∴ Energy Storage by an inductor

$$\boxed{W_L = \frac{1}{2}Li_L^2}$$

R-L Transients:-

The Storage phase:-



Storage phase for an inductor by closing the switch.

Charging phase:- KVL for b circuit

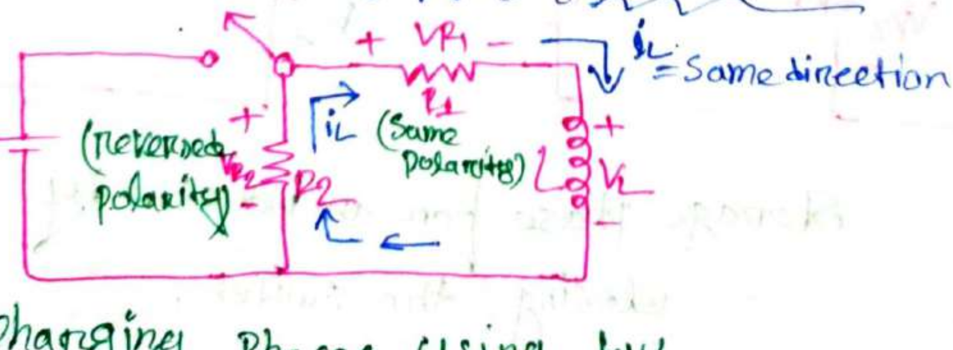
$$E = V_{R_1} + V_L$$
$$= R_1 i_L + V_L$$
$$= R_1 i_L + L\frac{di_L}{dt}$$
$$= L\frac{di_L}{dt} + R_1 i_L \quad \cdots\cdots (1)$$

from eq (1):-

	Unit
$\boxed{V_L = E e^{-t/\tau}}$	Volt - V
$\boxed{i_L = \frac{E}{R_1}\left(1 - e^{-t/\tau}\right)}$	Ampere - A
$\boxed{V_{R_1} = E\left(1 - e^{-t/\tau}\right)}$	Volt - V
$\boxed{\tau = \frac{L}{R_T} = \frac{L}{R_1}}$	Seconds - S
$\boxed{V_{R_2} = E}$	Volt - V

R-L Transients:-

The Release phase:-



Discharging phase:- Using KVL

$$V_L = -i_L R_2 + i_L R_1$$

$$L\frac{di_L}{dt} + (R_1 + R_2) i_L = 0 \quad \text{———} (10)$$

At the moment of switch in open $i_{L0} = \frac{E}{R_1}$ thus:-

$$\boxed{V_i = (R_1 + R_2) i_{L0} = \left(1 + \frac{R_2}{R_1}\right) E}$$

$$\boxed{V_{L0} = -V_i}$$

From eq (ii) :-

$$\boxed{V_L = -V_i e^{-t/\tau'} = \left(1 + \frac{R_2}{R_1}\right) E e^{-t/\tau'}}$$

$$\boxed{i_L = \frac{E}{R_1} e^{-t/\tau'}}$$

$$\boxed{V_{R_1} = E e^{-t/\tau'}}$$

$$\boxed{V_{R_2} = -\frac{R_2}{R_1} E e^{-t/\tau'}}$$

$$\boxed{\tau' = \frac{L}{R_1 + R_2}}$$

After time t:-

$i_L = I_m e^{-t/\tau}$

$V_i = (R_1 + R_2) I_m$

$I_m = \frac{E}{R_1}\left(1 - e^{-t/\tau}\right)$

t = After a time t

V_L = Voltage across to inductance

i_L = Current across to inductance

V_{R_1} = Voltage across to Resistor R_1

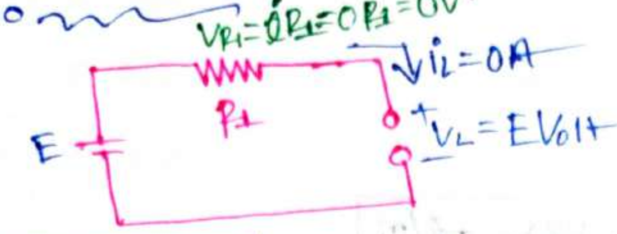
V_{R_2} = Voltage across to Resistor R_2

τ' = time constant

$t_{total} = 5\tau$ = total discharging time.

At the instance of Switch in closed:—

At time $t = 0$:—



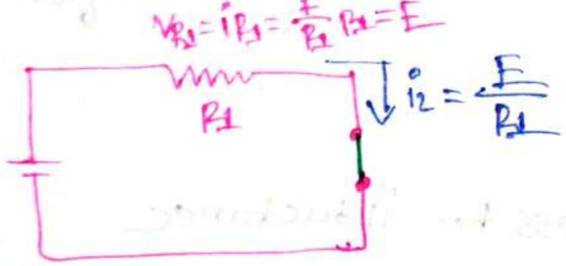
$$\boxed{i_L = 0\ A}$$

$$\boxed{V_L = E\ V}$$

$$\boxed{V_{R1} = 0\ V}$$

Steady-State Condition:—

At time $t = \infty$

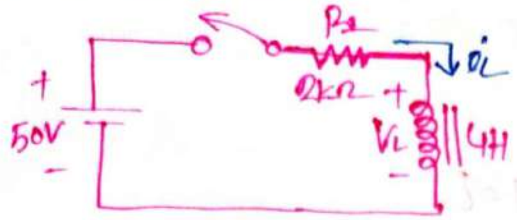


$$\boxed{i_L = \frac{E}{R_1}}$$

$$\boxed{V_L = 0}$$

$$\boxed{V_{R1} = E}$$

problem: 11.3:—



⇒ Find the mathematical expressions for i_L, V_L if the switch in closed at $t = 0s$. Sketch the resulting curves.

⇒ Ingiven:—

$E = 50V$

$R_1 = 2k\Omega$

$V_L = ??$

$L = 4H$

We know:—

$$V_L = Ee^{-t/\tau}$$

$$= 50e^{-t/2m} \, V$$

$$\tau = \frac{L}{R_1} = \frac{4H}{2k} = 2ms$$

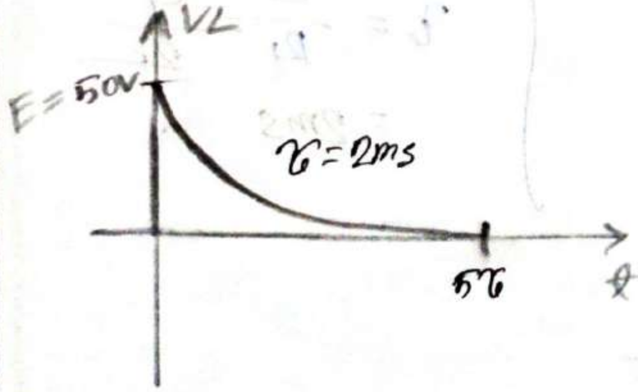
$$i_L = \frac{E}{R_1}(1 - e^{-t/\tau})$$

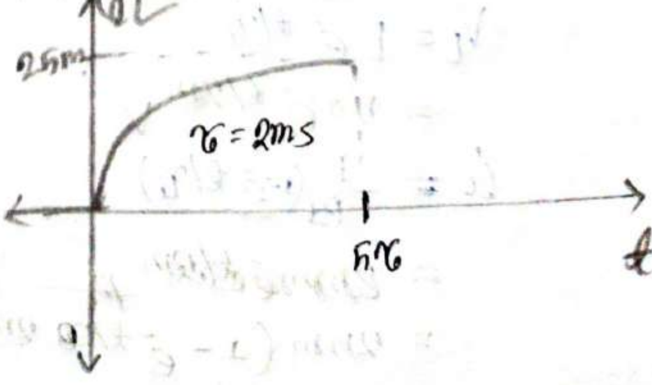
$$= \frac{50}{2k}(1 - e^{-t/2m})$$

$$= 25m(1 - e^{-t/2m}) \, A$$

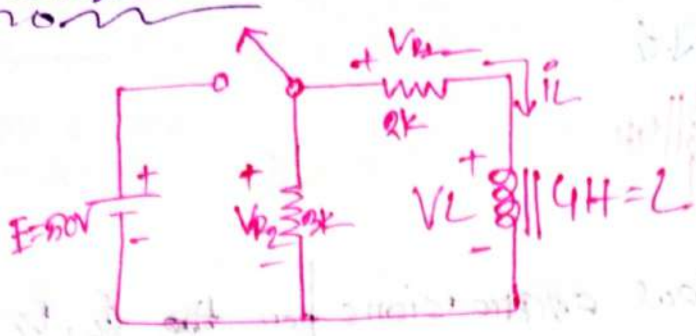
$$V_{R_1} = E(1 - e^{-t/\tau})$$

$$= 50(1 - e^{-t/2m}) \, V$$





Problem: 11.5:-



=> Resistor R_2 was added to the network

(a) Find the mathematical Expressions for i_L, V_L, V_{R1} and V_{R2} for five time constant of the storage phase.

(b) Find the mathematical Expression for i_L, V_L, V_{R1} & V_{R2} if the switch is opened after 5 constant of the storage phase.

(c) Sketch the waveforms for each voltage and current for both phases.

=> In given:-

$E = 50V$
$R_1 = 2k$
$R_2 = 3k$
$L = 4H$

(a)

We know:- For Storage Phase:-

$$V_L = Ee^{-t/\tau}$$
$$= 50e^{-t/2m}\ V$$
$$i_L = \frac{E}{R_1}(1 - e^{-t/\tau})$$

$$\tau = \frac{L}{R_1} = \frac{4}{2k} = 2ms$$

$$= 25m(1 - e^{-t/2m})\ A$$

$$V_{R_1} = E(1-e^{-t/\tau})$$
$$= 50(1-e^{-t/2m})\,V$$
$$V_{R_2} = E$$
$$= 50V$$

(b)

we know:- For discharging/Pelesase phase:-

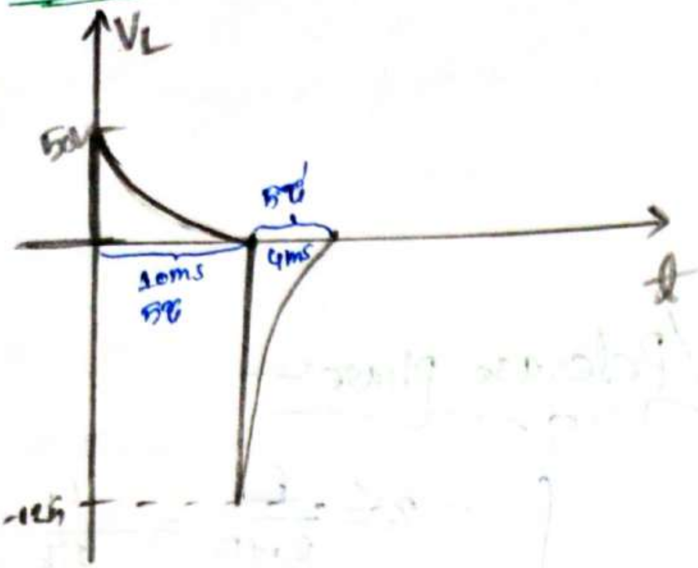
$$i_L = \frac{E}{R_1} e^{-t/\tau'}$$
$$= \left(\frac{50}{2} e^{-t/0.8m}\right) A$$

$$\tau' = \frac{L}{R_1+R_2} = \frac{4}{5k}$$
$$= 0.8ms$$

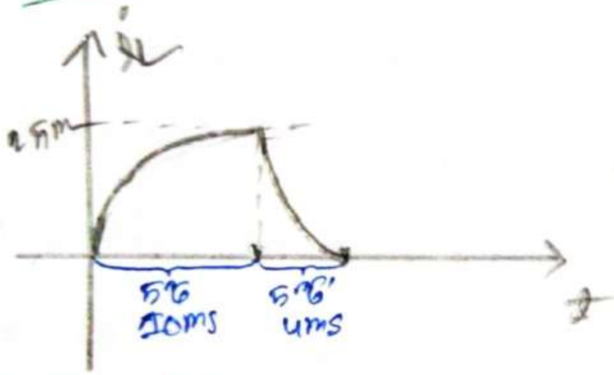
$$V_L = -V_i e^{-t/\tau'}$$
$$= -\left(1+\frac{R_2}{R_1}\right) E e^{-t/\tau'}$$
$$= -\left(1+\frac{3k}{2k}\right) 50 e^{-t/0.8m}$$
$$= \left(-125 e^{-t/0.8m}\right) V$$
$$V_{R_1} = E e^{-\frac{t}{\tau'}}$$
$$= \left(50 e^{-t/0.8m}\right) V$$
$$V_{R_2} = -\frac{R_2}{R_1} E e^{-\frac{t}{\tau'}}$$
$$= -\frac{3k}{2k} \times 50 e^{-t/0.8m}$$
$$= \left(-75 e^{-t/0.8m}\right) V$$

(d)

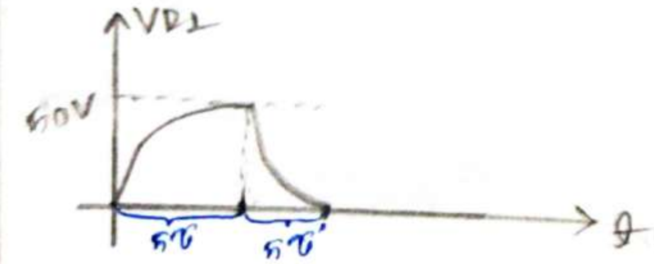
=> For V_L :-



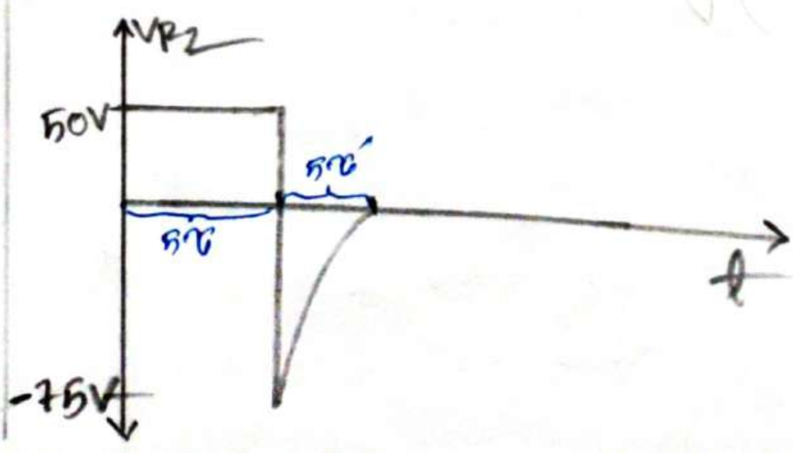
=> For i_L :-



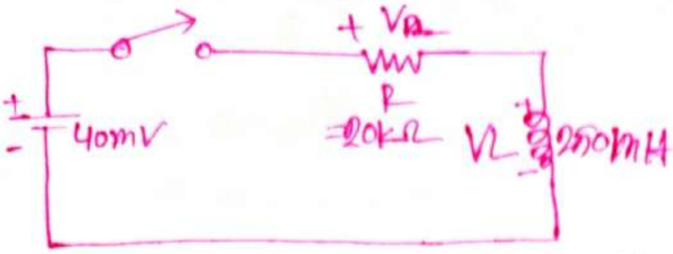
=> For V_{R1} :-



=> For V_{R2} :-



problems 12:



(a) Determine the time constant

(b) Write the Mathematical Expression for the i_L After Switch is closed.

(c) Repeat part (b) for V_L and V_R.

(d) Determine i_L and V_L at 1, 3, and 5 time constant

(e) Sketch the waveforms of i_L, V_L, and V_R.

⇒ In given:

$$L = 250\,mH,\quad R = 20k\Omega \quad E = 40mV$$

(a)

We know:

$$\tau = \frac{L}{R} = \frac{250m}{20k} = 12.5\times10^{-6} = 12.5\mu s$$

(b)

$$i_L = \frac{E}{R}\left(1-e^{-t/\tau}\right)$$
$$= \frac{40m}{20k}\left(1-e^{-t/12.5\mu}\right)$$
$$= \left(2\mu\left(1-e^{-t/12.5\mu}\right)\right)A$$

(c)

$$V_L = E e^{-\frac{t}{\tau}}$$

$$= \left(40m e^{-\frac{t}{12.5\mu}}\right) V$$

$$V_{R1} = E\left(1 - e^{-t/\tau}\right)$$

$$= 40m\left(1 - e^{-t/12.5\mu}\right)$$

(d)

at $t = \tau$:—

$$i_L = \frac{E}{R_1}\left(1 - e^{-\frac{t}{1\tau}}\right)$$

$$= 2\mu\left(1 - e^{-\frac{\tau}{\tau}}\right)$$

$$= 1.264\mu A$$

$$V_L = E e^{-t/\tau}$$

$$= 40m e^{-\tau/\tau}$$

$$= 14.715m\,V$$

At $t = 3\tau$:—

$$i_L = \frac{E}{R_1}\left(1 - e^{-t/\tau}\right)$$

$$= 2\mu\left(1 - e^{-3\tau/\tau}\right)$$

$$= 1.9\mu A$$

$$V_L = E e^{-3\tau/\tau}$$

$$= 40m e^{-3}$$

$$= 1.99\,mV$$

At $t = 5\tau$:—

$$i_L = \frac{E}{R_1}\left(1 - e^{-5\tau/\tau}\right)$$

$$= 2\mu\left(1 - e^{-5}\right)$$

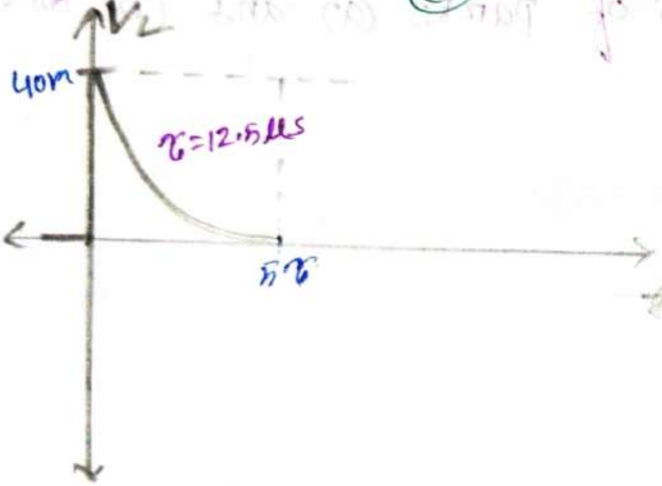
$$= 1.98 \times 10^{-6}$$

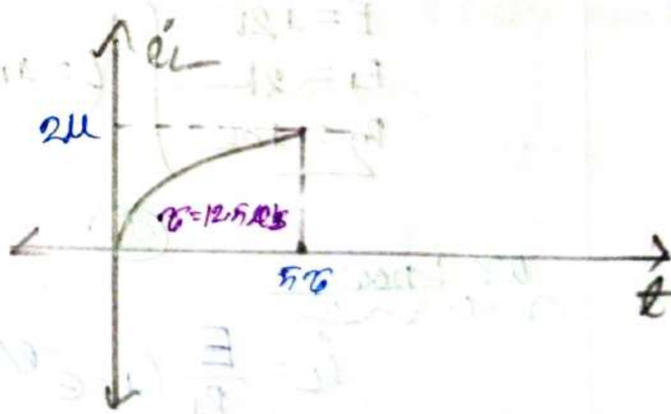
$$= 1.98\,\mu A$$

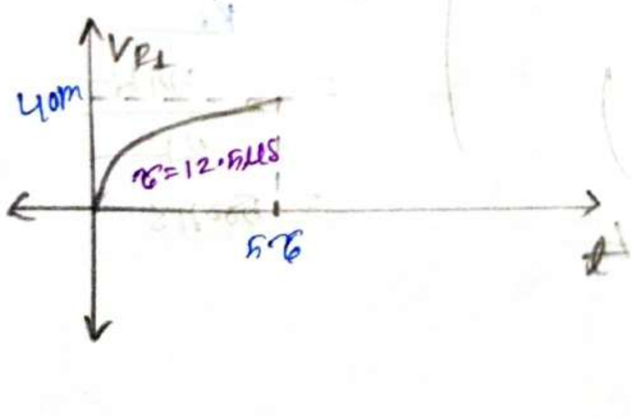
$$V_L = E e^{-5\tau/\tau}$$

$$= 40m\, e^{-5}$$

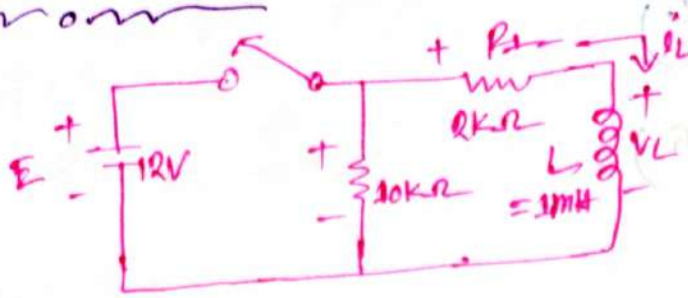
$$= 269.51\,\mu V$$







Problems 20:-



(a) Determine the mathematical expression for the current i_L and V_L following the closing of the switch.

(b) Repeat (a) part if the switch is opened at $t = 1\mu s$

(c) Sketch the waveforms of parts (a) and (b) on the same set of axes.

⇒ is given:-

$E = 12V$
$R_1 = 2k$
$R_2 = 10k$

$L = 1mH$

(a)

We know:-

$$i_L = \frac{E}{R_1}(1 - e^{-t/\tau})$$

$$= \frac{12}{2k}(1 - e^{-t/500n})$$

$$= 6m(1 - e^{-t/500n})\ A$$

$$\tau = \frac{L}{R_1} = \frac{1mH}{2k} = 500ns$$

$$V_L = E e^{-t/\tau}$$

$$= (12 e^{-t/500n})\ V$$

(b)

We know:—

$$I_m = \frac{E}{R_1}\left(1 - e^{-t/\tau}\right)$$

$$= 6m\left(1 - e^{-1\mu/500n}\right)$$

$$= 5.187mA$$

$\tau = 500ns$

$\tau' = \frac{1mH}{2k+10k}$

$= 83.33ns$

Now:— For Released ~~charge~~ Phase:—

$$i_L = I_m e^{-t/\tau'}$$

$$= 5.187m\, e^{-t/83.33n}$$

Again:— $V_L = -V_i e^{-t/\tau'}$

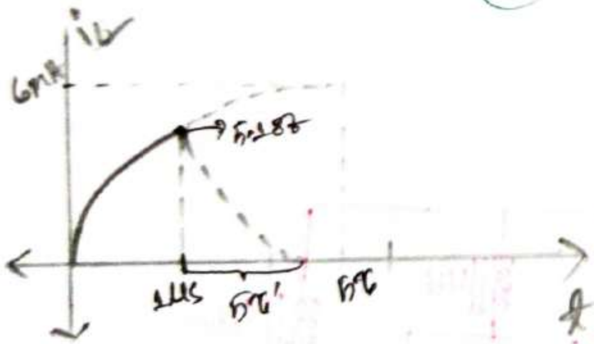
$$V_L = -62.244 \times e^{-t/83.33n}$$

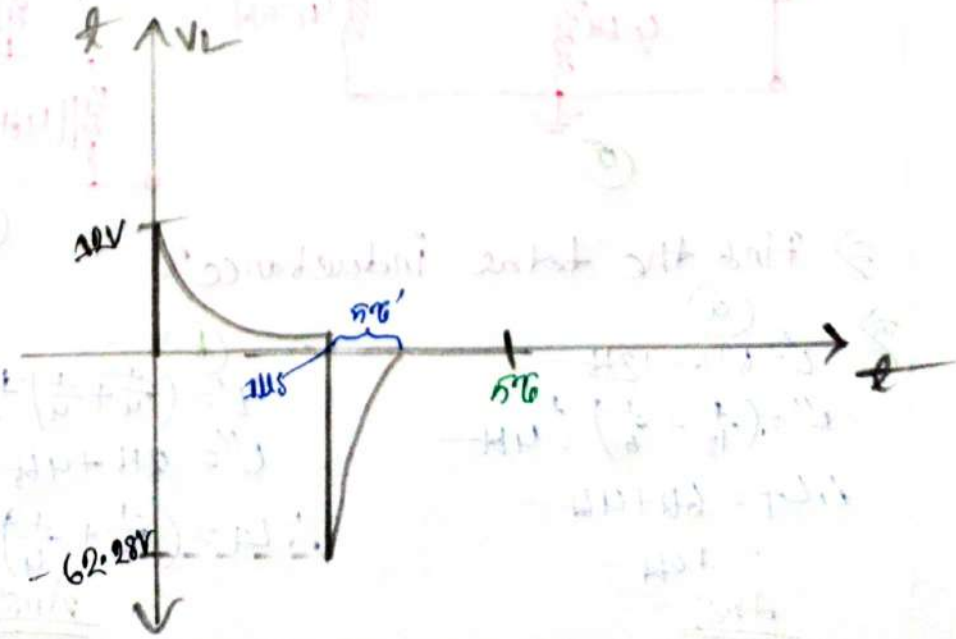
$V_i = (R_1 + R_2) I_m$

$= (10+2)k \times 5.187mA$

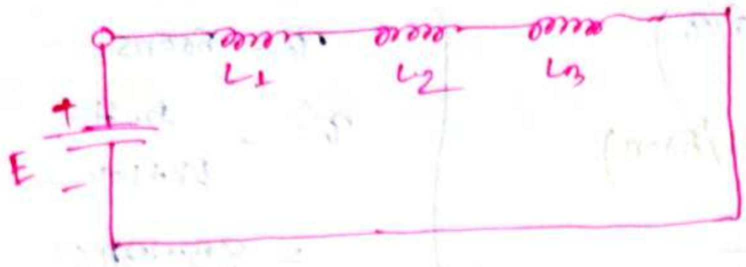
$= 62.244$

(c)



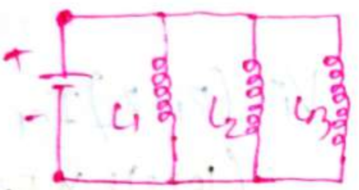


Inductors in Series



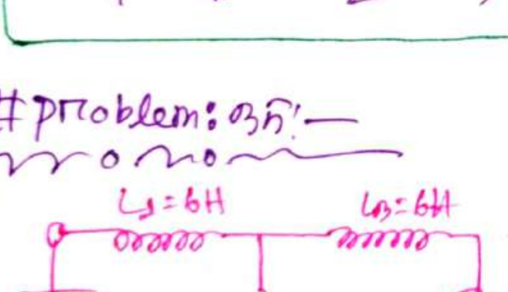
$$\boxed{L_T = L_1 + L_2 + L_3}$$

Inductors in parallel



$$\boxed{\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}}$$

problem: 35 :—



⇒ Find the total inductance?

⇒ (a)

$L' = 6+6 = 12H$

$L'' = \left(\frac{1}{12} + \frac{1}{6}\right)^{-1} = 4H$

$\therefore L_T = 6H + 4H$

$= 10H$

Ans :—

(b)

$L' = \left(\frac{1}{4} + \frac{1}{4}\right)^{-1} = 2H$

$L'' = 2H + 4H = 6H$

$\therefore L_T = \left(\frac{1}{6} + \frac{1}{4}\right)^{-1} = 2.4H$

Ans :—

Sinusoidal Alternative Waveforms:

Ch no. 13

Wave-Forms:—

⇒ A waveform is a graphical representation of a signal in the form of a wave. It shows how the amplitude (strength or intensity) of the signal varies with time.

Magnitude of Induced Voltage:—

$$\boxed{e(t) = Blv \sin(\alpha) V = E_m \sin(\alpha) V}$$

Where, $E_m = Blv$ is constant

E_m = maximum Voltage.

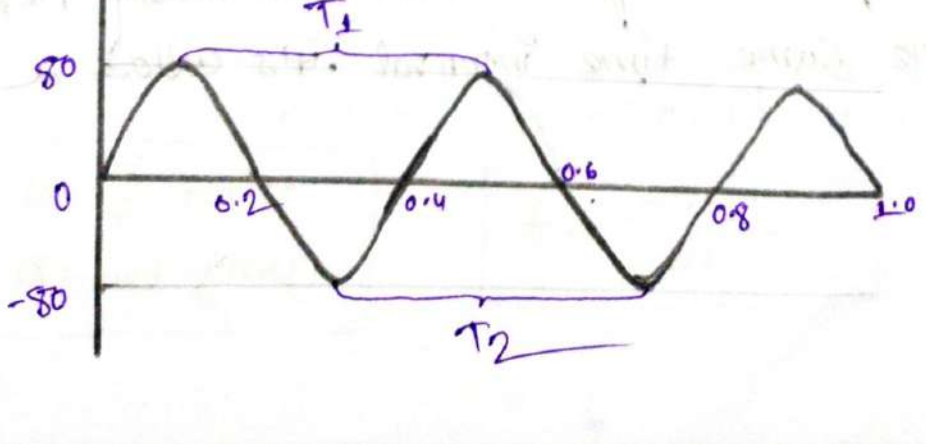
B = flux density

l = effective length of conductor

V = conductor velocity.

Note:- The average value of AC quantity is zero.

Some Definition Related To waveforms:—



Instantaneous Value:-

⇒ The magnitude of a waveform at any instant of time is called instantaneous value.

Peak or crest or Amplitude or Maximum Value:-

⇒ The maximum instantaneous value attained by an alternating quantity during positive and negative half-cycle is called it's amplitude or peak or crest maximum value. denote by E_m/ I_m/ V_m

here, $E_m = 80$ V

Peak-to-Peak Value:-

⇒ The full voltage between positive and negative peaks of the waveform, that is the peak-to-peak. denote by E_{p-p}, V_{p-p}, I_{p-p}

here, $E_{p-p} = 160$ V

Cycle:-

⇒ The portion of a waveform that continually repeats itself After the same time interval it's called a cycle.

Periodic waveform / time period:—

⇒ Time to complete one cycle. denote by T.

Frequency:—

⇒ Number of cycle per second. denote by f.

Relation between Frequency and Time period:—

$$\boxed{f = \frac{1}{T}\ Hz}$$

$$\boxed{T = \frac{1}{f}\ s}$$

Relation between Degree and Radian:—

$$\boxed{Radians = \left(\frac{\pi}{180^\circ}\right) \times (Degree)}$$

$$\boxed{Degree = \left(\frac{180^\circ}{\pi}\right) \times (radians)}$$

Angular Frequency or Velocity

$$\boxed{\begin{aligned} \omega &= \frac{\alpha}{t}\ rad/s \\ \alpha &= \omega t\ [rad] \end{aligned}}$$

$$\boxed{t = \frac{\alpha[rad]}{\omega}\ s}$$

Relation among Angular Frequency, Frequency and Time Period:—

$$\boxed{\omega = \frac{2\pi}{T} \text{ rad/s} = 2\pi f \text{ rad/s}}$$

$$\boxed{T = \frac{2\pi}{\omega} \text{ s}}$$

$$\boxed{f = \frac{\omega}{2\pi} \text{ Hz}}$$

Problem: 13.2

⇒ Find the period of periodic waveform with a frequency of (a) 60 Hz. (b) 1000 Hz.

⇒ (a)

In given:—

$f = 60\,Hz$

We know:—

$$T = \frac{1}{f} = \frac{1}{60} = 16.67\,ms$$

(b)

In given:—

$f = 1000\,Hz$

We know:—

$$T = \frac{1}{f} = \frac{1}{1000} = 1\,ms$$

General Format of Mathematical Equation For the Sinusoidal Voltage or Current with 0° Initial Angle:—

Induced EMF:—

$$\boxed{e(t) = E_m \sin\alpha = E_m \sin\omega t = E_m \sin 2\pi f \times t = E_m \sin \frac{2\pi}{T} \times t} \text{ V}$$

Voltage Drop:—

$$\boxed{v(t) = V_m \sin\alpha = V_m \sin\omega t = V_m \sin 2\pi f \times t = V_m \sin \frac{2\pi}{T} \times t} \text{ V}$$

Current:—

$$\boxed{i(t) = I_m \sin\alpha = I_m \sin\omega t = I_m \sin 2\pi f \times t = I_m \sin \frac{2\pi}{T} \times t} \text{ A}$$

problems:—

⇒ For the sinusoidal wave form

(a) What is the peak value?

(b) what is the instantaneous value of 15ms and at 20ms?

(c) what is the peak-to-peak value of the wave from?

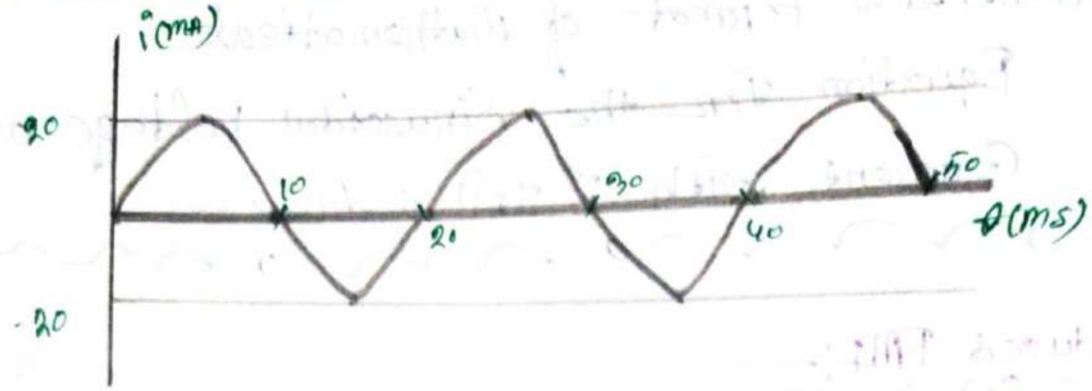
(d) what is the period of the waveform?

(e) How many cycle are shown.

(f) what is the frequency of wave from.

(g) What is the value of Angular frequency

(h) write the instantaneous equation of the waveform.



=)

(a)

The peak Value $I_m = 20mA$

(b)

In given:—

$$I_m = 20mA \quad T = 10ms$$
$$t_1 = 15ms$$
$$t_2 = 20ms$$

$$f = \frac{1}{T} = \frac{1}{20m} \quad \frac{100Hz}{2}$$

$$\omega = 2\pi f = 2\pi \times \frac{100Hz}{2}$$
$$= \frac{200\pi}{2}$$

We know:— $= 100\pi$

For t_1:—

$$i_1(t) = I_m \sin \omega t_1$$
$$= 20 \times \sin\left(200\pi \times \frac{180}{\pi} \times \frac{1}{2} \times 15 \times 10^{-3}\right)$$
$$= -20mA$$

For t_2:—

$$i_2(t) = I_m \sin \omega t_2$$
$$= 20 \times \sin\left(200\pi \times \frac{180}{\pi} \times \frac{1}{2} \times 20 \times 10^{-3}\right)$$
$$= 0mA$$

(c)

The peak to peak value is $I_{p-p} = 20 - (-20)$
$= 40 mA$

(d)

The period of waveform $T = 20ms$

(e)

There are 2.5 cycles.

(f)

We know:-

$$f = \frac{1}{T} = \frac{1}{20m} = 50Hz$$

$$\therefore T = \frac{1}{f} = \frac{1}{50} = 20mS$$

(g)

We know:-

$$\omega = 2\pi f = 2\times\pi\times 50 = 100\pi = 314.15 \text{ rad/s}$$

(h)

∴ The instantaneous:-

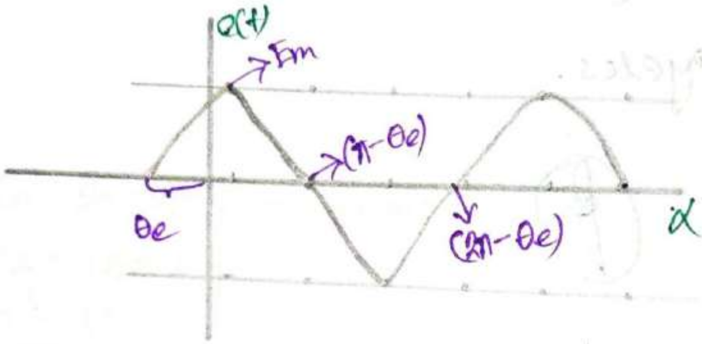
$$e(t) = 20 \sin 314.15 t$$

General Formate of Mathematical Equations for the Sinusoidal Voltage/Current with Phase Angle:—

Induced EMF:—

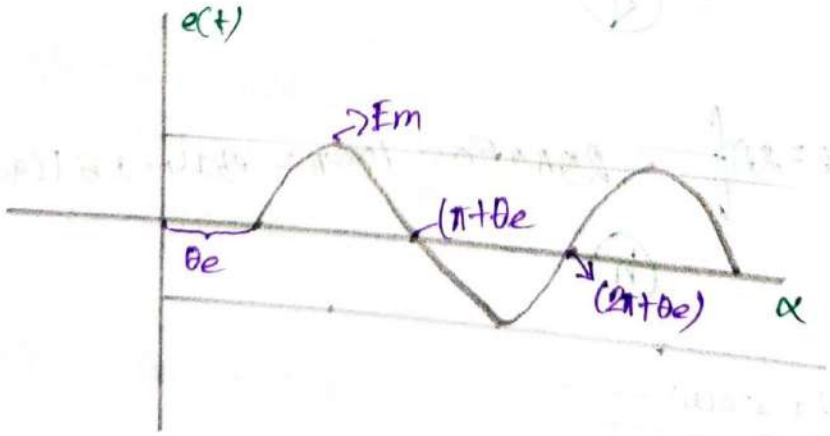
When θ_e Positive:—

$$\boxed{e(t) = E_m \sin(\alpha + \theta_e) = E_m \sin(\omega t + \theta_e)}\ V$$



When θ_e is negative:—

$$\boxed{e(t) = E_m \sin(\alpha - \theta_e) = E_m \sin(\omega t - \theta_e)}\ V$$



Note:— The other two are the same.

Initial Angle:-

$$\boxed{\text{Initial Angle} = -\text{ phase Angle}}$$

denote by θ_{e0}

$$\therefore \boxed{\theta_{e0} = -\theta_e}$$

Convention of $\sin\theta$:-

⇒ To identify the phase Angle from an equation at first $+\cos$, $-\cos$ and $-\sin$ convent to $+\sin$.

$$\boxed{\begin{aligned} \cos\alpha &= \sin(\alpha+90^\circ) \\ -\cos\alpha &= \sin(\alpha-90^\circ) \\ -\sin\alpha &= \sin(\alpha \pm 180^\circ) \end{aligned}}$$

Problem: 1:-

Identify the peak to peak value, angular frequency and phase angel and find find the initial angle for the waveform.

$$e(t) = -160\cos(400t - 20^\circ)\,V$$

⇒ 1st convent the waveform into sin :-

$$e(t) = 160\sin(400t - 20^\circ - 90^\circ)$$
$$= 160\sin(400t - 110^\circ)$$

$\therefore E_{p-p} = 2E_m = 160 \times 2 = 320\,V$

$\omega = 400$ rad/s

Phase Angle $\theta_e = -110^\circ$

Initial " $\theta_{e0} = -\theta_e = -(-110) = 110^\circ$

Phase Difference and Phase Relation Between Two waveforms:-

Conditions:-

1. The must be same type waveforms.
2. Their frequency or angular frequency or time period must be same.

$$\boxed{\text{Angle Difference} = \text{Voltage angle} - \text{Current Angle.}}$$

$$\boxed{\text{Phase Difference} = |\text{Angle Difference}|}$$

Cases:-

1) If Angle Difference = Voltage Angle − Current Angle = 0°

Phase Relation: v(t) and i(t) are in phse.

The starting points of v(t) and i(t) are same.

Case: 2:-

2) Angle Difference > 0°

Phase relation: V(t) leads i(t) or i(t) lags V(t)

Case 3:-

Angle Difference < 0°

phase relation:- V(t) lags i(t) or i(t) leads V(t)

problem 10.12

⇒ What is

(i) the phase difference.

(ii) the phase relationship between the sinusoidal waveforms of the following sets?

$$i(t) = -15\cos(\omega t + 150^\circ)\,A$$

$$v(t) = 10\sin(\omega t - 20^\circ)\,V$$

⇒ 1st Convert $i(t)$:-

$$i(t) = 15\sin(\omega t + 150 - 90^\circ)$$

$$= 15\sin(\omega t + 60^\circ)\,A$$

Is given:-

$$V(t) = 10\sin(\omega t - 20^\circ)\,V$$

(i)

We know:-

Angle Diff = Voltage Angle − Current Angle.

$$= -20^\circ - 60^\circ$$

$$= -80^\circ$$

Phase Diff = |Angle Diff|

$$= |-80^\circ|$$

$$= 80^\circ$$

(ii)

Angle Diff < 0°

So,

$v(t)$ lags $i(t)$ by 80° or

$i(t)$ leads $v(t)$ by 80°

Ans:-